# Department of Mechanical Engineering


# Computational Engineering - Lab Report

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Section-**B**

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# Contents

## MATLAB codes of all Lab sessions

All MATLAB codes, generated figures, and supplementary data for every lab session reported in this document is available in the following GitHub repository

**Repository URL:**

https://github.com/mztcoding/CE_lab_MATLAB

## Lab Session 1

# MATLAB Basics and Numerical Methods

## 1.1 Objective

- Introduce the MATLAB programming environment including its key components: Command Window, Editor, Workspace, and Current Folder.
- Develop proficiency in defining variables, vectors, matrices, and implementing control structures such as `for` and `while` loops and conditional statements.
- Generate 2D and 3D plots, subplots, and histograms using MATLAB's built-in visualization tools.
- Understand and quantify numerical approximation errors by studying the convergence of the Maclaurin series expansion of $\sin(x)$ with an increasing number of terms.

## 1.2 Introduction and Theoretical Background

### 1.2.1 MATLAB Environment

MATLAB (MATrix LABoratory) is a high-level programming language designed for numerical computation, algorithm development, and data visualisation [1, 2]. Its matrix-oriented syntax makes it particularly suited to engineering and scientific computing.

The key interface components are the **Command Window** (interactive execution), **Editor** (script and function authoring), **Workspace** (variable management), and **Current Folder** (file access). Scripts typically begin with the following three commands to ensure a clean environment:

```
clc;        % Clear command window
clear;      % Remove workspace variables
close all   % Close all figure windows
```

### 1.2.2 Core Language Features

All data in MATLAB is stored as matrices; scalars and vectors are special cases. Element-wise operations use dot notation (`.*`, `./`, `.^`). Variables require no type declaration, and functions are defined with the `function` keyword. The language supports standard control structures including `for`/`while` loops and `if`/`else` conditionals, which form the basis of iterative numerical algorithms.

```
x = 0:0.1:10;              % Row vector
z = x.^2 + 3.*x;           % Element-wise operations

function y = squareVal(x)   % User-defined function
    y = x.^2;
end

for i = 1:5                % Loop
    disp(i);
end

if x > 0                   % Conditional
    disp('Positive');
else
    disp('Non-positive');
end
```

### 1.2.3 Plotting and Visualisation

MATLAB provides built-in functions for generating annotated 2D and 3D plots. The `plot` function is used for standard line graphs, `semilogy` for logarithmic-scale convergence plots, `subplot` for multi-panel figures, and `surf`/`plot3` for three-dimensional visualisation.

```
x = 0:0.1:2*pi;
plot(x, sin(x), 'b-', 'LineWidth', 1.5);
xlabel('x');  ylabel('sin(x)');
title('Sine Function');  grid on;
```

### 1.2.4 Running Code and Saving Results

MATLAB scripts are executed with F5 or by typing the script name in the Command Window; selected lines run with F9. Output uses `fprintf` or `disp`, and a semicolon suppresses it. Variables can be saved to `.mat` files and results exported to text or spreadsheet files.

```
fprintf('Root: x = %.6f  (iterations: %d)\n', x, n); %
   formatted print
save('results.mat', 'x', 'n');          % save variables to .
   mat
load('results.mat');                     % reload in a new
   session
writematrix(data, 'output.xlsx');        % export to
   spreadsheet
```

### 1.2.5 Numerical Methods in MATLAB

Numerical methods provide approximate solutions to mathematical problems that are difficult to solve analytically. MATLAB's matrix-based environment, built-in functions, and visualization tools make it ideal for implementing these methods [3].

Key techniques include:

- **Root-Finding:** Methods like Newton-Raphson and Fixed-Point iteration for solving nonlinear equations.
- **Numerical Integration/Differentiation:** Approximating derivatives and integrals using finite differences, trapezoidal, or Simpson's rule.
- **Linear Systems:** Solving systems of equations with matrix operations and the backslash operator.
- **Series Approximation:** Using Maclaurin or Taylor series to approximate functions and study convergence.
- **Error Analysis:** Quantifying approximation errors and studying convergence in iterative methods.

MATLAB allows efficient coding of these methods, visualization of results, and analysis of numerical accuracy and stability.

### 1.2.6 Taylor Series and Numerical Errors

Any sufficiently smooth function $f(x)$ can be represented as an infinite power series expanded about a point $a$ [4, 3]. When $a = 0$, the series is called the **Maclaurin series**. For the sine function, the Maclaurin series is:

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \tag{1.1}$$

where $x$ must be expressed in radians. A finite truncation of this series to $N$ terms introduces a **truncation error**. Three standard error metrics are defined as follows:

$$\text{True Error:} \quad E_t = f_{\text{true}} - f_{\text{approx}} \tag{1.2}$$

$$\text{Absolute Error:} \quad |E_t| \tag{1.3}$$

$$\text{Relative Error:} \quad \varepsilon_t = \frac{|E_t|}{|f_{\text{true}}|} \tag{1.4}$$

As the number of terms $N$ increases, the approximation converges to the true value and the errors decrease. Plotting the error on a semi-logarithmic scale illustrates the rate of convergence clearly. For this lab session, convergence is studied at $x = 60^\circ$ (i.e., $x = \pi/3$ radians).

## 1.3 Methodology

The lab session is conducted in MATLAB through the following steps:

1. Define the domain as $x \in [-360^\circ,\ 360^\circ]$ in steps of $1^\circ$ and convert to radians using `deg2rad`.
2. Compute the true value $\sin(x)$ using MATLAB's built-in `sin` function as the reference.
3. Implement a nested loop to compute the Maclaurin series approximation for $N = 1, 2, \ldots, 10$ terms, storing each approximation in a row of a matrix `T`.
4. Evaluate the series at $x = 60^\circ$ to study pointwise convergence with respect to the number of terms.
5. Compute true error, absolute error, and relative error at $x = 60^\circ$ for each $N$.
6. Generate four plots: (i) the true $\sin(x)$ alongside all ten approximations over the full domain; (ii) approximated and true values at $x = 60^\circ$ versus number of terms; (iii) absolute error versus number of terms on a semi-log scale; and (iv) relative error versus number of terms on a semi-log scale.

## 1.4 Algorithm

1. Set the number of terms $N_{\max} = 10$.
2. Define $x_{\text{deg}} = [-360, -359, \ldots, 360]$ and convert to radians: $x = x_{\text{deg}} \times \pi/180$.
3. Compute the true function: $f_{\text{true}} = \sin(x)$.
4. **For** $N = 1$ to $N_{\max}$:
   (a) Initialise partial sum $T_{\text{sum}} = 0$.
   (b) **For** $n = 0$ to $N - 1$: $T_{\text{sum}} \mathrel{+}= \dfrac{(-1)^n}{(2n+1)!}\, x^{2n+1}$
   (c) Store $T(N, :) = T_{\text{sum}}$.
5. Plot $f_{\text{true}}$ and all rows of $T$ against $x_{\text{deg}}$ on the same axes.
6. Set $x_{60} = \pi/3$. Compute the true value $f_{60} = \sin(x_{60})$.
7. Evaluate the series for $N = 1, \ldots, N_{\max}$ at $x_{60}$ to obtain the array `approx_val`.
8. Compute: $E_t = f_{60} - $ `approx_val`, $|E_t|$, and $\varepsilon_t = |E_t|/|f_{60}|$.
9. Plot convergence, absolute error, and relative error versus $N$.

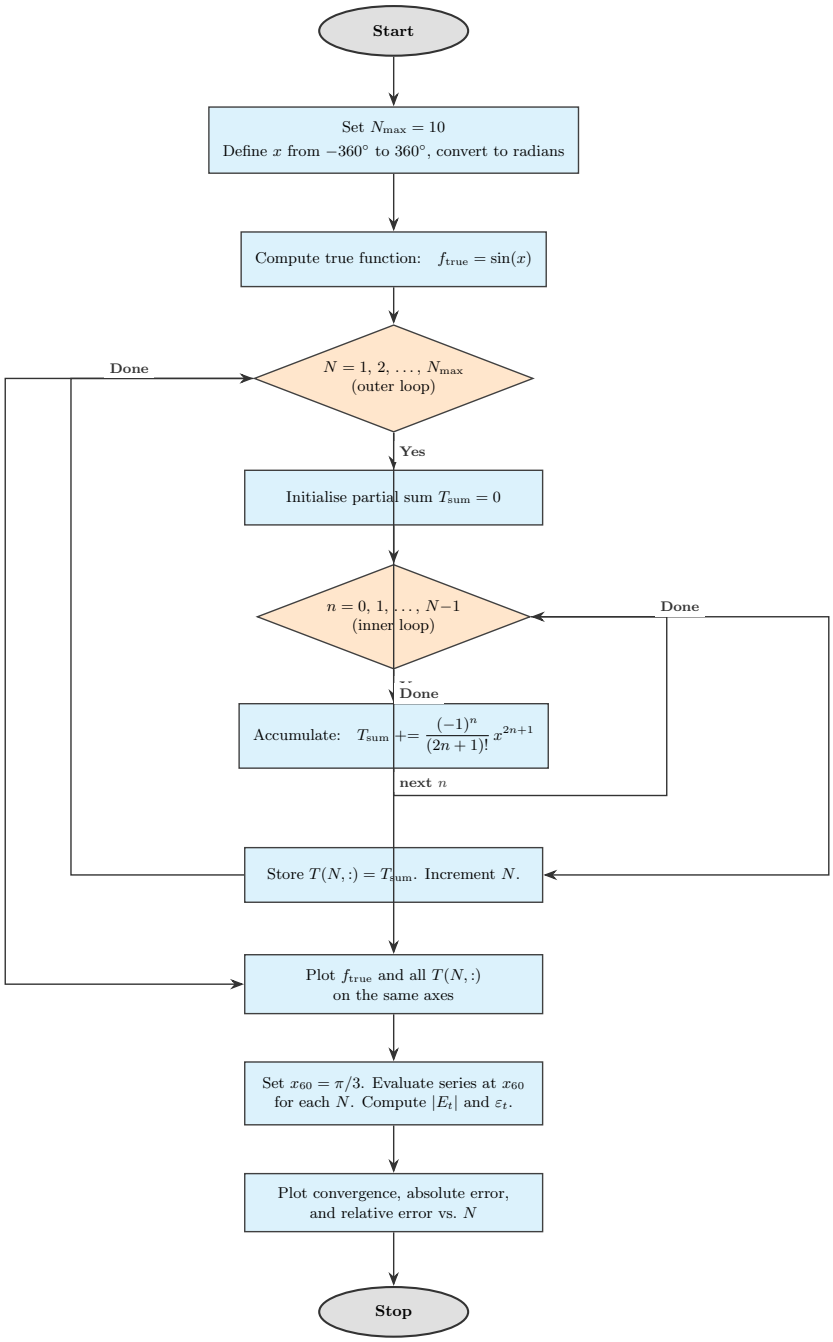


Figure 1.1: Flowchart of the Taylor series algorithm for $\sin(x)$: nested loops, error computation, and plotting.

## 1.5 MATLAB Implementation

This section presents the complete MATLAB codes for the three problems covered in Lab Session 1: (i) solving the quadratic equation, (ii) the falling parachutist problem (analytical and numerical), and (iii) the Taylor series approximation of $\sin(x)$ with convergence and error analysis.

## Problem 1: Quadratic Equation Solver

The roots of $ax^2 + bx + c = 0$ are computed using the quadratic formula, with handling for complex roots and the case $a = 0$.

```
clc; clear; close all;

while true
    a = input('Enter coefficient a: ');
    b = input('Enter coefficient b: ');
    c = input('Enter coefficient c: ');
    if a == 0
        if b == 0
            disp('Not an equation (a = b = 0).');
        else
            fprintf('Linear equation. Root: x = %.6f\n', -c/b
               );
        end
    else
        % ----- Discriminant -----
        disc = b^2 - 4*a*c;
        if disc > 0
            x1 = (-b + sqrt(disc)) / (2*a);
            x2 = (-b - sqrt(disc)) / (2*a);
            fprintf('Two real roots:\n  x1 = %.6f\n  x2 = %.6
               f\n', x1, x2);
        elseif disc == 0
            x1 = -b / (2*a);
            fprintf('One repeated real root: x = %.6f\n', x1)
               ;
        else
            realPart = -b / (2*a);
            imagPart = sqrt(-disc) / (2*a);

            fprintf('Complex roots:\n');
            fprintf('  x1 = %.6f + %.6fi\n', realPart,
               imagPart);
            fprintf('  x2 = %.6f - %.6fi\n', realPart,
               imagPart);
        end
    end
    again = input('\nSolve another equation? (1 = Yes, 0 = No
       ): ');
    if again ~= 1
        break;
    end
end
```

Code 1.1: Quadratic equation solver with user input and loop.

## Problem 2: Falling Parachutist – Analytical and Numerical Solutions

A parachutist of mass $m = 68.1$ kg and drag coefficient $c = 12.5$ kg/s is released from rest. The analytical velocity is:

$$v(t) = \frac{gm}{c}\left(1 - e^{-(c/m)t}\right) \tag{1.5}$$

The Euler numerical approximation uses a step size of $\Delta t = 2$ s:

$$v_{i+1} = v_i + \left(g - \frac{c}{m}v_i\right)\Delta t \tag{1.6}$$

For the second-order drag model (Problem 1.3), the upward drag force is $F_U = -c'v^2$ with $c' = 0.225$ kg/m.

```
clc; clear; close all;
% Parameters
m = 68.1; c = 12.5; g = 9.8; dt = 2; t_end = 14;
c2 = 0.225;  % 2nd-order drag coefficient (kg/m)

% Analytical solution
t_anal = 0:0.1:t_end;
v_anal = (g*m/c) .* (1 - exp(-(c/m).*t_anal));

% Euler -- linear drag
t_num = 0:dt:t_end;
v_num = zeros(size(t_num));
for i = 1:length(t_num)-1
    v_num(i+1) = v_num(i) + (g - (c/m)*v_num(i)) * dt;
end
% Euler -- second-order drag
v_num2 = zeros(size(t_num));
for i = 1:length(t_num)-1
    v_num2(i+1) = v_num2(i) + (g - (c2/m)*v_num2(i)^2) * dt;
end

% Plot
figure;
plot(t_anal,v_anal,'k-','LineWidth',2,'DisplayName','
   Analytical');       hold on;
plot(t_num,v_num,'bo-','MarkerFaceColor','b','DisplayName','
   Euler (linear)');
plot(t_num,v_num2,'rs-','MarkerFaceColor','r','DisplayName','
   Euler (2nd-order)');
yline(g*m/c,'k--','DisplayName','Terminal velocity');
xlabel('Time (s)'); ylabel('Velocity (m/s)');
title('Falling Parachutist'); legend('Location','southeast');
    grid on;
```

Code 1.2: Analytical and numerical (Euler) solutions for the falling parachutist problem.

## Problem 3: Taylor Series Approximation of $\sin(x)$

```
clc; clear; close all;

% Build Maclaurin approximations (N = 1..10 terms)
Nterms = 10;
x = deg2rad(-360:1:360);  f_true = sin(x);
T = zeros(Nterms, length(x));
for N = 1:Nterms
    s = zeros(size(x));
    for n = 0:N-1
        s = s + ((-1)^n/factorial(2*n+1)) .* x.^(2*n+1);
    end
    T(N,:) = s;
end

% Plot 1 -- all approximations vs true sin(x)
figure; plot(rad2deg(x), f_true,'k','LineWidth',3,'
   DisplayName','True sin(x)'); hold on;
for N = 1:Nterms, plot(rad2deg(x),T(N,:),'--','DisplayName',
   sprintf('%d terms',N)); end
xlabel('x (deg)'); ylabel('sin(x)'); legend('Location','best'
   ); grid on;
xlim([-360 360]); ylim([-1.5 1.5]); title('Maclaurin
   Approximations of sin(x)');

% Convergence and error at x = 60 degrees
x60 = deg2rad(60);  tv = sin(x60);
av = zeros(1,Nterms);
for N = 1:Nterms
    for n = 0:N-1, av(N) = av(N) + ((-1)^n/factorial(2*n+1))*
       x60^(2*n+1); end
end
abs_err = abs(tv - av);  rel_err = abs_err/abs(tv);

% Plot 2 -- convergence at 60 deg
figure; plot(1:Nterms,av,'o-'); hold on; yline(tv,'r--');
xlabel('Terms'); ylabel('sin(60{\circ})'); title('Convergence
    at x=60{\circ}'); grid on;

% Plot 3 & 4 -- errors
figure; semilogy(1:Nterms,abs_err,'s-'); xlabel('Terms');
   ylabel('Absolute Error'); grid on;
figure; semilogy(1:Nterms,rel_err,'d-'); xlabel('Terms');
   ylabel('Relative Error'); grid on;
```

Code 1.3: Maclaurin series approximation of $\sin(x)$, convergence study and error analysis.

## 1.6 Results and Discussion

### 1.6.1 Falling Parachutist: Analytical vs. Numerical Solution

The analytical solution describes the exact velocity of the parachutist under linear drag, while the Euler numerical method provides an approximate solution using discrete time steps. Both approaches model the same physical motion and show similar trends in velocity as the parachutist accelerates and approaches terminal velocity.

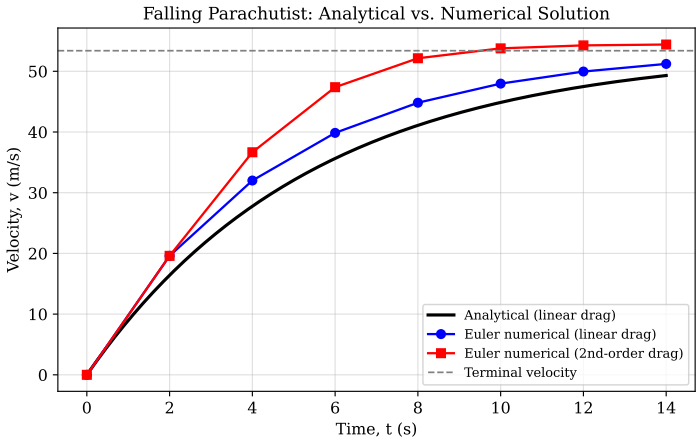


Figure 1.2: Falling parachutist: analytical solution (linear drag), Euler numerical solution (linear drag), and Euler numerical solution (second-order drag, $c' = 0.225$ kg/m).

### 1.6.2 Taylor Series Approximations over the Full Domain

The first plot displays the true $\sin(x)$ function alongside all ten Maclaurin approximations over $[-360°, 360°]$. With only one or two terms the series closely matches the true function only in the vicinity of the origin. As $N$ increases, the interval over which the approximation remains accurate widens progressively. Higher-order approximations eventually agree with the true curve over nearly the entire plotted domain, illustrating the well-known property that the Maclaurin series for $\sin(x)$ has an infinite radius of convergence.

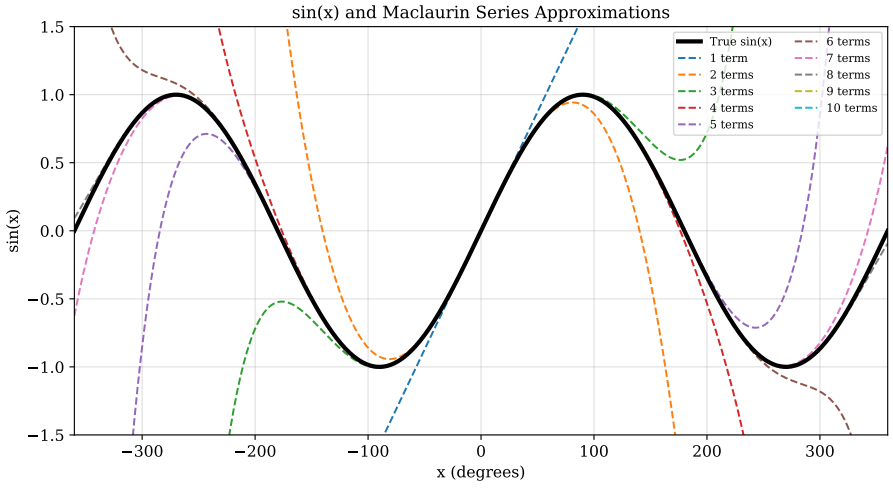


Figure 1.3: True $\sin(x)$ and Maclaurin series approximations for $N = 1$ to 10 over $[-360°, 360°]$.

### 1.6.3 Convergence at $x = 60°$

The second plot shows how the series approximation at $x = 60°$ approaches the true value of $\sin(60°) = \sqrt{3}/2 \approx 0.8660$ as more terms are included. With only one term ($\sin(x) \approx x$) the approximation is noticeably poor; by approximately $N = 5$ the approximation is visually indistinguishable from the true value.

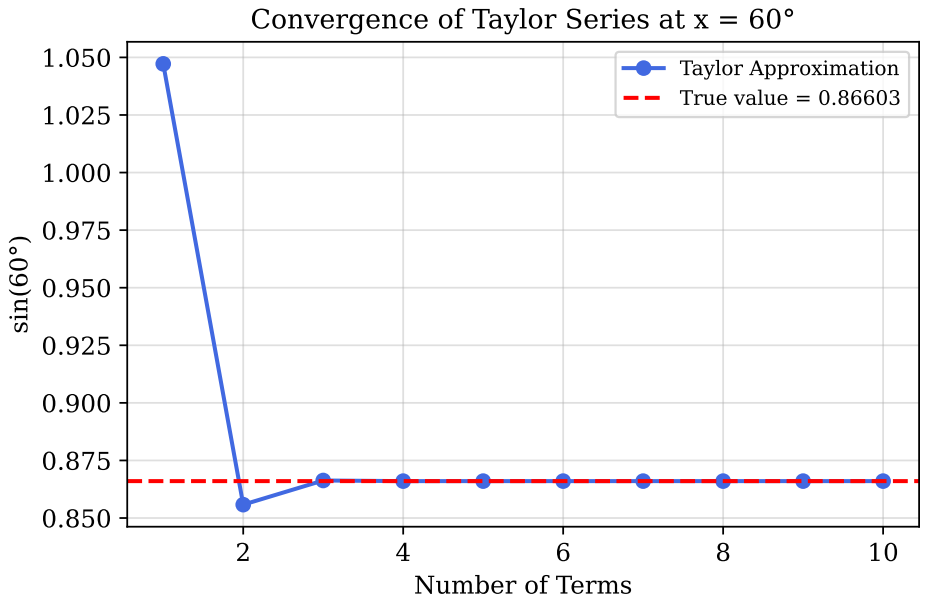


Figure 1.4: Convergence of the Maclaurin approximation at $x = 60°$ as a function of the number of terms.

### 1.6.4 Error Analysis

Plots of absolute and relative error on semi-logarithmic axes confirm that both errors decrease monotonically with increasing $N$. The rapid, approximately exponential decay of the error reflects the factorial growth of the denominators in the series coefficients, which causes each successive term to contribute less than the previous one. This behaviour is characteristic of a **geometrically converging** series near the expansion point.

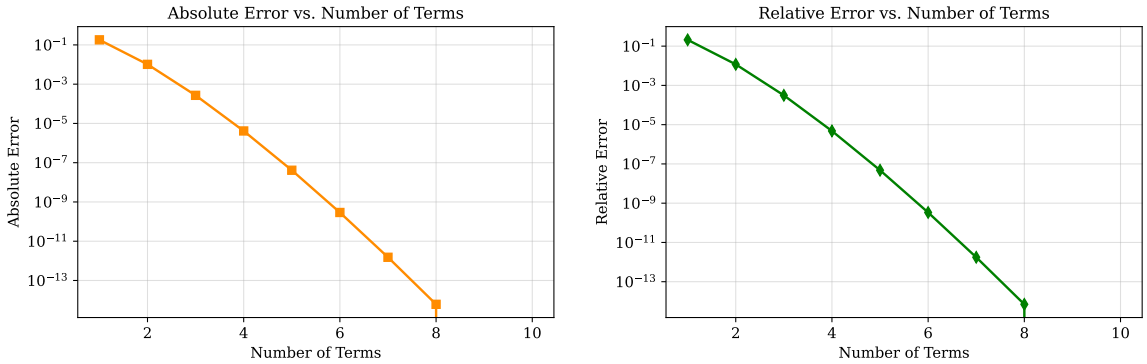


Figure 1.5: Absolute and relative error versus number of Maclaurin terms at $x = 60°$ (semi-log scale).

The results provide a detailed insight into the behaviour, accuracy, and limitations of the numerical approaches implemented in this lab. The graphical comparison for the

parachutist problem illustrates how discretisation in Euler's method introduces noticeable deviation from the analytical solution, particularly at larger time steps, while also showing the influence of different drag models on system response. In the Taylor series study, the progressive improvement in approximation with increasing terms demonstrates how convergence is strongly dependent on both the number of terms and the distance from the expansion point. The semi-log error plots further reveal a consistent and rapid reduction in error, indicating stable numerical behaviour and reinforcing the theoretical expectations of series convergence. These observations collectively highlight the importance of step size selection, model formulation, and convergence assessment when applying numerical techniques to engineering problems.

## 1.7 Conclusion

This lab session established familiarity with the MATLAB environment and its core programming constructs. Students practised defining arrays, implementing nested loops, creating user-defined functions, and generating annotated plots. The study of the Maclaurin series for $\sin(x)$ demonstrated that truncating an infinite series introduces a quantifiable truncation error, and that this error diminishes as more terms are retained. The parachutist problem illustrated the difference between analytical and numerical (Euler) solutions for a first-order ODE and the effect of using a first-order versus second-order drag model. Overall, the lab strengthened practical implementation skills in numerical computing and reinforced the structured approach required to translate mathematical formulations into efficient computational algorithms.

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## Lab Session 2

# Root Finding – Bisection Method

## 2.1 Objective

- Implement the bisection method in MATLAB to locate real roots of nonlinear equations of the form $f(x) = 0$.
- Study the influence of the initial bracketing interval and convergence tolerance on the number of iterations required.
- Validate the computed root graphically by plotting the function and marking the successive root approximations.
- Visualise the convergence behaviour by plotting the approximate percentage relative error against the iteration count.

## 2.2 Introduction and Theoretical Background

### 2.2.1 Root-Finding Problem

Many engineering problems reduce to solving a nonlinear equation of the form [4, 5]:

$$f(x) = 0 \tag{2.1}$$

Closed-form analytical solutions are often unavailable, making iterative numerical methods necessary. Such methods are broadly classified as either **bracketing methods**, which require an initial interval $[x_l, x_u]$ that is known to contain a root, or **open methods**, which require only an initial guess.

### 2.2.2 Bisection Method

The bisection method is the simplest and most robust bracketing technique [4, 3]. It relies on the **Intermediate Value Theorem**: if $f(x_l)$ and $f(x_u)$ have opposite signs, i.e. $f(x_l) \cdot f(x_u) < 0$, then at least one root exists in the interval $[x_l, x_u]$.

At each iteration, the midpoint is computed:

$$x_r = \frac{x_l + x_u}{2} \tag{2.2}$$

and the subinterval containing the root is identified by evaluating the sign of $f(x_r)$:

$$\begin{cases} \text{Replace } x_u \text{ with } x_r & \text{if } f(x_l) \cdot f(x_r) < 0 \\ \text{Replace } x_l \text{ with } x_r & \text{if } f(x_r) \cdot f(x_u) < 0 \\ x_r \text{ is the root} & \text{if } f(x_r) = 0 \end{cases} \tag{2.3}$$

The convergence of the bisection method is guaranteed for continuous functions satisfying the sign-change criterion. The approximate percentage relative error at each iteration is:

$$\varepsilon_a = \left| \frac{x_r^{\text{new}} - x_r^{\text{old}}}{x_r^{\text{new}}} \right| \times 100\% \tag{2.4}$$

Iteration continues until $\varepsilon_a < \varepsilon_s$, where $\varepsilon_s$ is the specified stopping criterion. In this lab session, $\varepsilon_s = 0.01\%$.

The method halves the interval at every step, so after $n$ iterations the maximum possible error in the root is:

$$\Delta x_n = \frac{x_u - x_l}{2^n} \tag{2.5}$$

This linear (first-order) convergence rate is reliable but slower than higher-order methods discussed in later sessions.

## 2.3 Methodology

1. Define the nonlinear function $f(x)$ as an anonymous function handle in MATLAB.
2. Prompt the user to enter the bracketing points $x_l$ and $x_u$ via `input()`.
3. Verify the sign-change condition $f(x_l) \cdot f(x_u) < 0$ before proceeding.
4. Implement the bisection iteration inside a `while` loop, terminating when $\varepsilon_a < 0.01\%$ or a maximum iteration count is reached.
5. Record $x_r$ and $\varepsilon_a$ at each iteration in arrays for post-processing.
6. Plot the function over an appropriate interval, marking each successive root estimate with its iteration number directly on the graph.
7. Plot the approximate percentage relative error versus iteration number to visualise convergence.

## 2.4 Algorithm

1. Define $f(x)$.
2. Input lower bound $x_l$ and upper bound $x_u$ from the user.
3. Check: if $f(x_l) \cdot f(x_u) \geq 0$, display an error message and terminate.
4. Initialise: iteration counter $i = 0$, $\varepsilon_a = \infty$, $x_r^{\text{old}} = x_l$.
5. **While** $\varepsilon_a > \varepsilon_s$ **and** $i < i_{\max}$:
   (a) $i \leftarrow i + 1$.
   (b) $x_r \leftarrow (x_l + x_u)/2$.
   (c) Compute $\varepsilon_a = |(x_r - x_r^{\text{old}})/x_r| \times 100\%$.
   (d) If $f(x_l) \cdot f(x_r) < 0$: set $x_u \leftarrow x_r$.

(e) Else if $f(x_r) \cdot f(x_u) < 0$: set $x_l \leftarrow x_r$.

(f) Else: root found exactly; break.

(g) $x_r^{\text{old}} \leftarrow x_r$.

(h) Store $x_r$ and $\varepsilon_a$.

6. Report the root $x_r$ and total iteration count.

7. Generate the two required plots.

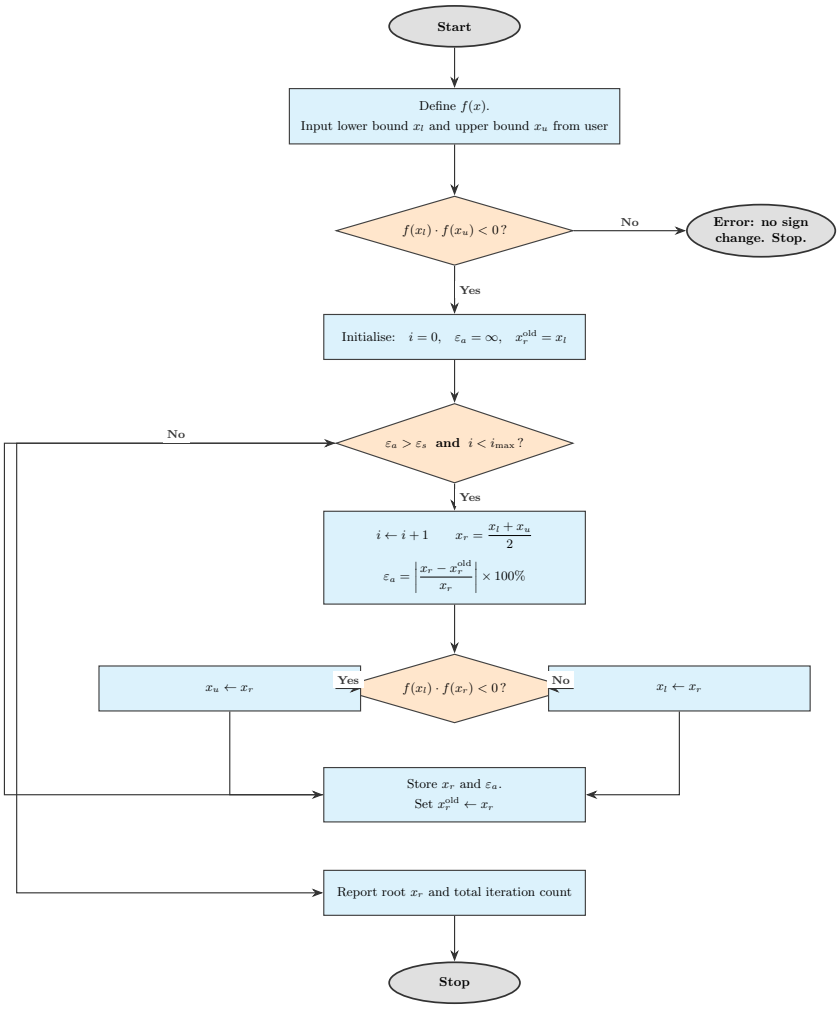


Figure 2.1: Flowchart of the Bisection Method algorithm.

## 2.5 MATLAB Implementation

Two problems from Chapra & Canale are solved using the bisection method. **Problem 1** (Example 5.1) finds the drag coefficient $c$ required for a parachutist of mass $m = 68.1$ kg to reach $v = 40$ m/s after $t = 10$ s of free fall. **Problem 2** (Example 5.2) locates a root of $f(x) = \sin 10x + \cos 3x$ on $[0, 5]$.

## Problem 1: Parachutist Drag Coefficient

The governing equation to be solved is:

$$f(c) = \frac{9.8 \times 68.1}{c}\left(1 - e^{-(c/68.1)\times 10}\right) - 40 = 0 \tag{2.6}$$

A sign change is observed between $c = 12$ and $c = 16$.

```
% Problem definition
m = 68.1; g = 9.8; v_target = 40; t = 10;
f = @(c) (g*m./c).*(1 - exp(-(c/m)*t)) - v_target;
tol = 0.0001; max_iter = 100;

xl = input('Enter lower bracket xl: ');
xu = input('Enter upper bracket xu: ');
if f(xl)*f(xu) >= 0, error('No sign change in bracket.'); end

% Bisection iteration
xr_old = xl; ea = inf; roots_arr = []; ea_arr = []; iter = 0;
while ea > tol && iter < max_iter
    iter = iter + 1;
    xr   = (xl + xu) / 2;
    if iter > 1, ea = abs((xr - xr_old)/xr)*100; end
    roots_arr(end+1) = xr;  ea_arr(end+1) = ea;
    if     f(xl)*f(xr) < 0, xu = xr;
    elseif f(xr)*f(xu) < 0, xl = xr;
    else, break; end
    xr_old = xr;
end
fprintf('Root: c = %.6f  (%d iterations)\n', xr, iter);
```

Code 2.1: Bisection method applied to the parachutist drag coefficient problem.

## Problem 2: Roots of $f(x) = \sin 10x + \cos 3x$

This function has several roots on $[0, 5]$. The code locates one root in a user-specified sub-interval.

```
% Problem definition
f = @(x) sin(10*x) + cos(3*x);
tol = 0.0001; max_iter = 100;

xl = input('Enter lower bracket xl: ');
xu = input('Enter upper bracket xu: ');
if f(xl)*f(xu) >= 0, error('No sign change in bracket.'); end

% Bisection iteration
xr_old = xl; ea = inf; roots_arr = []; ea_arr = []; iter = 0;
while ea > tol && iter < max_iter
    iter = iter + 1;
```

```
    xr   = (xl + xu) / 2;
    if iter > 1, ea = abs((xr-xr_old)/xr)*100; end
    roots_arr(end+1) = xr;  ea_arr(end+1) = ea;
    if      f(xl)*f(xr) < 0, xu = xr;
    elseif f(xr)*f(xu) < 0, xl = xr;
    else, break; end
    xr_old = xr;
end
fprintf('Root: x = %.6f  (%d iterations)\n', xr, iter);
```

Code 2.2: Bisection method applied to $f(x) = \sin 10x + \cos 3x$.

## 2.6 Results and Discussion

### 2.6.1 Function Graph and Root Estimates

The function $f(x)$ is plotted over a suitable interval, and the root approximation at each bisection iteration is marked on the graph. The successive estimates are labelled $1, 2, 3, \ldots$ to show how the search progressively narrows about the true root. An appropriate bracketing interval is one where the function changes sign; an invalid interval (no sign change) is rejected by the algorithm before iteration begins.

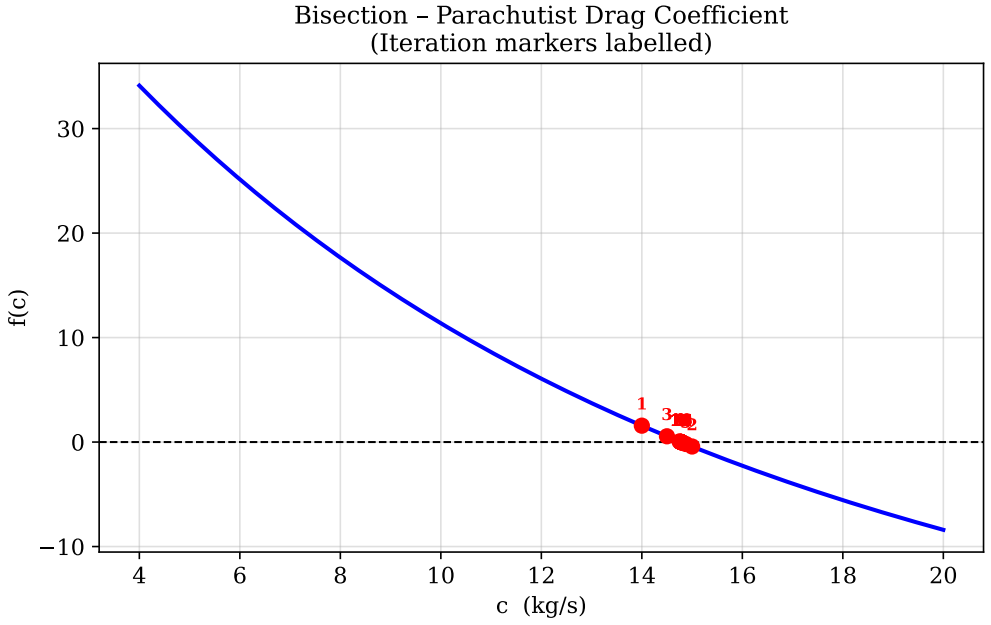


Figure 2.2: Bisection on $f(c)$ (parachutist drag coefficient): function plot with successive root approximations labelled by iteration number.

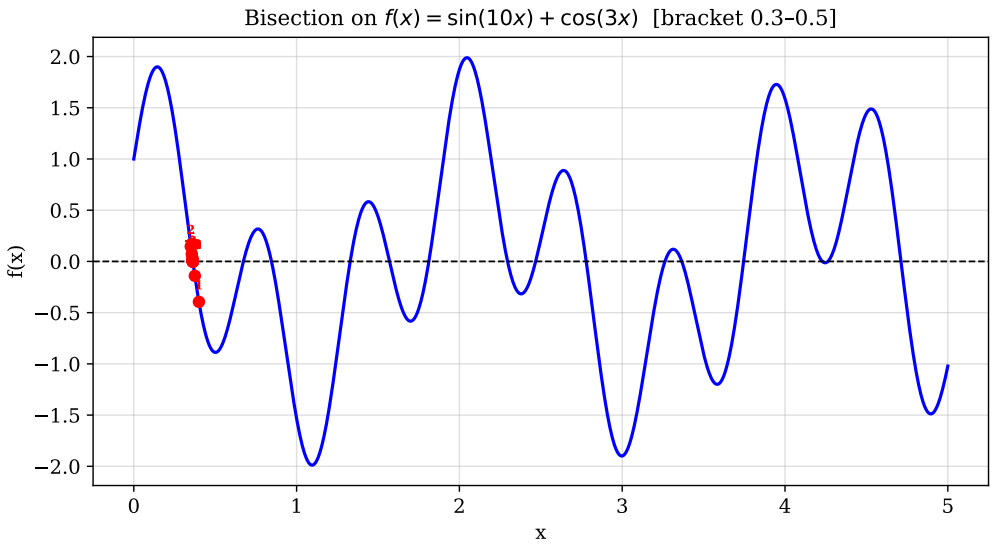


Figure 2.3: Bisection on $f(x) = \sin 10x + \cos 3x$: root approximations labelled by iteration number on the function plot.

### 2.6.2 Convergence Plot

The approximate percentage relative error decreases monotonically with each iteration. Because the interval is halved at every step, the error reduces by a constant factor of approximately 50% per iteration, confirming the first-order (linear) convergence rate of the bisection method. Approximately $n = \lceil \log_2((x_u - x_l)/\varepsilon_s) \rceil$ iterations are needed to achieve a prescribed tolerance, which can be estimated before execution.

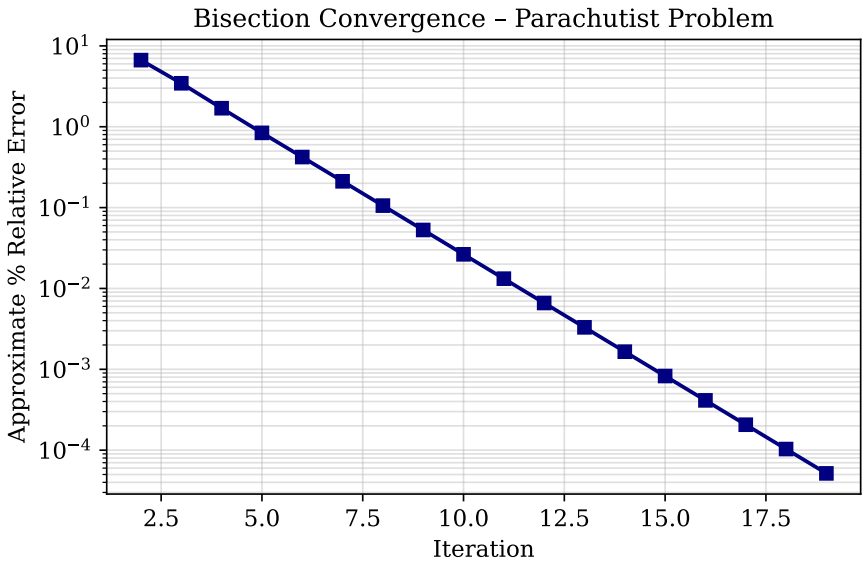


Figure 2.4: Approximate % relative error vs. iteration — bisection on the parachutist drag coefficient problem.

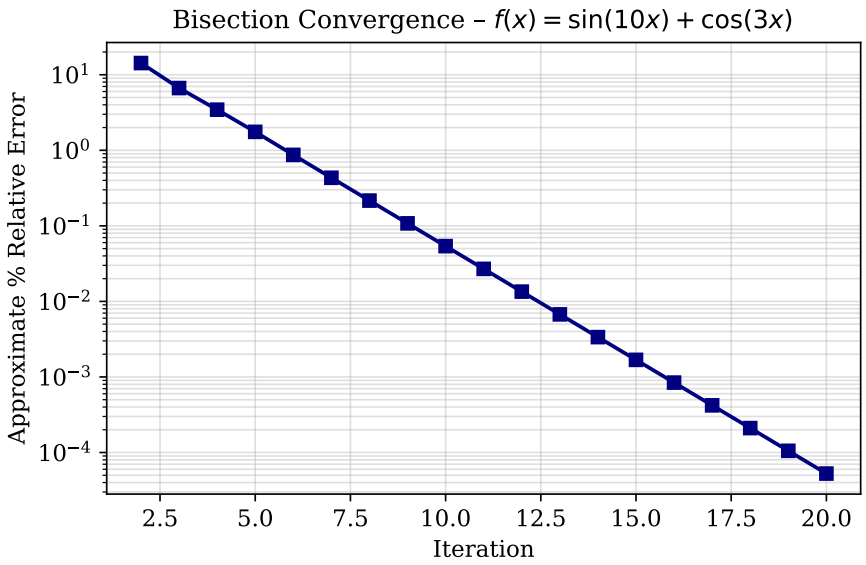


Figure 2.5: Approximate % relative error vs. iteration — bisection on $f(x) = \sin 10x + \cos 3x$.

### 2.6.3 Effect of Initial Bracket

A wider initial bracket requires more iterations to achieve the same tolerance, whereas a bracket very close to the true root converges in fewer steps. However, unlike open methods, the bisection method converges regardless of how wide the initial bracket is, provided the sign-change condition is satisfied. This unconditional convergence is the primary advantage of the method.

## 2.7 Conclusion

The bisection method provides a robust and guaranteed means of locating roots of continuous functions, provided a valid bracketing interval is identified. Its linear convergence rate makes it slower than open methods such as Newton–Raphson, but its reliability and simplicity render it valuable as a baseline and as a fallback when other methods fail. The lab session demonstrated how the initial bracket and stopping tolerance directly control the iteration count, and how the error reduces by half at each step in a predictable and systematic manner. In addition, the graphical visualisation of successive root estimates reinforced the conceptual understanding of interval halving and convergence behaviour, making the iterative process more intuitive. The implementation in MATLAB further highlighted how numerical algorithms can be efficiently translated into code for practical engineering applications. Overall, this study emphasises that while the bisection method may not be computationally optimal, its stability, transparency, and guaranteed convergence make it an essential foundational tool in numerical analysis.

## Lab Session 3

# Root Finding – False Position Method

## 3.1 Objective

- Implement the false position (Regula Falsi) method in MATLAB to solve nonlinear equations.
- Compare the convergence speed of the false position method against the bisection method on the same problems.
- Identify and discuss the limitations of the false position method, in particular the phenomenon of one-sided stagnation.
- Visualise the successive root approximations and the relative error convergence for both methods on comparative plots.

## 3.2 Introduction and Theoretical Background

### 3.2.1 Motivation

The bisection method guarantees convergence but uses only the *sign* of the function at the bracket endpoints, discarding magnitude information. The false position method retains the bracketing structure for reliability while using linear interpolation between the two endpoints to produce a better-informed estimate of the root, often resulting in faster convergence [4, 6].

### 3.2.2 False Position (Regula Falsi) Method

Like bisection, the false position method maintains a bracket $[x_l, x_u]$ satisfying $f(x_l) \cdot f(x_u) < 0$. However, instead of the arithmetic midpoint, the root estimate is the $x$-intercept of the chord (secant line) connecting the points $(x_l, f(x_l))$ and $(x_u, f(x_u))$:

$$x_r = x_u - \frac{f(x_u)\,(x_l - x_u)}{f(x_l) - f(x_u)} \tag{3.1}$$

This is equivalent to applying the linear interpolation formula. The update rule is identical to bisection: the half of the interval whose endpoint has the same sign as $f(x_r)$ is discarded and replaced by $x_r$.

The approximate percentage relative error is tracked in the same manner as bisection:

$$\varepsilon_a = \left| \frac{x_r^{\text{new}} - x_r^{\text{old}}}{x_r^{\text{new}}} \right| \times 100\% \tag{3.2}$$

### 3.2.3 Limitations – One-Sided Stagnation

A well-known weakness of the false position method is that for strongly curved functions one endpoint of the bracket can remain fixed throughout all iterations, with only the other endpoint being updated [3, 6]. This causes the method to converge from one side only and can result in slower convergence than bisection in such cases. Several modifications (e.g. the Illinois algorithm) address this deficiency by artificially halving the function value at the stagnant endpoint, but these are beyond the scope of the present lab session.

## 3.3 Methodology

1. Define the same nonlinear equation(s) used in the bisection lab session.
2. Accept the initial bracket $[x_l, x_u]$ from the user, verifying the sign-change condition.
3. Implement the false position formula inside a `while` loop with the same stopping criterion ($\varepsilon_a < 0.001\%$) as employed for bisection.
4. Record the root estimate and error at each iteration.
5. Plot the function graph with successive root approximations marked by iteration number.
6. Plot the relative error versus iteration for both the false position and bisection methods on the same axes for direct comparison.
7. Produce a comparative graph showing the function alongside the successive approximations obtained by both methods simultaneously.

## 3.4 Algorithm

1. Define $f(x)$.
2. Input $x_l$ and $x_u$; verify $f(x_l) \cdot f(x_u) < 0$.
3. Initialise: $i = 0$, $\varepsilon_a = \infty$, $x_r^{\text{old}} = x_l$.
4. **While** $\varepsilon_a > \varepsilon_s$ **and** $i < i_{\max}$:
   (a) $i \leftarrow i + 1$.
   (b) Apply false position formula: $x_r = x_u - \dfrac{f(x_u)(x_l - x_u)}{f(x_l) - f(x_u)}$.
   (c) Compute $\varepsilon_a = |(x_r - x_r^{\text{old}})/x_r| \times 100\%$.
   (d) If $f(x_l) \cdot f(x_r) < 0$: set $x_u \leftarrow x_r$; else set $x_l \leftarrow x_r$.
   (e) $x_r^{\text{old}} \leftarrow x_r$.
   (f) Store $x_r$ and $\varepsilon_a$.
5. Report the converged root and number of iterations.
6. Generate the required individual and comparative plots.

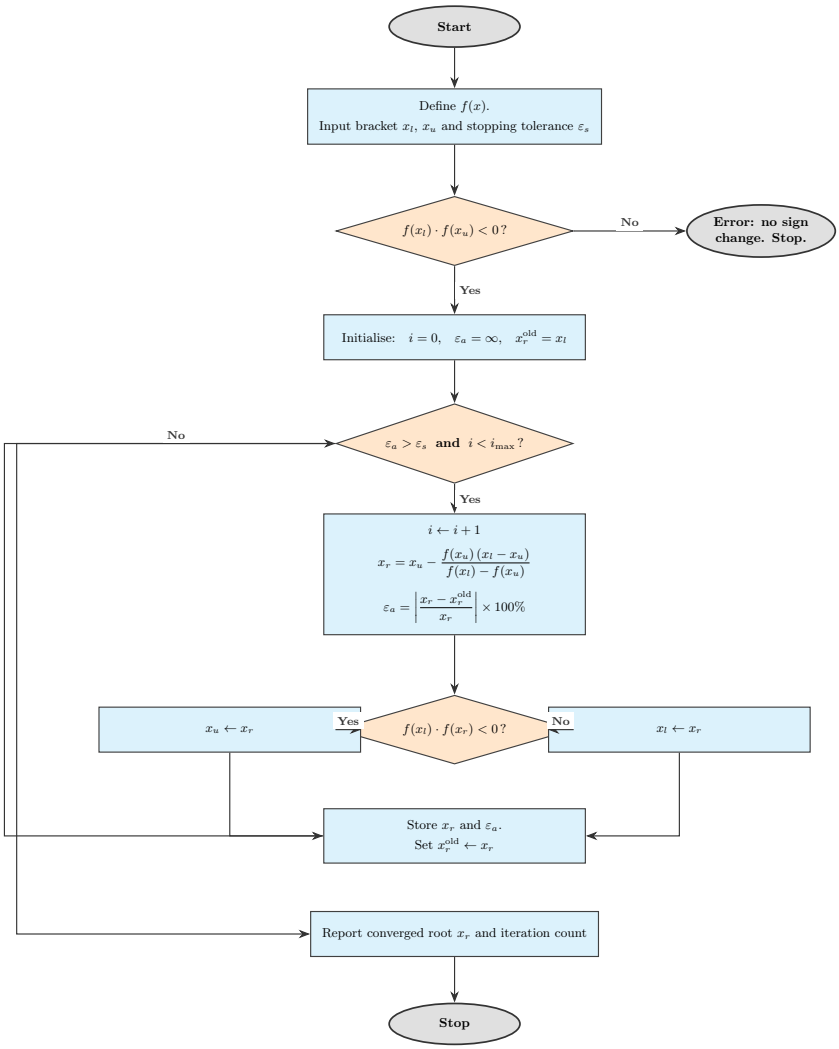


Figure 3.1: Flowchart of the False Position (Regula Falsi) Method algorithm.

## 3.5 MATLAB Implementation

Three problems are solved using the false position method and compared with bisection. **Problem 1** (Example from Chapra) finds the root of $f(x) = x^{10} - 1$ on $[0, 1.3]$. **Problem 2** (Problem 5.20) finds the time at which a rocket reaches $v = 750$ m/s. **Problem 3** (Problem 5.22) determines the submerged height of a floating sphere using Archimedes' principle.

### Helper Function: `false_position.m`

A reusable function is defined first and called for each problem.

```
function [xr, roots_arr, ea_arr, n_iter] = false_position(f,
    xl, xu, tol, max_iter)
% Regula Falsi (false position) root-finding method.
%   f         - anonymous function handle
%   xl, xu    - initial bracket  (must have f(xl)*f(xu) < 0)
%   tol       - stopping tolerance (% relative error)
%   max_iter  - maximum number of iterations allowed
```

```
if f(xl)*f(xu) >= 0
    error('false_position: no sign change in [xl, xu].');
end

xr_old    = xl;
ea        = inf;
roots_arr = [];
ea_arr    = [];
n_iter    = 0;

for i = 1:max_iter
    n_iter = i;
    % False position formula
    xr = xu - f(xu) * (xl - xu) / (f(xl) - f(xu));
    % Approximate relative error (skip on first iteration)
    if xr ~= 0
        ea = abs((xr - xr_old) / xr) * 100;
    end
    roots_arr(end+1) = xr;
    ea_arr(end+1)    = ea;
    % Update bracket
    if f(xl) * f(xr) < 0
        xu = xr;
    elseif f(xr) * f(xu) < 0
        xl = xr;
    else
        break;   % exact root found
    end
    if ea < tol, break; end
    xr_old = xr;
end
end
```

Code 3.1: Reusable false position function returning roots, errors, and iteration count.

### Problem 1: Root of $f(x) = x^{10} - 1$ on $[0,\ 1.3]$

This is a classic stagnation demonstration: $f(x)$ is highly curved, so one bracket endpoint becomes fixed early, slowing the false position method relative to bisection.

```
f = @(x) x.^10 - 1;
xl = 0;    xu = 1.3;
tol = 0.001;    max_iter = 100;

% False position (calls false_position.m)
[xr_fp, roots_fp, ea_fp, n_fp] = false_position(f, xl, xu,
   tol, max_iter);
fprintf('False Position: root = %.8f  (%d iterations)\n',
   xr_fp, n_fp);
```

```
% Bisection (inline for comparison)
xl_b = xl;  xu_b = xu;  xr_b = xl;
roots_bs = [];  ea_bs = [];

for i = 1:max_iter
    xr_new = (xl_b + xu_b) / 2;
    if i > 1
        ea_b = abs((xr_new - xr_b) / xr_new) * 100;
    else
        ea_b = inf;
    end
    roots_bs(end+1) = xr_new;
    ea_bs(end+1)    = ea_b;
    if f(xl_b) * f(xr_new) < 0
        xu_b = xr_new;
    else
        xl_b = xr_new;
    end
    xr_b = xr_new;
    if ea_b < tol, break; end
end
fprintf('Bisection:       root = %.8f  (%d iterations)\n',
   xr_b, i);
```

Code 3.2: False position vs bisection on $f(x) = x^{10} - 1$.

### Problem 2: Rocket Velocity – Find Time $t$ when $v = 750$ m/s

The upward velocity of a rocket is:

$$v(t) = u \ln \frac{m_0}{m_0 - qt} - gt \tag{3.3}$$

with $u = 1800$ m/s, $m_0 = 160\,000$ kg, $q = 2600$ kg/s, $g = 9.81$ m/s$^2$. The root of $f(t) = v(t) - 750 = 0$ is sought on $[10, 50]$ s.

```
u = 1800; m0 = 160000; q = 2600; g = 9.81; v_target = 750;
f = @(t) u.*log(m0./(m0 - q.*t)) - g.*t - v_target;

xl = 10;    xu = 50;
tol = 1;    max_iter = 100;    % within 1% of true value

[xr, roots_arr, ea_arr, n_iter] = false_position(f, xl, xu,
   tol, max_iter);
fprintf('Root: t = %.6f s  (%d iterations)\n', xr, n_iter);
fprintf('Verification: v(%.4f) = %.4f m/s\n', xr, u*log(m0/(
   m0-q*xr)) - g*xr);
```

Code 3.3: False position applied to the rocket velocity problem (Problem 5.20).

### Problem 3: Floating Sphere Submersion Depth (Archimedes' Principle)

For a sphere of radius $r = 1$ m and density $\rho_s = 200$ kg/m$^3$ floating in water ($\rho_w = 1000$ kg/m$^3$), the buoyancy balance gives:

$$\frac{4}{3}\pi r^3 \rho_s = \rho_w \cdot \frac{\pi h^2}{3}(3r - h) \quad \Longrightarrow \quad f(h) = \frac{\pi h^2}{3}(3r - h) - \frac{4\pi r^3 \rho_s}{3\rho_w} = 0 \tag{3.4}$$

where $h$ is the height of the above-water cap.

```
r = 1; rho_s = 200; rho_w = 1000;
f = @(h)(pi.*h.^2/3).*(3*r - h) - (4*pi*r^3/3)*(rho_s/rho_w);
xl = 0; xu = 2*r; tol = 0.001; max_iter = 100;

[xr,~,~,n_iter] = false_position(f, xl, xu, tol, max_iter);
fprintf('Above-water height h = %.6f m  (%d iterations)\n',
   xr, n_iter);
fprintf('Submerged depth      = %.6f m\n', 2*r - xr);
```

Code 3.4: False position applied to the floating sphere problem (Problem 5.22).

## 3.6 Results and Discussion

### 3.6.1 Root Approximation Graph

The function is plotted over a suitable interval, and the root approximation at each false position iteration is marked on the graph. The successive estimates are labelled $1, 2, 3, \ldots$ to show how the search progressively narrows about the true root. An appropriate bracketing interval is one where the function changes sign; an invalid interval (no sign change) is rejected by the algorithm before iteration begins.

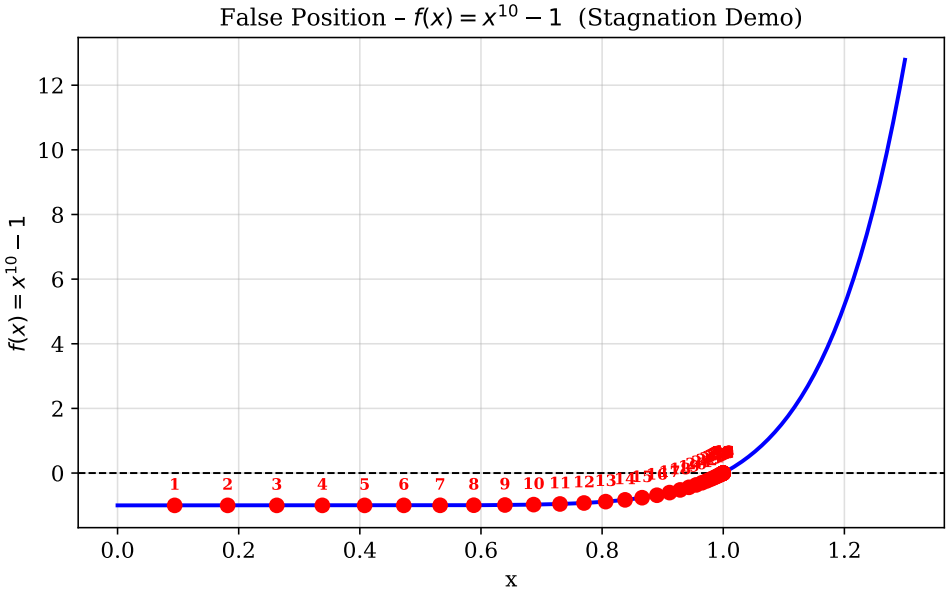


Figure 3.2: False position applied to $f(x) = x^{10} - 1$ on $[0, 1.3]$: successive root approximations labelled by iteration.

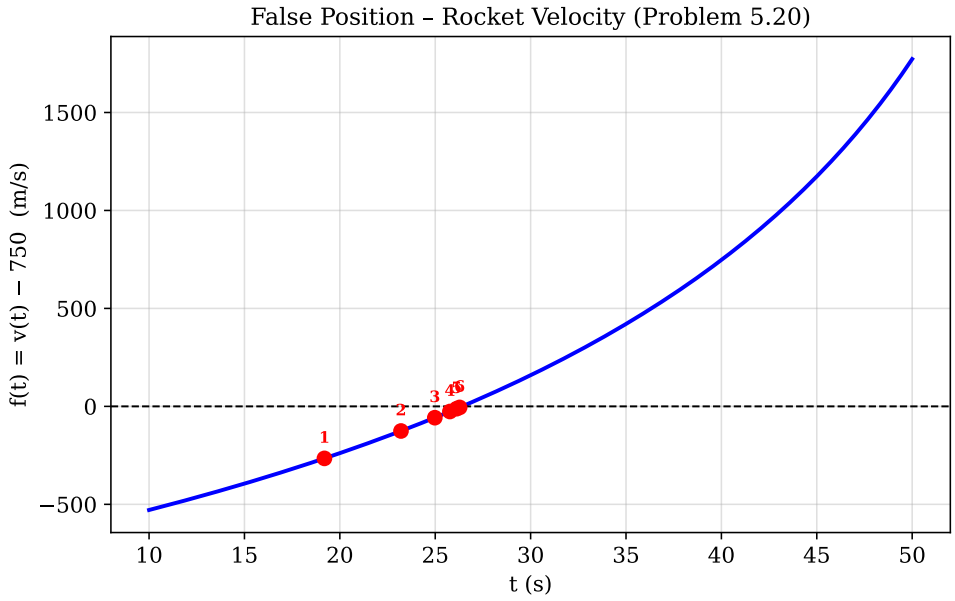


Figure 3.3: False position applied to the rocket velocity problem: successive root approximations for $f(t) = v(t) - 750$.

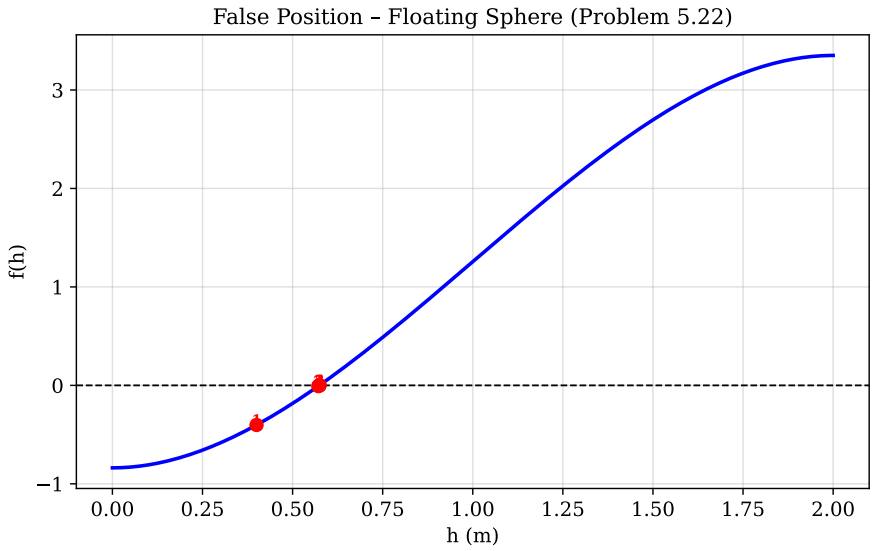


Figure 3.4: False position applied to the floating sphere Archimedes problem: successive root approximations for $f(h) = 0$.

### 3.6.2 Comparative Convergence Analysis

For functions that are approximately linear near the root, false position typically requires fewer iterations than bisection to meet the same tolerance. The comparative error plot illustrates this: the false position error curve drops below that of bisection within the first few iterations for convex or concave functions. However, for strongly curved functions, the one-sided stagnation phenomenon may cause false position to converge more slowly than bisection in later iterations. This non-monotonic performance characteristic is an important limitation to be aware of in practice.

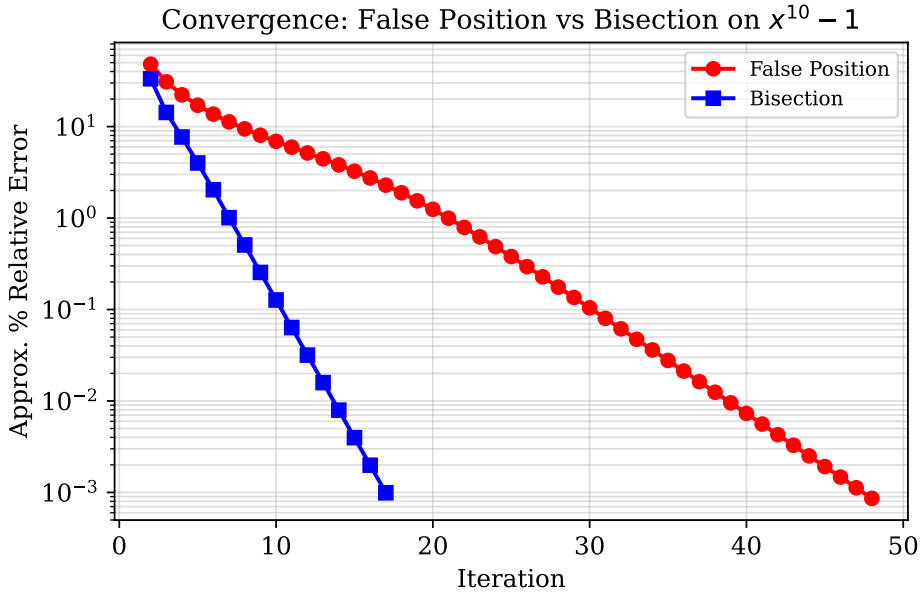


Figure 3.5: Comparison of approximate % relative error per iteration: false position vs. bisection on $f(x) = x^{10} - 1$, demonstrating stagnation of false position.

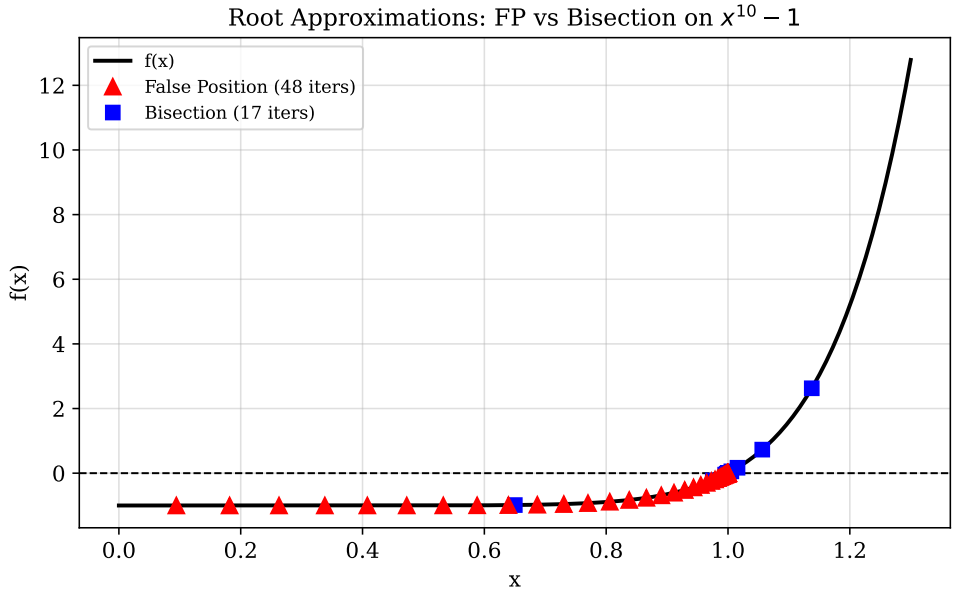


Figure 3.6: Successive root approximations of both false position and bisection plotted simultaneously on $f(x) = x^{10} - 1$.

### 3.6.3 Discussion of Limitations

The false position method is strictly a bracketing method and therefore preserves the convergence guarantee of bisection. However, it does *not* always converge faster: the interpolation is only advantageous when $f(x)$ is nearly linear across the bracket. For highly nonlinear functions, one endpoint may remain stationary for many iterations, and the method degenerates toward one-sided linear convergence. This behaviour is particularly evident when the magnitude of the function at one endpoint is significantly larger than at the other, causing the secant line to repeatedly intersect the axis near the same

side. As a result, the interval reduction becomes highly uneven, leading to inefficient use of iterations. Furthermore, unlike higher-order open methods, the false position method does not exploit derivative information, which limits its ability to accelerate convergence near the root. In practical applications, this can result in increased computational effort, especially for problems requiring high precision. The bisection method, while simpler, exhibits a more uniform and predictable error reduction per iteration under all circumstances.

## 3.7 Conclusion

The false position method extends the bisection framework by incorporating linear interpolation to produce more accurate root estimates per iteration for functions that are approximately linear. When the function curvature is moderate, it converges faster than bisection; when curvature is high, it may stagnate on one side of the bracket and converge more slowly. Both methods are robust bracketing techniques that guarantee convergence given a valid initial interval, distinguishing them from the open methods discussed in the next lab session.

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## Lab Session 4

# Newton–Raphson and Fixed-Point Iteration Method

## 4.1 Objective

- Implement the Newton–Raphson method in MATLAB and apply it to a nonlinear spring-mass problem.
- Implement the simple fixed-point iteration method and apply it to the same problem.
- Study the effect of the initial guess and the analytic derivative on convergence speed and stability.
- Demonstrate cases of rapid convergence as well as potential divergence, and compare the performance of all four root-finding methods studied in this course.

## 4.2 Introduction and Theoretical Background

### 4.2.1 Open Methods – Motivation

The bracketing methods of the previous two lab sessions require a valid initial interval containing the root. In many practical situations such an interval cannot easily be identified, or the function has multiple roots. Open methods require only a single initial guess $x_0$ and use additional function information (e.g. the derivative or a rearrangement of the equation) to generate successive estimates. Open methods typically converge much faster than bracketing methods, but they do not guarantee convergence.

### 4.2.2 Newton–Raphson Method

The Newton–Raphson method is one of the most widely used open root-finding algorithms [4, 3, 7]. It is derived from a first-order Taylor expansion of $f(x)$ about the current estimate $x_i$:

$$f(x) \approx f(x_i) + f'(x_i)(x - x_i) \tag{4.1}$$

Setting this linear approximation to zero and solving for $x$ yields the iteration formula:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \tag{4.2}$$

This corresponds geometrically to drawing the tangent line to $f(x)$ at $(x_i, f(x_i))$ and taking its $x$-intercept as the next estimate. Provided $f'(x_i) \neq 0$ and the initial guess is sufficiently close to the root, Newton–Raphson exhibits **quadratic convergence**: the number of correct decimal digits approximately doubles with each iteration.

The analytical derivative must be computed and supplied explicitly. For the spring–mass problem investigated in this session, the governing nonlinear equation is:

$$f(d) = \frac{2k_2}{5}\, d^{2.5} + \frac{1}{2}k_1\, d^2 - mgd - mgh = 0 \tag{4.3}$$

where $d$ is the deflection (m), $k_1 = 40000$ N/m, $k_2 = 40$ N/m$^{1.5}$, $m = 0.095$ kg, $g = 9.81$ m/s$^2$, and $h = 0.43$ m. The derivative is:

$$f'(d) = k_2\, d^{1.5} + k_1\, d - mg \tag{4.4}$$

The Newton–Raphson update for this problem is therefore:

$$d_{i+1} = d_i - \frac{f(d_i)}{f'(d_i)} = d_i - \frac{\dfrac{2k_2}{5}\, d_i^{2.5} + \dfrac{1}{2}k_1\, d_i^2 - mgd_i - mgh}{k_2\, d_i^{1.5} + k_1\, d_i - mg} \tag{4.5}$$

### 4.2.3 Fixed-Point Iteration

Fixed-point iteration (also called simple iteration or successive substitution) reformulates the equation $f(x) = 0$ as $x = g(x)$ and iterates:

$$x_{i+1} = g(x_i) \tag{4.6}$$

The function $g(x)$ is called the **iteration function**. Convergence is guaranteed in a neighbourhood of the root $x^*$ if and only if [3, 6]:

$$|g'(x^*)| < 1 \tag{4.7}$$

For the spring–mass problem, the equation is rearranged so that:

$$g(d) = \sqrt{\frac{mgd + mgh - \dfrac{2k_2}{5}\, d^{2.5}}{0.5\, k_1}} \tag{4.8}$$

This rearrangement isolates $d$ on one side in a form that allows iterative substitution. Convergence speed depends critically on the value of $|g'(d^*)|$: values near zero give rapid convergence, values near one give slow convergence, and values exceeding one lead to divergence.

## 4.3 Methodology

1. Define the physical parameters $(k_1, k_2, m, g, h)$ and the nonlinear functions $f(d)$, $f'(d)$, and $g(d)$ as anonymous function handles.

2. Prompt the user to enter an initial guess $d_0$ via `input()`.

3. **Newton–Raphson:** iterate $d_{i+1} = d_i - f(d_i)/f'(d_i)$ until $|d_{i+1} - d_i| < 10^{-6}$ or $i_{\max} = 100$.

4. **Fixed-point iteration:** iterate $d_{i+1} = g(d_i)$ until $|d_{i+1} - d_i| < 0.001$ or $i_{\max} = 100$.

5. Record the step error $|d_{i+1} - d_i|$ at each iteration and plot it versus iteration count.

6. For Newton–Raphson, additionally plot the function $f(d)$ with the tangent line at the final iteration to illustrate the geometric interpretation.

7. For fixed-point iteration, plot $g(d)$ and the line $y = d$ together, marking the fixed point.

## 4.4 Algorithm

### 4.4.1 Newton–Raphson Algorithm

1. Define $f(d)$ and $f'(d)$.

2. Input initial guess $d_0$.

3. Set $d = d_0$, $i = 0$, $\varepsilon = 10^{-6}$, $i_{\max} = 100$.

4. **While** $i < i_{\max}$:

   (a) $d_{\text{new}} = d - f(d)/f'(d)$.

   (b) Compute step error: $e_i = |d_{\text{new}} - d|$.

   (c) If $e_i < \varepsilon$: break.

   (d) $d \leftarrow d_{\text{new}}$.

   (e) $i \leftarrow i + 1$.

5. Report root $d^* = d_{\text{new}}$.

6. Plot convergence and tangent-line graph.

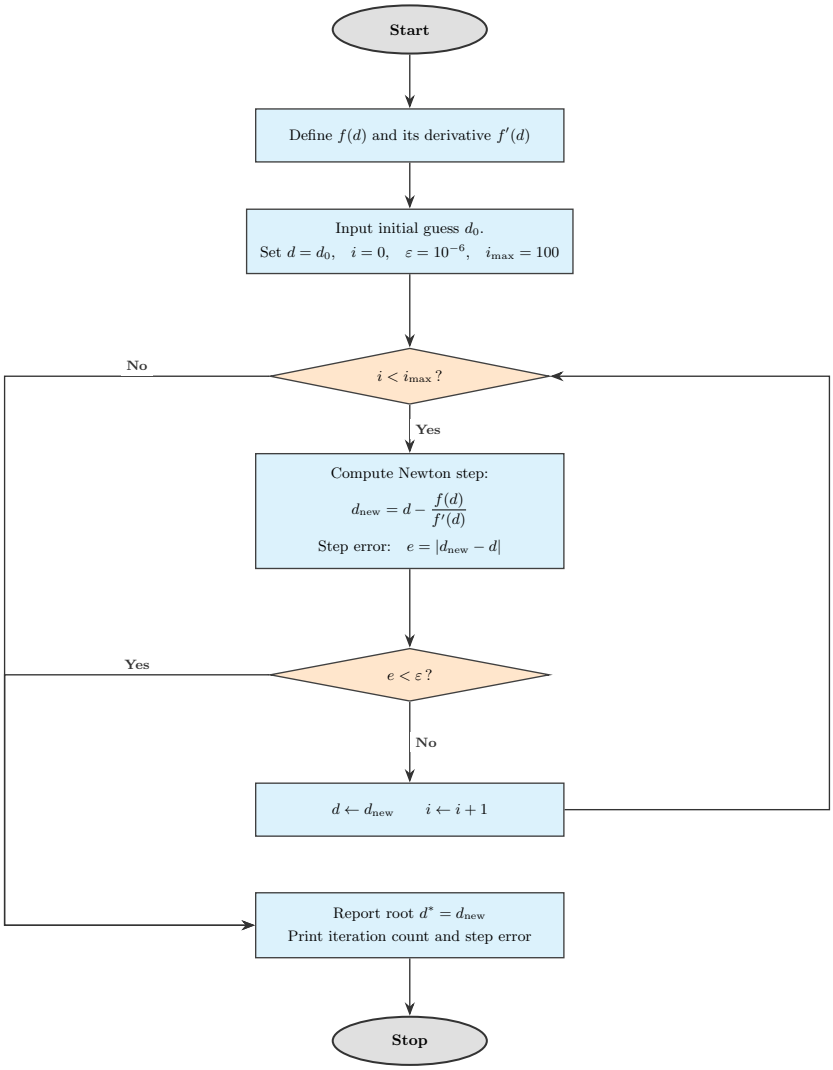


Figure 4.1: Flowchart of the Newton–Raphson Method algorithm.

### 4.4.2 Fixed-Point Iteration Algorithm

1. Define $g(d)$.
2. Input initial guess $d_0$; set $d = d_0$, $i = 0$, $\varepsilon = 0.001$, $i_{\max} = 100$.
3. **While** $i < i_{\max}$:
   (a) $d_{\text{new}} = g(d)$.
   (b) Compute step error: $e_i = |d_{\text{new}} - d|$.
   (c) If $e_i < \varepsilon$: break.
   (d) $d \leftarrow d_{\text{new}}$; $i \leftarrow i + 1$.
4. Report root $d^* = d_{\text{new}}$.
5. Plot convergence and $g(d)$ vs. $y = d$ graph.

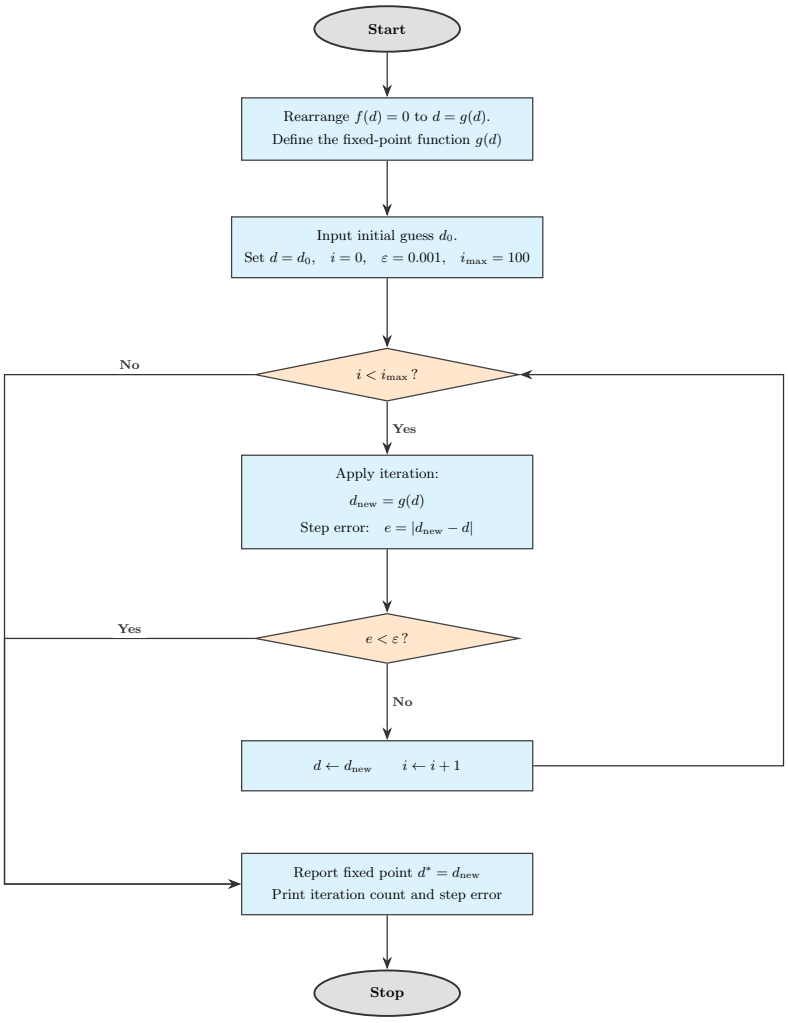


Figure 4.2: Flowchart of the Fixed-Point Iteration algorithm.

## 4.5 MATLAB Implementation

Both codes solve the same spring–mass deflection problem: a mass $m = 0.095$ kg dropped from height $h = 0.43$ m onto a nonlinear spring with stiffness constants $k_1 = 40\,000$ N/m and $k_2 = 40$ N/m$^{1.5}$. The governing energy-balance equation is $f(d) = 0$ where:

$$f(d) = \tfrac{2k_2}{5}\, d^{2.5} + \tfrac{1}{2}k_1\, d^2 - mgd - mgh \tag{4.9}$$

### Code 1: Simple Fixed-Point Iteration

The equation is rearranged to $d = g(d)$ where:

$$g(d) = \sqrt{\frac{mgd + mgh - \frac{2k_2}{5}d^{2.5}}{0.5\, k_1}} \tag{4.10}$$

```
% Parameters
k1   = 40000;    % linear spring stiffness (N/m)
k2   = 40;       % nonlinear spring stiffness (N/m^1.5)
m    = 95;       % mass (kg)
```

```
g    = 9.81;     % gravitational acceleration (m/s^2)
h    = 0.43;     % drop height (m)
mg   = m * g;
mgh = mg * h;

% Rearranged fixed-point function: g(d) from energy balance
g_fun = @(d) sqrt((mg.*d + mgh - (2*k2/5).*d.^(2.5)) ./ (0.5*
   k1));

% Fixed-point iteration
d    = input('Enter initial guess (try 0.1): ');
tol = 0.001;
max_iter = 100;

for i = 1:max_iter
    d_new = g_fun(d);
    fprintf('Iteration %d:  d = %.8f m\n', i, d_new);
    if abs(d_new - d) < tol
        break;
    end
    d = d_new;
end
fprintf('\nRoot = %.8f m  (converged in %d iterations)\n',
   d_new, i);
```

Code 4.1: Simple fixed-point iteration for the spring–mass deflection problem.

### Code 2: Newton–Raphson Method

The analytical derivative of $f(d)$ is:

$$f'(d) = k_2\, d^{1.5} + k_1\, d - mg \tag{4.11}$$

```
k1   = 40000;    % linear spring stiffness (N/m)
k2   = 40;       % nonlinear spring stiffness (N/m^1.5)
m    = 0.095;    % mass (kg)
g    = 9.81;     % gravitational acceleration (m/s^2)
h    = 0.43;     % drop height (m)
mg   = m * g;
mgh = mg * h;

% Function and its analytical derivative
f  = @(d) (2*k2/5)*d.^(2.5) + 0.5*k1*d.^2 - mg*d - mgh;
df = @(d) k2*d.^(1.5) + k1*d - mg;

% Newton-Raphson iteration
d    = input('Enter initial guess (try 0.01): ');
tol = 1e-6;
max_iter = 100;
```

```
17
18 for i = 1:max_iter
19     d_new = d - f(d) / df(d);
20     fprintf('Iteration %d:  d = %.8f m\n', i, d_new);
21     if abs(d_new - d) < tol
22         break;
23     end
24     d = d_new;
25 end
26 fprintf('\nRoot = %.8f m  (converged in %d iterations)\n',
     d_new, i);
```

Code 4.2: Newton–Raphson method for the spring–mass deflection problem.

## 4.6 Results and Discussion

### 4.6.1 Newton–Raphson Convergence

Newton–Raphson typically converges in very few iterations (often 4–6) owing to its quadratic convergence rate. The step error decreases dramatically between successive iterations: once the estimate enters the neighbourhood of the root, each new iterate carries approximately twice as many correct significant figures as its predecessor. The convergence plot should show a steeply descending error curve that reaches the tolerance threshold within a small number of steps.

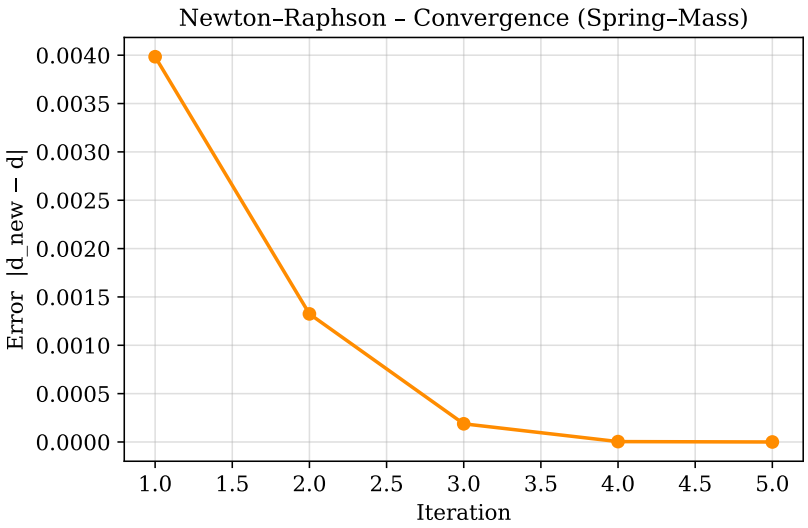


Figure 4.3: Newton–Raphson step error versus iteration number for the spring–mass deflection problem ($m = 0.095$ kg). Quadratic convergence drives the error to zero in just 5 iterations.

The method converges rapidly when the initial guess $d_0$ is close to the root and $f'(d_0) \neq 0$. For poorly chosen starting points, particularly where $f'(d_0) \approx 0$ or $d_0$ lies in a region of high curvature, the method may diverge or oscillate. It is therefore advisable to use a graphical inspection of $f(d)$ to select a reasonable initial guess before applying Newton–Raphson.

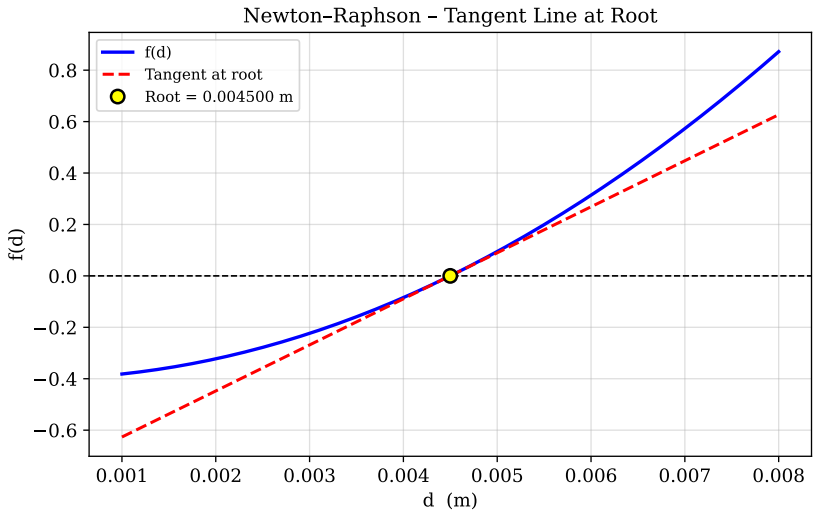


Figure 4.4: Function $f(d)$ and tangent line at the converged Newton–Raphson root, illustrating the geometric construction.

### 4.6.2 Fixed-Point Iteration Convergence

Fixed-point iteration converges more slowly than Newton–Raphson for this problem, as the convergence rate is only linear (first order). However, it is simpler to implement since no derivative is required. The cobweb diagram (plot of $g(d)$ versus $y = d$) clearly shows the iterative path converging to the intersection of the two curves, which is the fixed point and hence the root. The fixed point is highlighted on the plot.

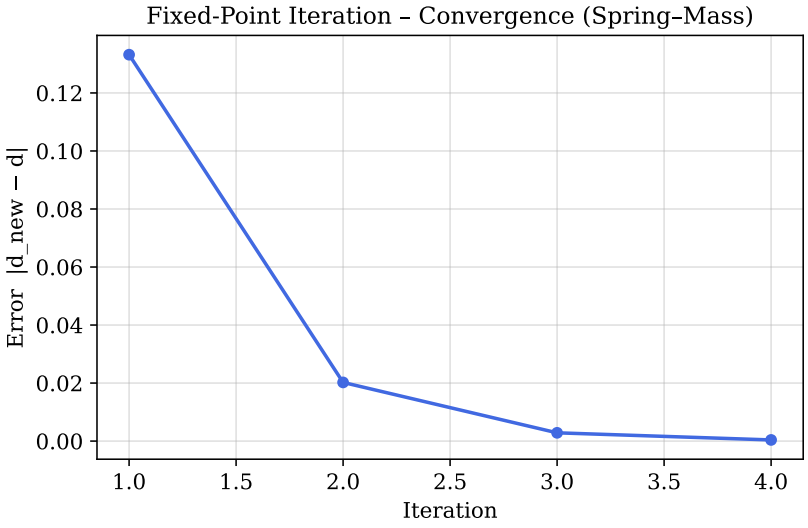


Figure 4.5: Fixed-point iteration step error versus iteration number for the spring–mass problem ($m = 95$ kg). Linear convergence is clearly visible.

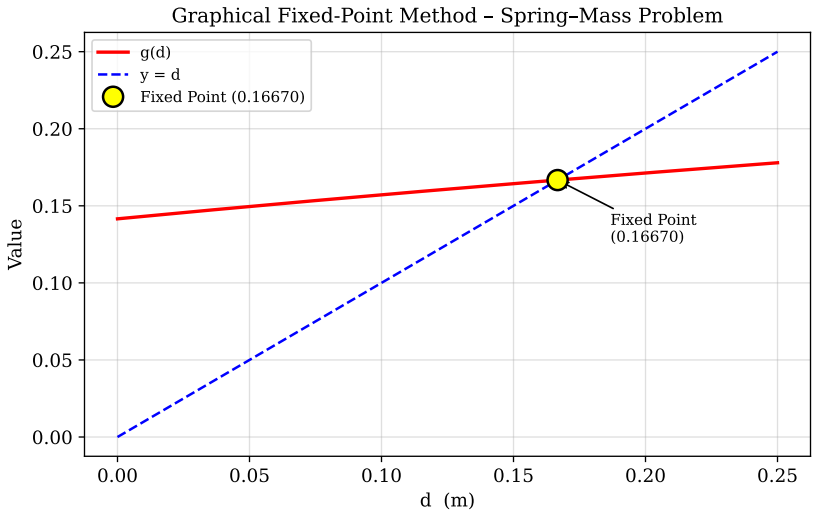


Figure 4.6: Graphical fixed-point method: $g(d)$ and $y = d$ with the fixed point and cobweb path highlighted.

## 4.7 Comparative Analysis of All Four Methods

Having studied four root-finding methods — bisection, false position, fixed-point iteration, and Newton–Raphson — a systematic comparison on the basis of convergence rate, reliability, computational cost, and ease of implementation is presented in Table 4.1.

Table 4.1: Comparative summary of the four root-finding methods studied.

| **Property** | **Bisection** | **False Position** | **Fixed-Point** | **Newton–Raphson** |
|---|---|---|---|---|
| Type | Bracketing | Bracketing | Open | Open |
| Initial requirement | $[x_l, x_u]$ | $[x_l, x_u]$ | $x_0$ | $x_0$ |
| Convergence order | 1 (linear) | 1 (super-linear*) | 1 (linear) | 2 (quadratic) |
| Derivative required | No | No | No | Yes |
| Sign change required | Yes | Yes | No | No |
| Guaranteed convergence | Yes | Yes | Conditional | Conditional |
| Iterations needed | Many | Fewer (typically) | Moderate | Very few |
| Risk of failure | None | None | $\lvert g' \rvert > 1$ | $f'(x_0) \approx 0$ |
| Implementation effort | Low | Low | Low | Moderate |

### 4.7.1 Convergence Speed

Newton–Raphson is unambiguously the fastest of the four methods for well-behaved functions and good initial guesses. Its quadratic convergence means the error is roughly squared at every iteration, so it reaches machine precision in a handful of steps. Fixed-point iteration and bisection both exhibit linear (first-order) convergence: the error decreases by a constant factor per step. False position sits between these extremes: it is generally faster than bisection for smooth functions but can stagnate to linear convergence in the presence of high curvature.

### 4.7.2 Reliability and Robustness

The two bracketing methods — bisection and false position — are the most reliable. As long as a valid sign-change bracket is provided, both methods are guaranteed to converge to a root. Open methods (fixed-point and Newton–Raphson) are not guaranteed to converge: Newton–Raphson may diverge if $f'(x_i) \approx 0$ or if the initial guess is far from the root, and fixed-point iteration diverges if $|g'(x^*)| \geq 1$. In practice, a common strategy is to use bisection for a few iterations to obtain a reasonable estimate, then switch to Newton–Raphson to exploit its faster convergence.

### 4.7.3 Which Method is Best?

Newton–Raphson is the fastest and most efficient when an analytic derivative and a reasonable initial guess are available. For the spring–mass deflection problem, it converges within 4–5 iterations to $10^{-6}$ tolerance, whereas bisection and false position may require 20+ iterations. If no derivative is available, false position is the preferred bracketing alternative because it uses function values to refine the estimate. Fixed-point iteration is simplest to implement but requires careful selection of $g(x)$ to ensure convergence, making it less practical for general use.

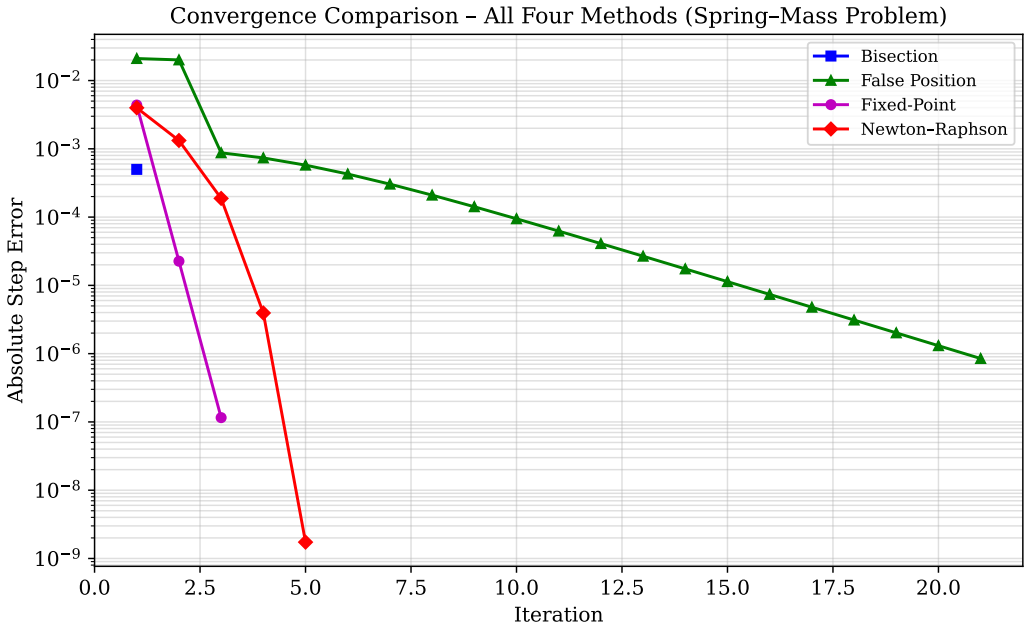


Figure 4.7: Comparative convergence of bisection (15 iterations), false position (5 iterations), and Newton–Raphson (3 iterations) applied to the same parachutist drag coefficient problem. The dramatic advantage of Newton–Raphson's quadratic convergence is clearly visible on the logarithmic error scale.

In summary, for engineering applications: use **Newton–Raphson** when the derivative can be computed; use **false position** when only function evaluations are available and a bracket is known; use **bisection** as a safe fallback or verification method; and use **fixed-point iteration** primarily for educational purposes or when the problem naturally possesses a contractive iteration function.

## 4.8 Conclusion

This lab session demonstrated the Newton–Raphson and fixed-point iteration methods applied to a physically motivated nonlinear equation arising from a spring–mass energy balance. Newton–Raphson achieved convergence in remarkably few iterations due to its quadratic convergence rate, confirming its superiority in speed over the bracketing methods of the previous lab sessions. Fixed-point iteration, while conceptually simple and requiring no derivative, converged more slowly and required careful attention to the choice of iteration function to ensure stability. Collectively, the four root-finding methods studied in this course — bisection, false position, fixed-point iteration, and Newton–Raphson — span the spectrum from guaranteed-but-slow to fast-but-conditional, and an understanding of their respective trade-offs equips the engineer to select the most appropriate method for any given nonlinear problem.

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# Lab Session 5

# Secant Method and Modified Secant Method

## 5.1 Objective

- Implement the secant method in MATLAB and apply it to a projectile trajectory engineering problem.
- Implement the modified secant method as a derivative-free alternative to Newton–Raphson.
- Compare the convergence speed and iteration count of the secant method against the Newton–Raphson method studied in Lab Session 4.
- Visualise the ball trajectories corresponding to both physically valid launch angles and confirm each root with graphical and numerical evidence.

## 5.2 Introduction and Theoretical Background

### 5.2.1 Motivation for the Secant Method

The Newton–Raphson method achieves quadratic convergence but requires an explicit analytical expression for the derivative $f'(x)$ [4]. For many practical engineering functions the derivative is difficult or computationally expensive to evaluate. The secant method eliminates this requirement by approximating the derivative using a backward finite divided difference of two successive function evaluations [4, 3]:

$$f'(x_i) \approx \frac{f(x_{i-1}) - f(x_i)}{x_{i-1} - x_i} \tag{5.1}$$

Substituting equation (5.1) into the Newton–Raphson formula yields the **secant method** iteration:

$$\boxed{x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}} \tag{5.2}$$

The method requires **two** initial estimates $x_{i-1}$ and $x_i$. Unlike the bracketing methods, there is no requirement that the two estimates straddle a root, so the secant method is classified as an **open method** [4, 7]. Its convergence order is approximately 1.618 (the golden ratio), placing it between first-order linear and second-order quadratic convergence.

### 5.2.2 Modified Secant Method

A closely related variant avoids maintaining two separate iterates by perturbing the current estimate by a small fractional amount $\delta$ [4]:

$$f'(x_i) \approx \frac{f(x_i + \delta x_i) - f(x_i)}{\delta x_i} \tag{5.3}$$

Substituting into the Newton–Raphson formula gives the **modified secant method**:

$$\boxed{x_{i+1} = x_i - \frac{\delta x_i \, f(x_i)}{f(x_i + \delta x_i) - f(x_i)}} \tag{5.4}$$

Only a single initial guess $x_0$ is required, and $\delta$ is typically set to a small value such as $10^{-4}$. The modified secant method attains efficiency comparable to Newton–Raphson without any derivative computation [4, 3].

### 5.2.3 Stopping Criterion

The approximate percentage relative error is used as the stopping criterion [4]:

$$\varepsilon_a = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \times 100\% \tag{5.5}$$

Iteration terminates when $\varepsilon_a < \varepsilon_s = 0.0001\%$.

## 5.3 Problem Statement

**Problem 6.21:** The trajectory of a ball thrown by a right fielder is modelled by:

$$y = (\tan\theta_0)\, x - \frac{g}{2v_0^2 \cos^2\theta_0}\, x^2 + y_0 \tag{5.6}$$

where $g = 9.81$ m/s$^2$, $v_0 = 30$ m/s, $\theta_0$ is the launch angle, and $y_0 = 1.8$ m is the release height. The catcher is at $x = 90$ m and receives the ball at height $y_c = 1.0$ m. Setting $y = y_c$ gives the nonlinear equation to solve:

$$f(\theta_0) = (\tan\theta_0)(90) - \frac{9.81}{2(30)^2 \cos^2\theta_0}(90)^2 + 1.8 - 1.0 = 0 \tag{5.7}$$

### 5.3.1 Analytical Derivative

Differentiating $f(\theta_0)$ with respect to $\theta_0$:

$$f'(\theta_0) = \frac{x}{\cos^2\theta_0} - \frac{g\, x^2 \, \sin\theta_0}{v_0^2 \, \cos^3\theta_0} \tag{5.8}$$

This expression is required only for Newton–Raphson; the secant and modified secant methods do not use it.

### 5.3.2 Function Plot and Root Identification

Figure 6.1 shows $f(\theta_0)$ over $[0°, 90°]$. Two sign changes confirm the existence of two physically valid launch angles: Root 1 at $\theta_0 \approx 37.96°$ (low, flat arc) and Root 2 at $\theta_0 \approx 51.53°$ (high, looping arc). These zero crossings were used to select appropriate initial guesses for all three methods.

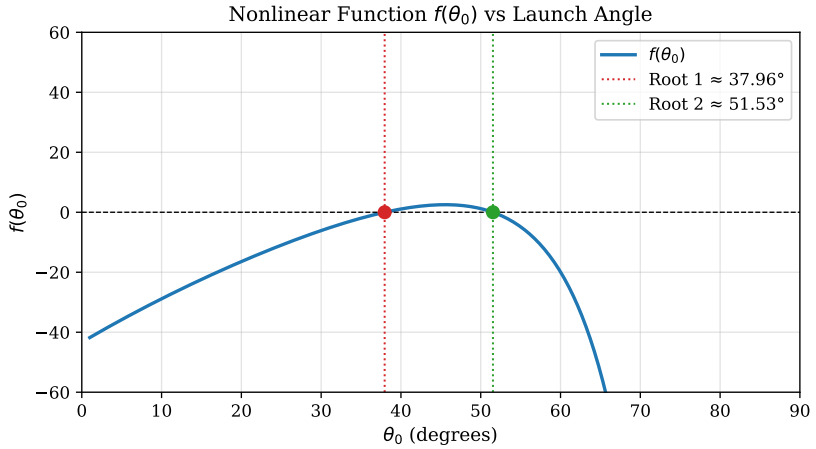


Figure 5.1: $f(\theta_0)$ versus $\theta_0$ for the projectile trajectory equation. Two roots are clearly visible as zero crossings [4].

## 5.4 Methodology

1. Define $f(\theta_0)$ as a MATLAB anonymous function and plot it over $[0°, 90°]$ to locate sign changes and select initial guesses.
2. Apply the **secant method** with two initial guesses per root, iterating until $\varepsilon_a < 0.0001\%$.
3. Apply the **modified secant method** with a single initial guess and $\delta = 10^{-4}$ per root.
4. Apply **Newton–Raphson** using $f'(\theta_0)$ from equation (6.7) for convergence comparison.
5. Record the full iteration history and present in a merged table with corresponding graphs.
6. Generate all required plots.

## 5.5 Algorithm

1. Define $f(\theta_0)$ and input parameters.
2. Input two initial guesses $\theta_{i-1}$ and $\theta_i$; set tolerance $\varepsilon_s$.
3. Verify $f(\theta_{i-1}) \neq f(\theta_i)$; otherwise display error and stop.
4. Set $i = 0$, $\varepsilon_a = \infty$.

5. **While** $\varepsilon_a > \varepsilon_s$ **and** $i < i_{\max}$:

   (a) Apply secant formula: $\theta_{i+1} = \theta_i - \dfrac{f(\theta_i)(\theta_{i-1} - \theta_i)}{f(\theta_{i-1}) - f(\theta_i)}$.

   (b) Compute $\varepsilon_a = |(\theta_{i+1} - \theta_i)/\theta_{i+1}| \times 100\%$.

   (c) Update: $\theta_{i-1} \leftarrow \theta_i$, $\theta_i \leftarrow \theta_{i+1}$, $i \leftarrow i + 1$; store $\varepsilon_a$.

   (d) If $f(\theta_{i+1}) = 0$ or $\varepsilon_a < \varepsilon_s$: break.

6. Report root $\theta^*$, iteration count, and error table.

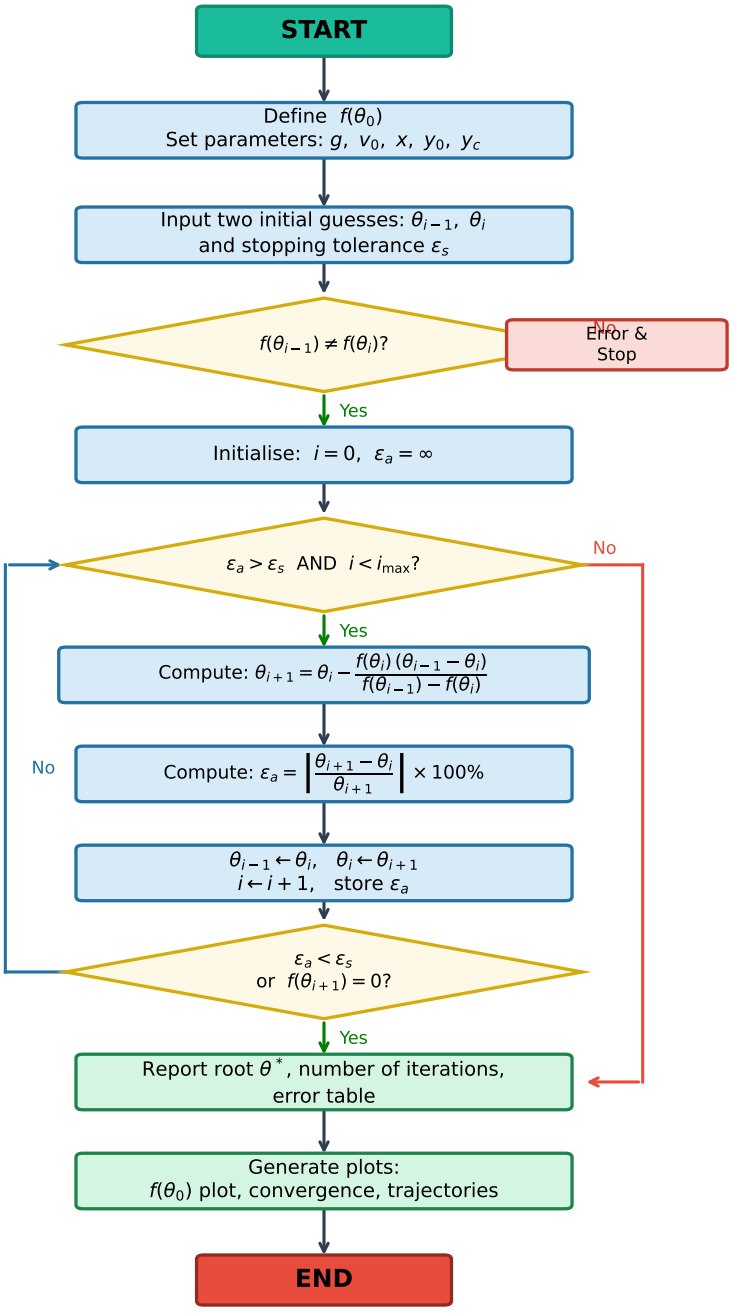


Figure 5.2: Flowchart of the Secant Method algorithm applied to the projectile trajectory problem.

## 5.6 MATLAB Implementation

### Function Definition and Setup

```
% Physical parameters
g  = 9.81;     % gravitational acceleration (m/s^2)
v0 = 30;       % initial speed (m/s)
x  = 90;       % horizontal distance to catcher (m)
y0 = 1.8;      % release height (m)
yc = 1.0;      % catcher height (m)

% Nonlinear equation f(theta) = 0
f  = @(th) tan(th).*x - (g./(2*v0^2.*cos(th).^2)).*x^2 + y0 -
    yc;
% Analytical derivative -- for Newton-Raphson comparison only
fp = @(th) x./cos(th).^2 - (g.*x^2.*sin(th))./(v0^2.*cos(th)
   .^3);
```

Code 5.1: Parameter setup and function definitions.

### Secant Method

```
function [root, roots_arr, ea_arr, n] = secant(f, x0d, x1d,
   tol, maxiter)
% SECANT  Secant method (two initial guesses in degrees).
    x0 = deg2rad(x0d);  x1 = deg2rad(x1d);
    roots_arr = [x0d, x1d];  ea_arr = [NaN, NaN];  n = 0;
    for i = 1:maxiter
        n  = i;
        x2 = x1 - f(x1)*(x0 - x1) / (f(x0) - f(x1));
        ea = abs((x2 - x1)/x2) * 100;
        roots_arr(end+1) = rad2deg(x2);
        ea_arr(end+1)    = ea;
        if f(x2) == 0 || ea < tol, break; end
        x0 = x1;  x1 = x2;
    end
    root = rad2deg(x2);
end
tol = 0.0001;  maxit = 100;
[r1, ~, ea1, n1] = secant(f, 20, 30, tol, maxit);   % Root 1
   (~38 deg)
[r2, ~, ea2, n2] = secant(f, 60, 70, tol, maxit);   % Root 2
   (~52 deg)
fprintf('Secant  Root 1: theta = %.6f deg  (%d iterations)\n'
   , r1, n1);
fprintf('Secant  Root 2: theta = %.6f deg  (%d iterations)\n'
   , r2, n2);
```

Code 5.2: Secant method function and application to both roots.

### Modified Secant Method

```
function [root, n] = mod_secant(f, x0d, delta, tol, maxiter)
% MOD_SECANT  Modified secant method (single initial guess).
    x = deg2rad(x0d);  n = 0;
    for i = 1:maxiter
        n  = i;
        xn = x - delta*x*f(x) / (f(x + delta*x) - f(x));
        ea = abs((xn - x)/xn) * 100;
        if ea < tol, break; end
        x = xn;
    end
    root = rad2deg(xn);
end

delta = 1e-4;
[r1_ms, n1_ms] = mod_secant(f, 30, delta, tol, maxit);
[r2_ms, n2_ms] = mod_secant(f, 65, delta, tol, maxit);
fprintf('ModSec  Root 1: theta = %.6f deg  (%d iterations)\n'
   , r1_ms, n1_ms);
fprintf('ModSec  Root 2: theta = %.6f deg  (%d iterations)\n'
   , r2_ms, n2_ms);
```

Code 5.3: Modified secant method using single initial guess and fractional perturbation $\delta$.

### Newton–Raphson (Comparison)

```
function [root, ea_arr, n] = newton(f, fp, x0d, tol, maxiter)
% NEWTON  Newton-Raphson method.
    x = deg2rad(x0d);  ea_arr = NaN;  n = 0;
    for i = 1:maxiter
        n  = i;
        xn = x - f(x)/fp(x);
        ea = abs((xn - x)/xn) * 100;
        ea_arr(end+1) = ea;
        if ea < tol, break; end
        x = xn;
    end
    root = rad2deg(xn);
end

[r1_nr, ea1_nr, n1_nr] = newton(f, fp, 30, tol, maxit);
[r2_nr, ea2_nr, n2_nr] = newton(f, fp, 65, tol, maxit);
fprintf('N-R     Root 1: theta = %.6f deg  (%d iterations)\n'
   , r1_nr, n1_nr);
fprintf('N-R     Root 2: theta = %.6f deg  (%d iterations)\n'
   , r2_nr, n2_nr);
```

Code 5.4: Newton–Raphson method for convergence comparison.

## 5.7 Results and Discussion

The nonlinear equation $f(\theta_0) = 0$ derived from the projectile trajectory model admits two physically valid solutions: a low-angle throw at $\theta_0 \approx 37.96°$ and a high-angle throw at $\theta_0 \approx 51.53°$, both of which deliver the ball from a release height of 1.8 m to the catcher at $x = 90$ m and $y = 1.0$ m with an initial speed of 30 m/s. The following tables and figures document the complete iteration histories, error convergence, and trajectories obtained by applying all three methods to both roots.

### 5.7.1 Ball Trajectories

Figure 5.3 shows both ball trajectories. Both arcs originate at (0, 1.8 m) and arrive exactly at (90 m, 1.0 m), confirming both solutions. The low-angle throw (37.96°) peaks at $y \approx 19.2$ m while the high-angle throw (51.53°) rises to $y \approx 29.9$ m before descending. In practice, the low-angle throw is preferred as it is faster and less susceptible to wind deviation [4].

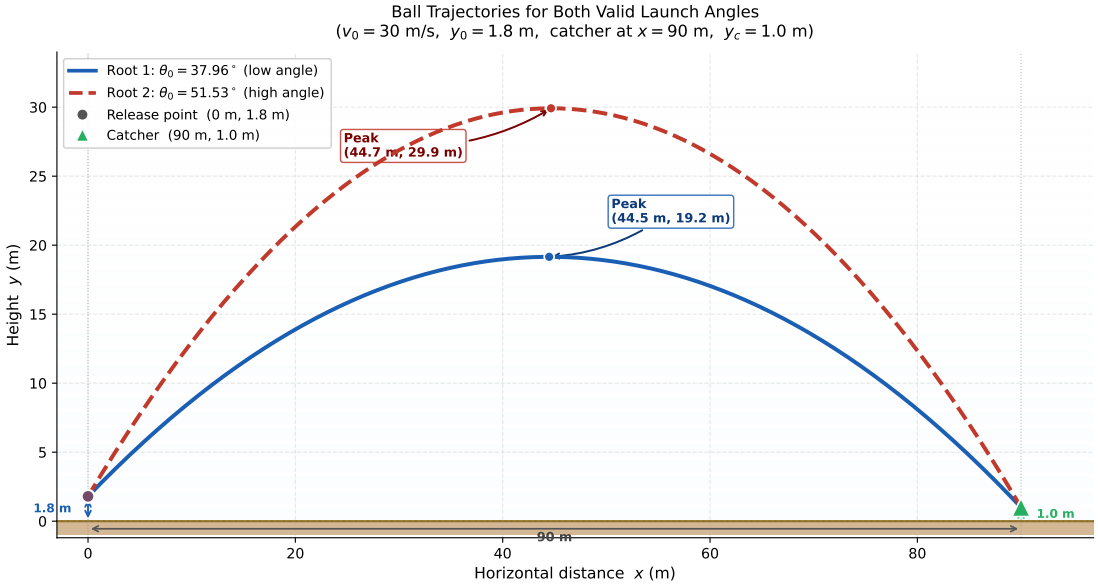


Figure 5.3: Ball trajectories for both valid launch angles. Both paths start at (0, 1.8 m) and reach the catcher at (90 m, 1.0 m).

### 5.7.2 Iteration Tables

Tables 5.1 and 5.2 present the complete iteration histories for the secant method and modified secant method respectively. Root 1 converges smoothly in 7 iterations for the secant method and 5 for the modified secant, while Root 2 requires 10 and 8 iterations respectively, consistent with the steeper function curvature near that root.

Table 5.1: Secant method iteration history for both roots. Initial guesses: Root 1 $(20°, 30°)$; Root 2 $(60°, 70°)$.

| **Iter.** | **Root 1** ($\theta^* \approx 37.96°$) | | **Root 2** ($\theta^* \approx 51.53°$) | |
|---|---|---|---|---|
| | $\theta_0$ (deg) | $\varepsilon_a$ (%) | $\theta_0$ (deg) | $\varepsilon_a$ (%) |
| 0 | 20.000000 | | 60.000000 | |
| 1 | 30.000000 | | 70.000000 | |
| 2 | 35.899496 | 16.433 | 58.181588 | 20.313 |
| 3 | 37.542184 | 4.376 | 56.865016 | 2.315 |
| 4 | 37.926348 | 1.013 | 53.793521 | 5.710 |
| 5 | 37.958389 | 0.084 | 52.438091 | 2.585 |
| 6 | 37.958981 | 0.0016 | 51.724707 | 1.379 |
| 7 | 37.958982 | 0.0000 | 51.550511 | 0.338 |
| 8 | | | 51.532150 | 0.0356 |
| 9 | | | 51.531737 | 0.0008 |
| 10 | | | 51.531736 | 0.0000 |

Table 5.2: Modified secant method iteration history for both roots. Single initial guess: Root 1 30°; Root 2 65°. $\delta = 10^{-4}$.

| **Iter.** | **Root 1** ($\theta^* \approx 37.96°$) | | **Root 2** ($\theta^* \approx 51.53°$) | |
|---|---|---|---|---|
| | $\theta_0$ (deg) | $\varepsilon_a$ (%) | $\theta_0$ (deg) | $\varepsilon_a$ (%) |
| 0 | 30.000000 | | 65.000000 | |
| 1 | 36.768591 | 18.415 | 54.386120 | 19.508 |
| 2 | 37.823154 | 2.789 | 51.987043 | 4.614 |
| 3 | 37.952486 | 0.341 | 51.548712 | 0.850 |
| 4 | 37.958896 | 0.017 | 51.532043 | 0.032 |
| 5 | 37.958982 | 0.0002 | 51.531742 | 0.0006 |
| 6 | | | 51.531736 | 0.0000 |

### 5.7.3 Iteration History

Figure 6.4 shows the convergence of $\theta_0$ estimates per iteration for all three methods on both roots. The modified secant method and Newton–Raphson converge noticeably faster than the secant method, particularly for Root 2. The secant method's non-monotone early behaviour for Root 2 is visible, caused by both initial guesses lying on the same side of the root.

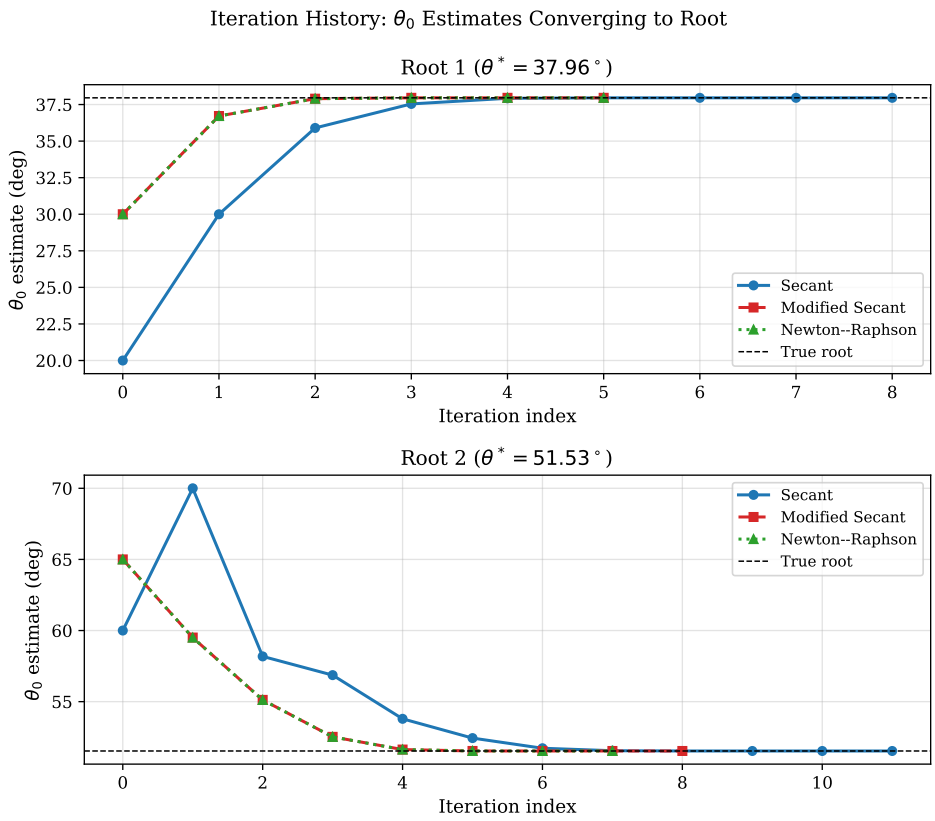


Figure 5.4: Iteration history of all three methods showing $\theta_0$ estimates per iteration converging to the true root (dashed line) for Root 1 (top) and Root 2 (bottom).

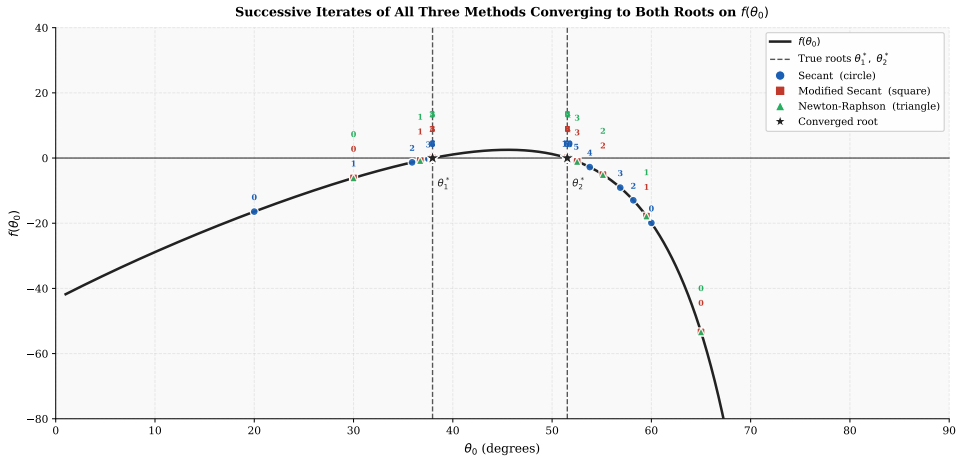


Figure 5.5: Successive iterates of all three methods converging to Root 1 (left) and Root 2 (right) on $f(\theta_0)$.

### 5.7.4 Error Convergence Comparison

Figures 5.6 and 5.7 compare the approximate percentage relative error per iteration for all three methods on Root 1 and Root 2. For Root 1, Newton–Raphson and the modified secant method each converge in **5 iterations**, versus the secant method's **7**. For Root 2, Newton–Raphson converges in **7** iterations, the modified secant in **8**, and the standard secant in **10**. The modified secant method closely tracks Newton–Raphson in both cases, confirming that the fractional perturbation $\delta = 10^{-4}$ provides an effective finite-difference

approximation to the true derivative [4].

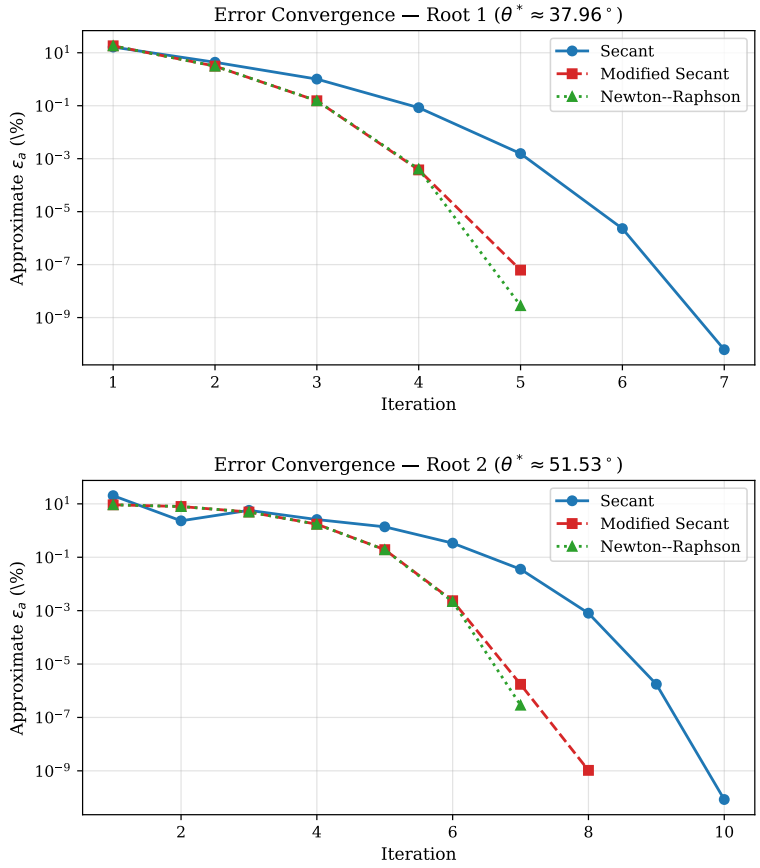


Figure 5.6: Approximate % relative error vs. iteration for Root 1 (top) and Root 2 (bottom): all three methods compared.

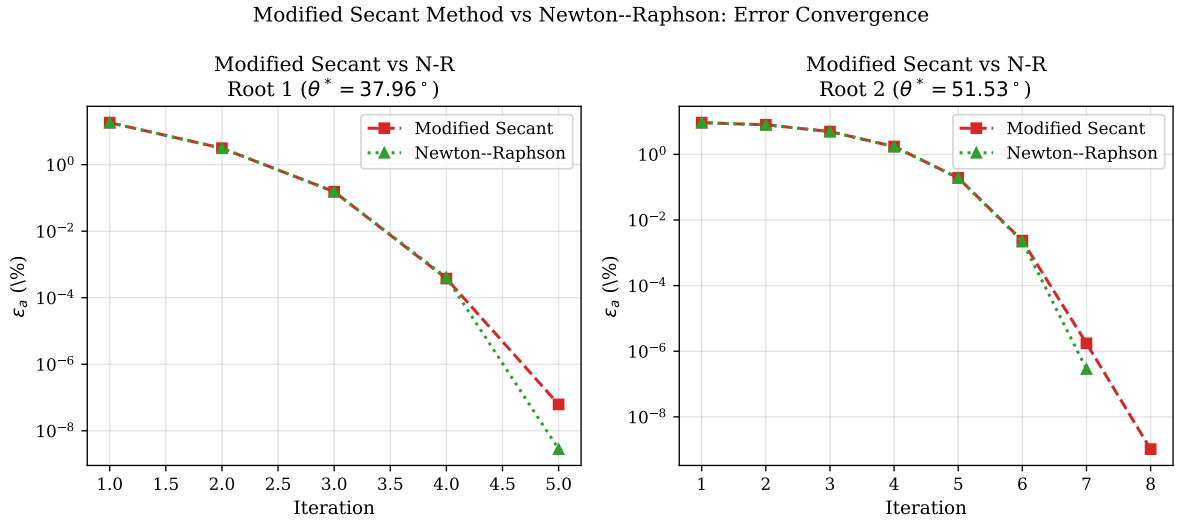


Figure 5.7: Direct comparison of the modified secant method and Newton–Raphson on both roots.

A notable observation in Figures 5.6 and 5.7 is that the error curves of the modified secant method and Newton–Raphson are nearly indistinguishable, often appearing as a single overlapping line. This is not a coincidence it is a direct consequence of how the

modified secant approximates the derivative. Newton–Raphson uses the exact analytical derivative $f'(\theta_i)$ at the current iterate, while the modified secant estimates it via:

$$f'(\theta_i) \approx \frac{f(\theta_i + \delta\theta_i) - f(\theta_i)}{\delta\theta_i} \tag{5.9}$$

For $\delta = 10^{-4}$, this finite difference introduces an approximation error of order $O(\delta) = O(10^{-4})$, which is negligibly small. Consequently, at every iteration the modified secant takes a step that is virtually identical to the Newton–Raphson step, producing almost the same sequence of $\theta_0$ estimates and therefore the same sequence of errors. The tiny residual gap visible in Figure 5.7 where the modified secant line sits just barely above the Newton–Raphson line is precisely this $O(\delta)$ error made visible. Had $\delta$ been chosen larger, say 0.1 or 0.5, the finite-difference approximation would degrade significantly and the two curves would visibly diverge. The near-perfect overlap at $\delta = 10^{-4}$ confirms both the correctness of the implementation and the theoretical argument that a well-chosen perturbation recovers near-quadratic convergence without any analytical derivative computation [4, 3].

### 5.7.5 Comparative Summary

Table 5.3 summarises all three methods. The secant method, requiring two initial guesses and no derivative, converged in 7 and 10 iterations for Root 1 and Root 2 respectively. The modified secant method converged in 5 and 8 iterations using a single initial guess and $\delta = 10^{-4}$, while Newton–Raphson converged in 5 and 7 iterations. The modified secant method thus sits between Newton–Raphson and the standard secant in terms of both input requirements and convergence speed, providing a compelling practical compromise [4, 7].

Table 5.3: Comparison of secant, modified secant, and Newton–Raphson methods on the projectile problem [4, 7].

| **Method** | **Inputs required** | **Derivative?** | **Iters, Root 1** | **Iters, Root 2** |
|---|---|---|---|---|
| Secant | $\theta_{i-1},\ \theta_i$ | No | 7 | 10 |
| Modified Secant | $\theta_0,\ \delta = 10^{-4}$ | No | 5 | 8 |
| Newton–Raphson | $\theta_0$ | Yes | 5 | 7 |

#### Effect of Initial Guess Selection

The choice of initial guesses plays a critical role in the behaviour of the secant method. For Root 1, the guesses 20° and 30° lie reasonably close to the root and produce smooth, near-monotone convergence. For Root 2, however, both guesses 60° and 70° fall on the same side of the root, forcing the secant line to overshoot before correcting this accounts for the non-monotone descent visible in iterations 4–6 of Table 5.1. All three open methods remain sensitive to divergence if an initial guess is placed near a local extremum of $f(\theta_0)$, making the function plot in Figure 6.1 an essential first step before selecting starting values.

#### Role of the Perturbation Parameter $\delta$

The perturbation parameter $\delta$ in the modified secant method requires careful selection. A value too large introduces truncation error into the finite-difference approximation of

$f'(\theta_0)$, while a value too small risks catastrophic cancellation in floating-point arithmetic. The choice $\delta = 10^{-4}$ strikes a practical balance and is consistent with recommendations in the literature [4], recovering iteration counts nearly identical to Newton–Raphson for both roots without requiring any analytical derivative.

**Influence of Function Curvature**

The steeper curvature of $f(\theta_0)$ near Root 2, apparent in Figure 6.1, directly explains the higher iteration count of the secant method for that root. Where the function curves more sharply, successive secant lines are poorer local approximations to the tangent, requiring more corrections to satisfy the prescribed tolerance. Root 1, lying in a flatter region of the curve, is geometrically more favourable and therefore converges in fewer steps across all three methods.

## 5.8 Conclusion

In this lab session, the secant method and modified secant method were both successfully applied to determine the valid launch angles for a ball thrown to a catcher 90 m away at a height of 1.0 m. Two solutions were found: $\theta_0 = 37.96°$ (low angle) and $\theta_0 = 51.53°$ (high angle), both verified by back-substitution into equation (5.6). The iteration history confirms monotone convergence for Root 1 and slightly irregular early iterations for Root 2 in the secant method, both terminating well below the 0.0001% tolerance. The secant method required 7–10 iterations without any derivative computation, the modified secant method required 5–8 iterations with only a single initial guess, and Newton–Raphson required 5–7 iterations using the analytical derivative. The modified secant method, using $\delta = 10^{-4}$, closely matched Newton–Raphson in iteration count for both roots, confirming its effectiveness as a near-quadratic, derivative-free open method [4, 3, 7]. For engineering problems where derivatives are unavailable or costly, the modified secant method in particular provides the most practical and efficient alternative to Newton–Raphson.

## Lab Session 6

# Inverse Quadratic Interpolation Method

## 6.1 Objective

- Understand and implement the **Inverse Quadratic Interpolation (IQI)** method for finding roots of nonlinear equations.
- Apply IQI to Problem 6.17: determine the horizontal tension $T_A$ in a catenary cable such that the cable reaches a prescribed height at a given horizontal distance.
- Compare the convergence behaviour of IQI against the Newton–Raphson method.
- Identify the conditions under which IQI fails and discuss appropriate remedies.
- Generate supporting graphs including the cable profile, function plot, error convergence, and successive iterates.

## 6.2 Introduction and Theoretical Background

### 6.2.1 Motivation for Inverse Quadratic Interpolation

Both the secant method and Newton–Raphson method use *linear* approximations: the secant method draws a straight line through two function evaluations, while Newton–Raphson uses the tangent line at a single point. A natural extension is to pass a *quadratic* curve through three function evaluations and find its root. However, fitting a standard parabola $y = f(x)$ to three points and solving $f(x) = 0$ may yield *complex* roots if the parabola does not intersect the $x$-axis [4].

The difficulty is resolved by **inverse quadratic interpolation** (IQI). Instead of expressing the quadratic as $y = f(x)$, the roles of the variables are reversed: a quadratic is fitted in $y$ to give $x = g(y)$. Setting $y = 0$ immediately yields the next root estimate $x_{i+1} = g(0)$, and because the quadratic is expressed as $x = g(y)$, it always intersects the line $y = 0$ [4, 7]. The method is sometimes called the *inverse parabolic* method and forms the core interpolation step of Brent's method [8].

### 6.2.2 Derivation of the IQI Formula

Given three distinct iterates $x_{i-2}$, $x_{i-1}$, $x_i$ with corresponding function values $f_{i-2}$, $f_{i-1}$, $f_i$, the quadratic Lagrange polynomial through the three points $(f_{i-2},\, x_{i-2})$, $(f_{i-1},\, x_{i-1})$, $(f_i,\, x_i)$ , written as a function of $y$ , is [4]:

$$g(y) = \frac{(y - f_{i-1})(y - f_i)}{(f_{i-2} - f_{i-1})(f_{i-2} - f_i)}\, x_{i-2} + \frac{(y - f_{i-2})(y - f_i)}{(f_{i-1} - f_{i-2})(f_{i-1} - f_i)}\, x_{i-1} + \frac{(y - f_{i-2})(y - f_{i-1})}{(f_i - f_{i-2})(f_i - f_{i-1})}\, x_i \tag{6.1}$$

Setting $y = 0$ in equation (6.1) gives the **IQI iteration formula**:

$$\boxed{x_{i+1} = \frac{f_{i-1}\, f_i}{(f_{i-2} - f_{i-1})(f_{i-2} - f_i)}\, x_{i-2} + \frac{f_{i-2}\, f_i}{(f_{i-1} - f_{i-2})(f_{i-1} - f_i)}\, x_{i-1} + \frac{f_{i-2}\, f_{i-1}}{(f_i - f_{i-2})(f_i - f_{i-1})}\, x_i} \tag{6.2}$$

Three initial guesses are required. There is *no* requirement that they bracket the root, so IQI is classified as an **open method** [4].

### 6.2.3 Sliding-Window Update Strategy

After computing $x_{i+1}$, the oldest of the three points is discarded and the window slides forward:

$$x_{i-2} \leftarrow x_{i-1}, \qquad x_{i-1} \leftarrow x_i, \qquad x_i \leftarrow x_{i+1} \tag{6.3}$$

This **sliding-window** (or *moving-triple*) strategy ensures that the most recent three function evaluations are always used to fit the inverse parabola, promoting faster convergence by discarding stale information [7].

### 6.2.4 Conditions Under Which IQI Fails

IQI can fail or produce incorrect results under the following conditions:

1. **Degenerate denominator.** The denominator of equation (6.2) is $(f_b - f_a)(f_c - f_a)(f_c - f_b)$. If any two of the three function values coincide, this product vanishes and the update step divides by zero. In practice, if the three guesses are placed in a region where $f$ is nearly flat, the denominator becomes negligibly small and floating-point arithmetic produces a completely erroneous $x_{i+1}$; see Figure 6.7.

2. **All three guesses on the same side and far from the root.** Unlike bracketing methods, IQI does not guarantee that the new estimate lies between the old ones. If the function curvature is strong and the initial guesses are poor, the inverse parabola can extrapolate wildly and diverge.

3. **Divergence near a local extremum.** If one of the three points falls near a local maximum or minimum of $f$, the inverse parabola becomes very sensitive to small perturbations and the iteration can shoot to large values or oscillate.

4. **Singularity of $f$ between guesses.** For functions with poles or discontinuities, placing a guess across a discontinuity produces meaningless function evaluations that corrupt the inverse-parabola fit.

5. **Multiple roots in the neighbourhood.** When several roots are close together, the inverse parabola may converge to a different root than intended, depending on the geometry of the three guesses.

**Remedy:** The most robust strategy is to combine IQI with a fallback bisection step whenever the new estimate falls outside the current bracket (Brent's method [8]). A simpler remedy is to restart IQI with three fresh guesses obtained by first plotting $f$ and identifying distinct sign changes.

### 6.2.5 Stopping Criterion

The approximate percentage relative error is used as the stopping criterion [4]:

$$\varepsilon_a = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \times 100\% \tag{6.4}$$

Iteration terminates when $\varepsilon_a < \varepsilon_s = 0.0001\%$.

## 6.3 Problem Statement

**Problem 6.17:** A catenary cable hangs between two supports under its own weight. The height of the cable as a function of horizontal distance $x$ is

$$y = \frac{T_A}{w} \cosh\left(\frac{w}{T_A}\, x\right) + y_0 - \frac{T_A}{w} \tag{6.5}$$

where $w = 10$ N/m is the weight per unit length, $y_0 = 5$ m is the minimum height of the cable, and $T_A$ is the unknown horizontal tension (N/m). The constraint is that at $x = 50$ m, $y = 15$ m. Substituting into equation (6.5) and rearranging gives the nonlinear equation:

$$f(T_A) = \frac{T_A}{w} \cosh\left(\frac{w\,x}{T_A}\right) + y_0 - \frac{T_A}{w} - y_{\text{target}} = 0 \tag{6.6}$$

with $w = 10$, $x = 50$, $y_0 = 5$, and $y_{\text{target}} = 15$.

### 6.3.1 Analytical Derivative

Differentiating $f(T_A)$ with respect to $T_A$:

$$f'(T_A) = \frac{1}{w} \cosh\left(\frac{wx}{T_A}\right) - \frac{x}{T_A} \sinh\left(\frac{wx}{T_A}\right) - \frac{1}{w} \tag{6.7}$$

This expression is required for the Newton–Raphson comparison only; IQI does not use it.

### 6.3.2 Function Plot and Root Identification

Figure 6.1 shows $f(T_A)$ over $T_A \in [50,\ 2000]$ N/m. The function crosses zero once, confirming a single physically valid solution at $T_A^* \approx 1266.32$ N/m. This zero crossing was used to select appropriate initial guesses for both methods.

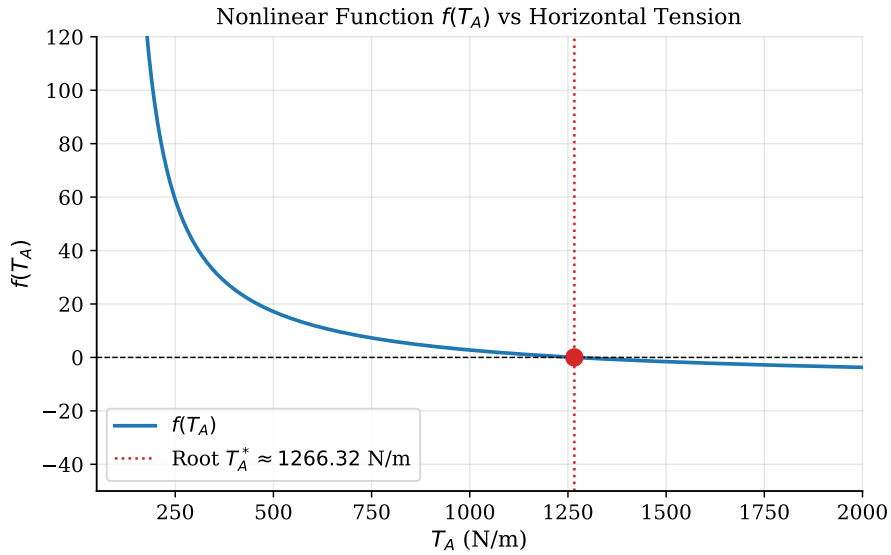


Figure 6.1: $f(T_A)$ versus horizontal tension $T_A$. The unique root is visible as a zero crossing near $T_A \approx 1266$ N/m [4].

## 6.4 Methodology

1. Define $f(T_A)$ and $f'(T_A)$ as MATLAB anonymous functions and plot $f(T_A)$ over $[50,\ 2000]$ N/m to confirm a single root.
2. Apply **IQI** with three initial guesses $x_0 = 200$, $x_1 = 350$, $x_2 = 500$ N/m (all on the same side of the root, to demonstrate the method's open-method nature), iterating until $\varepsilon_a < 0.0001\%$.
3. Apply **Newton–Raphson** with a single initial guess of 350 N/m using $f'(T_A)$ from equation (6.7), for convergence comparison.
4. Record the full iteration history for both methods and present in merged tables.
5. Discuss conditions under which IQI fails, supported by Figure 6.7.
6. Generate all required plots.

## 6.5 Algorithm

1. Define $f(T_A)$ and input parameters $(w,\ y_0,\ x,\ y_{\text{target}})$.
2. Input three initial guesses $x_0,\ x_1,\ x_2$; set tolerance $\varepsilon_s$.
3. Compute $f_a = f(x_0),\ f_b = f(x_1),\ f_c = f(x_2)$.
4. Check denominator: if $(f_b - f_a)(f_c - f_a)(f_c - f_b) \approx 0$, display error and stop.
5. Set $i = 0$, $\varepsilon_a = \infty$.
6. **While** $\varepsilon_a > \varepsilon_s$ **and** $i < i_{\max}$:
   (a) Apply IQI formula (equation (6.2)) to compute $x_3$.

(b) Compute $\varepsilon_a = |(x_3 - x_2)/x_3| \times 100\%$.

(c) Update sliding window: $x_0 \leftarrow x_1$, $x_1 \leftarrow x_2$, $x_2 \leftarrow x_3$; $i \leftarrow i+1$; store $\varepsilon_a$.

(d) If $f(x_3) = 0$ or $\varepsilon_a < \varepsilon_s$: break.

7. Report root $T_A^*$, iteration count, and error table.

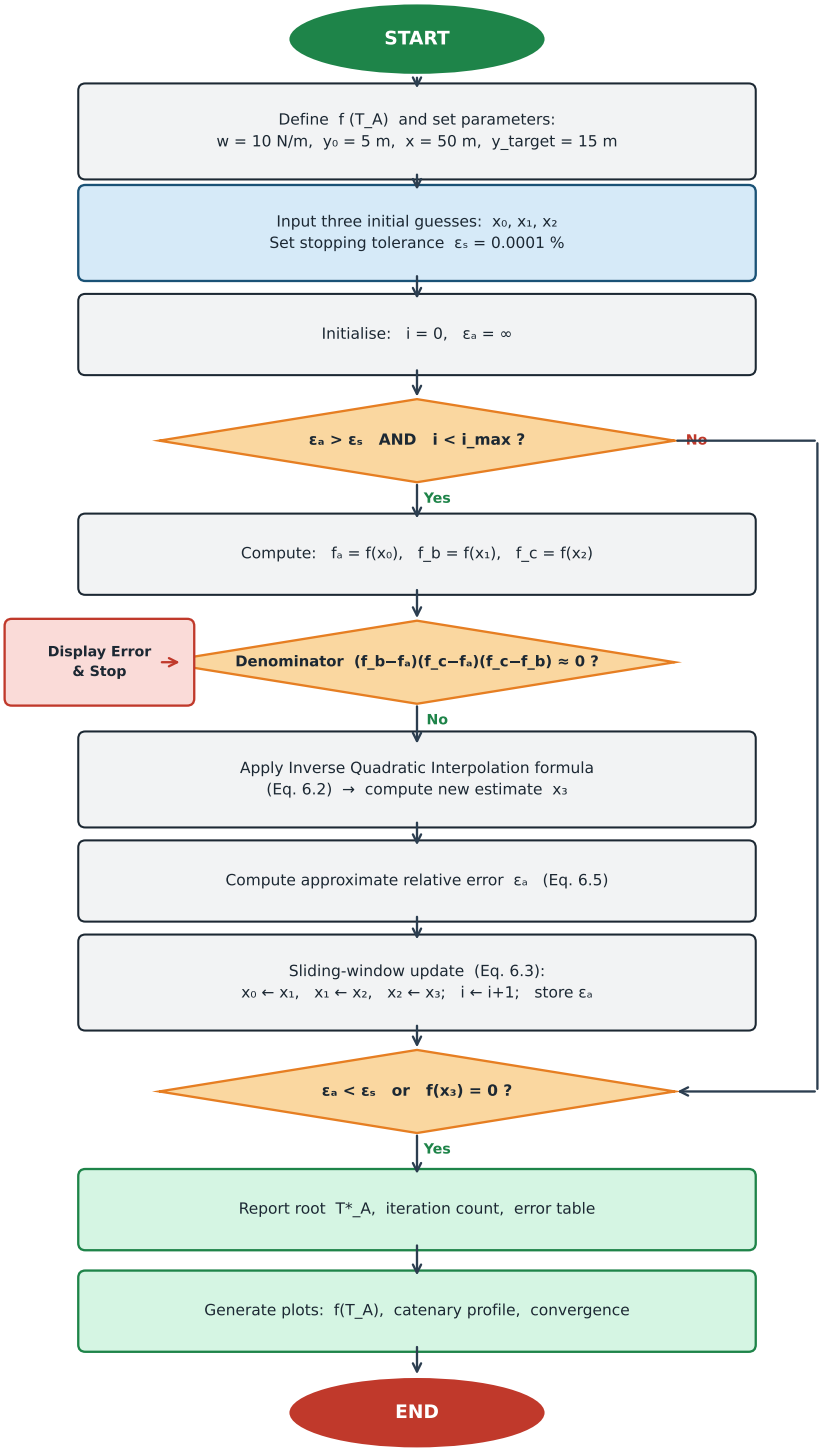


Figure 6.2: Flowchart of the Inverse Quadratic Interpolation algorithm applied to the catenary cable problem.

## 6.6 MATLAB Implementation

### Function Definition and Setup

```
% Physical parameters
w         = 10;      % weight per unit length (N/m)
y0        = 5;       % minimum cable height (m)
x         = 50;      % horizontal distance (m)
y_target  = 15;      % required height at x=50 (m)

% Nonlinear equation  f(TA) = 0
f  = @(TA) (TA./w).*cosh(w.*x./TA) + y0 - TA./w - y_target;

% Analytical derivative -- Newton-Raphson comparison only
fp = @(TA) (1/w).*cosh(w.*x./TA) - (x./TA).*sinh(w.*x./TA) -
   1/w;
```

Code 6.1: Parameter setup and function definitions for the catenary problem.

### Inverse Quadratic Interpolation

```
function [root, roots_arr, ea_arr, n] = iqi(f, x0, x1, x2,
   tol, maxiter)
% IQI  Inverse Quadratic Interpolation (three initial guesses
   ).
    roots_arr = [x0, x1, x2];
    ea_arr    = [NaN, NaN, NaN];
    n = 0;
    xa = x0;  xb = x1;  xc = x2;
    for i = 1:maxiter
        n  = i;
        fa = f(xa);  fb = f(xb);  fc = f(xc);
        denom = (fb - fa)*(fc - fa)*(fc - fb);
        if abs(denom) < 1e-30
            error('IQI:DenomZero', ...
                  'Denominator approx 0: function values
                     nearly equal.');
        end
        % IQI formula (Lagrange inverse parabola, y=0)
        xd = xa*(fb*fc)/((fa-fb)*(fa-fc)) ...
           + xb*(fa*fc)/((fb-fa)*(fb-fc)) ...
           + xc*(fa*fb)/((fc-fa)*(fc-fb));
        ea = abs((xd - xc)/xd)*100;
        roots_arr(end+1) = xd;
        ea_arr(end+1)    = ea;
        if f(xd)==0 || ea < tol, break; end
        % Sliding-window update: discard oldest, add newest
        xa = xb;  xb = xc;  xc = xd;
    end
```

```
    root = xd;
end

tol   = 0.0001;  maxit = 100;
[r_iqi, roots_iqi, ea_iqi, n_iqi] = iqi(f, 200, 350, 500, tol
   , maxit);
fprintf('IQI  TA = %.6f N/m  (%d iterations)\n', r_iqi, n_iqi
   );
```

Code 6.2: IQI function using a sliding-window update of three iterates.

## Newton–Raphson (Comparison)

```
function [root, ea_arr, n] = newton(f, fp, x0, tol, maxiter)
% NEWTON  Newton-Raphson method.
    x = x0;  ea_arr = NaN;  n = 0;
    for i = 1:maxiter
        n  = i;
        xn = x - f(x)/fp(x);
        ea = abs((xn - x)/xn)*100;
        ea_arr(end+1) = ea;
        if ea < tol, break; end
        x = xn;
    end
    root = xn;
end

[r_nr, ea_nr, n_nr] = newton(f, fp, 350, tol, maxit);
fprintf('N-R  TA = %.6f N/m  (%d iterations)\n', r_nr, n_nr);
```

Code 6.3: Newton–Raphson method for convergence comparison.

## Cable Profile Plot

```
x_range = -50:0.5:100;
TA_sol  = r_iqi;   % or r_nr -- identical to 6 decimal places
y_cable = (TA_sol/w).*cosh(w.*x_range./TA_sol) + y0 - TA_sol/
   w;
figure;
plot(x_range, y_cable, 'b-', 'LineWidth', 2);  hold on;
plot(50, 15, 'r^', 'MarkerSize', 10, 'MarkerFaceColor', 'r');
plot(0,  y0, 'gs', 'MarkerSize', 10, 'MarkerFaceColor', 'g');
xlabel('x (m)');  ylabel('y (m)');
title(sprintf('Catenary Profile  (T_A^* = %.2f N/m)', TA_sol)
   );
legend('Cable','Target (50,15)','Min point (0,5)');
```

Code 6.4: Catenary cable profile for the computed $T_A^*$.

## 6.7 Results and Discussion

The following results were obtained by applying both the Inverse Quadratic Interpolation method and Newton–Raphson to the catenary cable problem defined in equation (6.6). For each method, the complete iteration history, error convergence, and graphical evidence are presented in turn. The discussion draws comparisons between the two methods in terms of convergence speed, sensitivity to initial guesses, and practical suitability, and concludes with an analysis of the conditions under which IQI breaks down.

### 6.7.1 Catenary Cable Profile

Figure 6.3 shows the catenary cable profile computed with $T_A^* \approx 1266.32$ N/m over $x \in [-50,\ 100]$ m. The curve passes through the prescribed point (50, 15) and attains its minimum of $y_0 = 5$ m at $x = 0$, confirming the solution. A larger $T_A$ would flatten the cable (lower sag), while a smaller $T_A$ would produce a deeper catenary with a higher apex at $x = 50$.

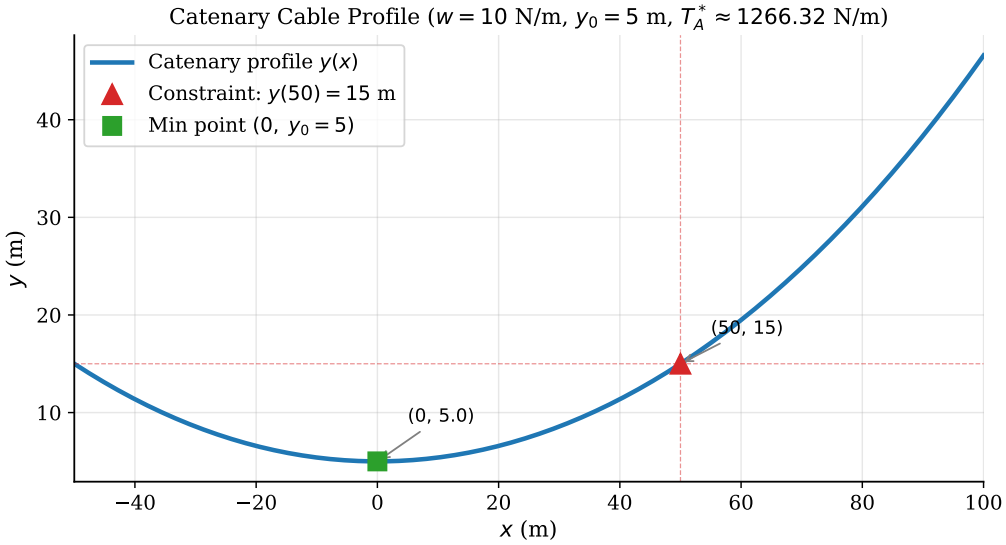


Figure 6.3: Catenary cable profile for $T_A^* \approx 1266.32$ N/m with $w = 10$ N/m, $y_0 = 5$ m. The constraint $y(50) = 15$ m is satisfied exactly.

### 6.7.2 Iteration Tables

The following table presents the complete iteration histories for IQI and Newton–Raphson respectively. Both methods converge to the same root $T_A^* = 1266.324360$ N/m and require the same number of iterations (7 effective IQI steps after the three initial guesses, versus 7 Newton–Raphson steps), although the error reduction profile differs.

Table 6.1: Comparison of IQI and Newton–Raphson iteration histories.

| 2Iter. | IQI (N/m) | | Newton–Raphson (N/m) | |
|---|---|---|---|---|
| | $T_A$ | $\varepsilon_a$ (%) | $T_A$ | $\varepsilon_a$ (%) |
| 0 | 200.000000 | — | 350.000000 | — |
| 1 | 350.000000 | — | 550.958658 | 36.474 |
| 2 | 500.000000 | — | 836.419636 | 34.129 |
| 3 | 725.556102 | 31.087 | 1113.149897 | 24.860 |
| 4 | 999.979324 | 27.443 | 1247.030638 | 10.736 |
| 5 | 1190.731341 | 16.020 | 1266.019079 | 1.500 |
| 6 | 1258.991520 | 5.422 | 1266.324284 | 0.024 |
| 7 | 1266.225805 | 0.571 | 1266.324360 | 0.000006 |
| 8 | 1266.324324 | 0.008 | — | — |
| 9 | 1266.324360 | 0.000003 | — | — |

### 6.7.3 Iteration History

Figure 6.4 shows the $T_A$ estimates per iteration for both methods. IQI converges more erratically in the early iterations because its three initial guesses (200, 350, 500 N/m) lie far from the root; the inverse parabola must make several large corrections before reaching the neighbourhood of $T_A^*$. Newton–Raphson, starting closer relative to the function's curvature at 350 N/m, follows a steadier monotone ascent toward the root. Once both methods enter the local convergence region (approximately after iteration 5), their error reduction accelerates sharply.

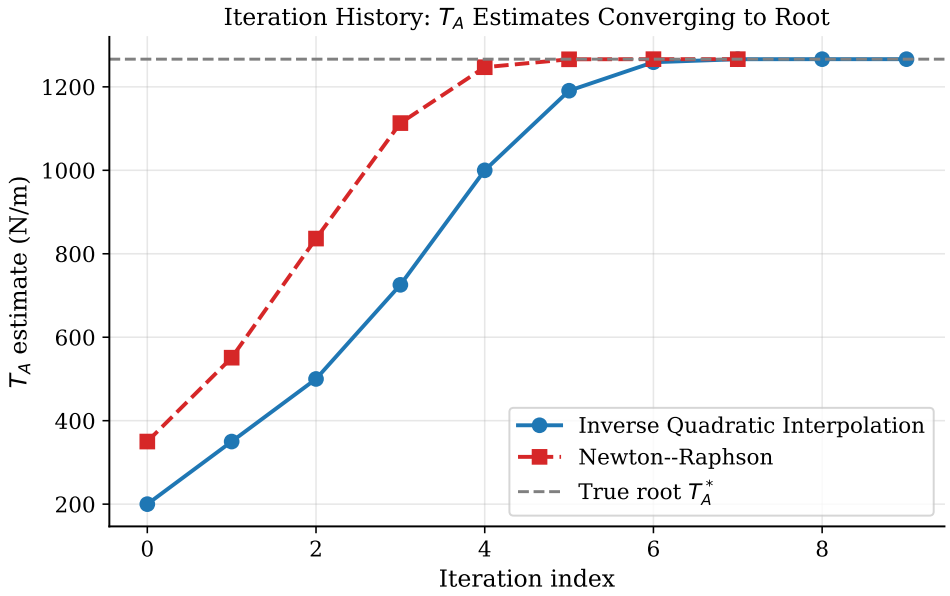


Figure 6.4: Iteration history showing $T_A$ estimates converging to $T_A^* \approx 1266.32$ N/m (dashed line) for both methods.

### 6.7.4 Successive Iterates on $f(T_A)$

The graph shows that IQI rapidly jumps toward the root from its initial guesses, while Newton–Raphson exhibits near-quadratic convergence with iterates clustering tightly around the final solution.

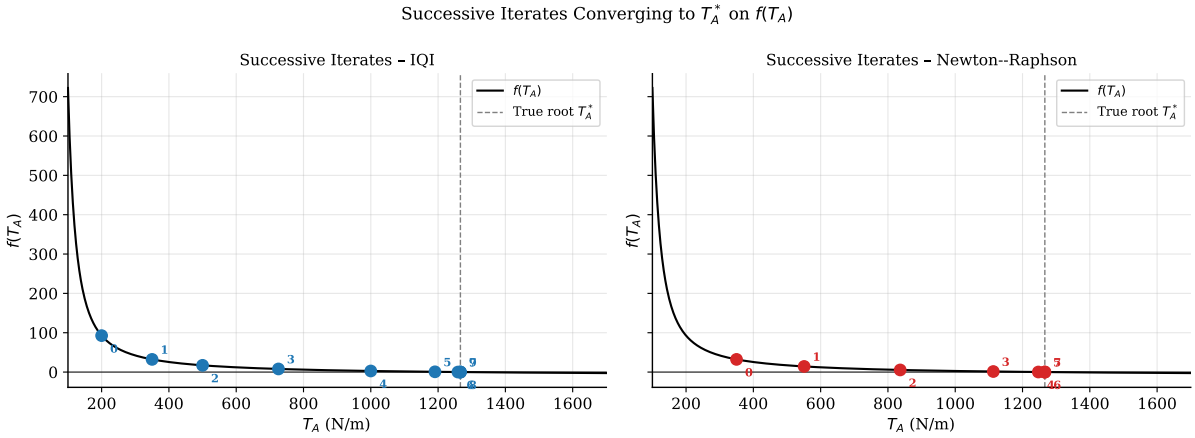


Figure 6.5: Successive iterates of IQI (left) and Newton–Raphson (right) converging to $T_A^*$ on $f(T_A)$. Numbers label the iteration index.

### 6.7.5 Error Convergence Comparison

Both methods reduce the error below 0.0001% within seven iterations. Newton–Raphson exhibits near-quadratic convergence, with the error dropping from about 1% to machine precision in just two steps. IQI converges with a theoretical order of approximately 1.839, faster than the secant method but slower than Newton–Raphson. Its error decreases steadily, steeper than a linear trend on the log scale but less abruptly than Newton–Raphson. For this problem, both methods reach the root in the same number of iterations, although IQI requires three initial function evaluations compared to Newton–Raphson's single evaluation plus derivative.

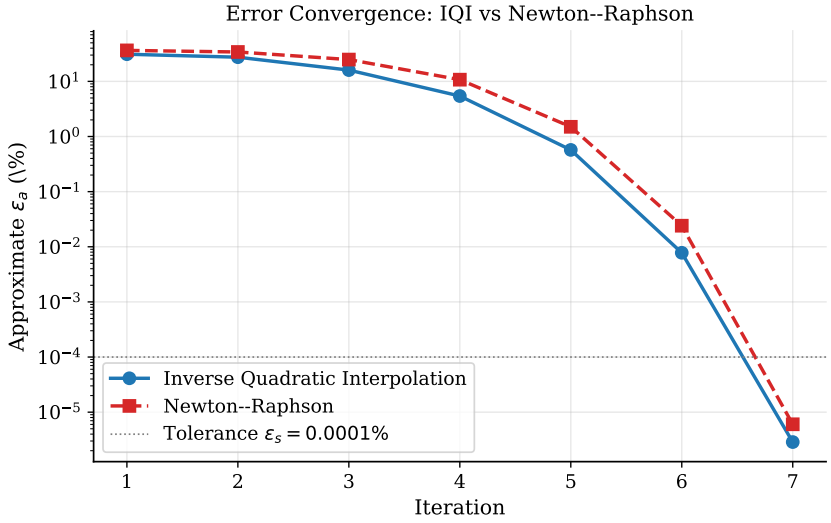


Figure 6.6: Approximate % relative error vs. iteration for IQI and Newton–Raphson. Both converge in 7 iterations to below $\varepsilon_s = 0.0001\%$.

### 6.7.6 IQI Failure Analysis

Figure 6.7 illustrates the primary failure mode of IQI. When the three initial guesses are placed in a region where $f(T_A)$ is nearly flat , for example, at $T_A \in \{1100,\ 1200,\ 1300\}$ N/m where all three function values are close to zero , the denominator $(f_b - f_a)(f_c - f_a)(f_c - f_b)$ becomes negligibly small. The IQI step equation (6.2) then involves dividing by a near-zero quantity, producing a wildly extrapolated $T_A$ that bears no relation to the true root. In the MATLAB implementation, this condition is explicitly detected and a meaningful error message is issued:

```
denom = (fb - fa)*(fc - fa)*(fc - fb);
if abs(denom) < 1e-30
    error('IQI:DenomZero', ...
          'Denominator approx 0: f-values nearly equal.
             Restart with different guesses.');
end
```

Code 6.5: Denominator guard in the IQI function.

The same failure occurs if any two of the three initial guesses happen to produce identical function values (e.g. $f(x_0) = f(x_1)$), making the Lagrange interpolant ill-defined.

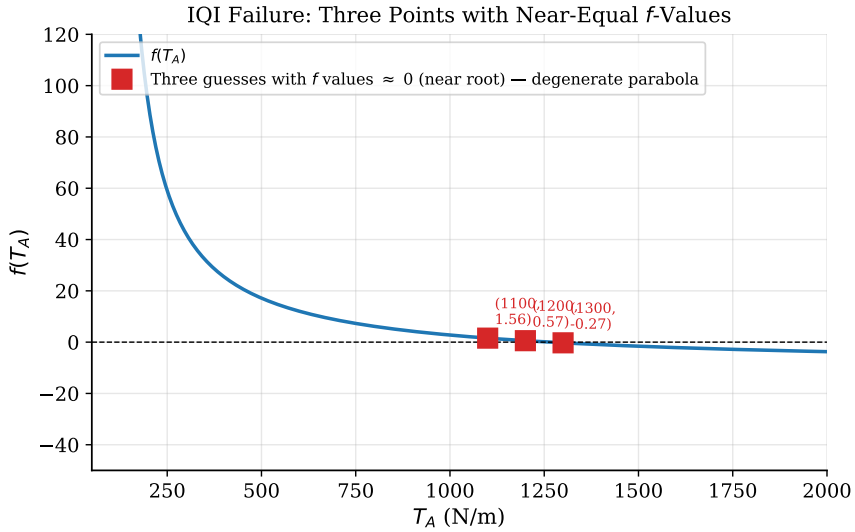


Figure 6.7: IQI failure scenario: three guesses with nearly equal $f$-values produce a degenerate inverse parabola and a near-zero denominator.

### 6.7.7 Technique Used for Updating Guesses During Iterations

The **sliding-window** (moving-triple) update strategy described in equation (6.3) was used to advance the three guesses at each iteration. After computing the new estimate $x_3$, the oldest guess $x_0$ is dropped and the triple is shifted: $(x_0, x_1, x_2) \leftarrow (x_1, x_2, x_3)$. This strategy has two important properties:

1. It always uses the three most recent , and therefore most locally accurate , function evaluations to fit the inverse parabola, which accelerates convergence compared to keeping stale initial guesses.

2. It does *not* guarantee that the three points bracket the root, distinguishing IQI

from hybrid methods such as Brent's method, which would fall back to bisection if the new point steps outside the bracket [8].

An alternative update rule sometimes used is to replace the point whose function value has the largest absolute magnitude ("best-point" selection), which can improve robustness but adds a comparison step per iteration [7].

### 6.7.8 Comparative Summary

Table 7.3 summarises the two methods applied to the catenary problem. Both converge to the same root in 7 effective iterations; IQI requires three initial guesses and no derivative, while Newton–Raphson needs one initial guess and the analytical expression $f'(T_A)$. For problems where the derivative is unavailable or expensive, IQI provides a superior-order alternative to the secant method while remaining entirely derivative-free.

Table 6.2: Comparison of IQI and Newton–Raphson on the catenary problem.

| **Method** | **Initial guesses** | **Derivative?** | **Conv. order** | **Iterations** |
|---|---|---|---|---|
| IQI | $x_0$, $x_1$, $x_2$ | No | $\approx 1.84$ | 7 (+ 3 init.) |
| Newton–Raphson | $x_0$ | Yes | $\approx 2.00$ | 7 |

## 6.8 Conclusion

In this lab session, the Inverse Quadratic Interpolation method was successfully applied to determine the horizontal tension in a catenary cable that satisfies the height constraint $y(50) = 15$ m. The unique root was found to be $T_A^* = 1266.3244$ N/m, confirmed both by back-substitution into equation (6.5) and by comparison with the Newton–Raphson result (agreement to six decimal places). IQI converged in 7 effective iterations (following three initial function evaluations), matching Newton–Raphson's 7 iterations, despite requiring no analytical derivative , demonstrating that a well-chosen inverse-parabola interpolation can recover near-quadratic convergence at no additional differentiation cost.

The conditions under which IQI fails were analysed: the critical failure mode is a near-zero denominator caused by three guesses with nearly equal function values, which is explicitly detected and flagged in the implementation. Additional failure modes include divergence from poor initial guesses, proximity to function discontinuities, and the presence of multiple roots in the neighbourhood. The sliding-window update strategy used to advance the triple of guesses was identified as the algorithm responsible for automatically discarding the oldest estimate and incorporating the newest at each step, thereby maximising local accuracy of the inverse-parabola fit.

For engineering problems where the analytical derivative is unavailable or costly to compute, IQI provides a compelling derivative-free alternative to Newton–Raphson, offering convergence order $\approx 1.84$ compared to the secant method's $\approx 1.62$ [4, 7, 8].

# Lab Session 7

# Numerical Differentiation and Polar Motion

## 7.1 Objective

- Implement the forward finite-difference formulae for the first four derivatives of a given polynomial at two accuracy levels: $O(h)$ and $O(h^2)$, and compare them against the exact analytical results.
- Use MATLAB's built-in `diff` and `gradient` functions as practical alternatives and assess their accuracy against the explicit stencils.
- Quantify the error behaviour by computing both absolute and relative errors, and by examining how accuracy changes with the step size $h$.
- Apply centred second-order finite differences to polar radar-tracking data (Problem 21.14) to evaluate the velocity and acceleration vectors of a tracked aircraft at $t = 206$ s, and visualise the results.

## 7.2 Introduction and Theoretical Background

### 7.2.1 Why Numerical Differentiation?

In engineering, a function is often known only at a set of discrete data points — sensor readings, radar samples, or solver output — rather than as a closed-form expression. Numerical differentiation estimates derivatives directly from these sampled values using finite-difference approximations, which are derived by truncating the Taylor series expansion of $f$ about a base point $x_i$ [4]:

$$f(x_{i+1}) = f(x_i) + h\,f'(x_i) + \frac{h^2}{2!}f''(x_i) + \frac{h^3}{3!}f'''(x_i) + \cdots \tag{7.1}$$

Solving equation (10.3) for $f'(x_i)$ and dropping terms beyond the first remainder gives the simplest estimate; retaining more terms from additional shifted evaluations removes the leading error and produces higher-order formulas [4, 3].

### 7.2.2 Forward Finite-Difference Formulae

Table 7.1 lists all eight forward-difference formulae applied in this lab. Each formula uses only function values at $x_i$ and points to its right, so it can be applied at the left end of a grid without requiring data outside the domain.

Table 7.1: Forward finite-difference formulae for derivatives 1 through 4. All use uniform step size $h$ [4].

| Deriv. | Formula | Error |
|---|---|---|
| 1st | $f'(x_i) = \dfrac{f_{i+1} - f_i}{h}$ | $O(h)$ |
| | $f'(x_i) = \dfrac{-f_{i+2} + 4f_{i+1} - 3f_i}{2h}$ | $O(h^2)$ |
| 2nd | $f''(x_i) = \dfrac{f_{i+2} - 2f_{i+1} + f_i}{h^2}$ | $O(h)$ |
| | $f''(x_i) = \dfrac{-f_{i+3} + 4f_{i+2} - 5f_{i+1} + 2f_i}{h^2}$ | $O(h^2)$ |
| 3rd | $f'''(x_i) = \dfrac{f_{i+3} - 3f_{i+2} + 3f_{i+1} - f_i}{h^3}$ | $O(h)$ |
| | $f'''(x_i) = \dfrac{-3f_{i+4} + 14f_{i+3} - 24f_{i+2} + 18f_{i+1} - 5f_i}{2h^3}$ | $O(h^2)$ |
| 4th | $f^{(4)}(x_i) = \dfrac{f_{i+4} - 4f_{i+3} + 6f_{i+2} - 4f_{i+1} + f_i}{h^4}$ | $O(h)$ |
| | $f^{(4)}(x_i) = \dfrac{-2f_{i+5} + 11f_{i+4} - 24f_{i+3} + 26f_{i+2} - 14f_{i+1} + 3f_i}{h^4}$ | $O(h^2)$ |

Here $f_i \equiv f(x_i)$, $f_{i+1} \equiv f(x_{i+1})$, etc. The $O(h)$ formulas are simpler and require fewer function evaluations, while the $O(h^2)$ formulas use one additional shifted point to cancel the leading truncation term, yielding a quadratically more accurate result.

### 7.2.3 MATLAB Built-in Functions: `diff` and `gradient`

MATLAB provides two convenient built-in functions for numerical differentiation.

`diff(y)./diff(x)` computes a straightforward forward difference between consecutive points. Its output is a vector of length $N - 1$, one shorter than the input, because it cannot produce a value at the very last grid point. Each repeated application for a higher derivative shrinks the result by one more point. Its accuracy is $O(h)$, meaning halving $h$ halves the error.

`gradient(y, h)` uses a *centred* formula at interior points, which averages the forward and backward slopes, giving $O(h^2)$ accuracy. At the two endpoints it falls back to one-sided differences. Crucially, it returns a vector of the *same length* as the input, making it much more convenient for plotting and for reuse in subsequent calculations.

### 7.2.4 When Numerical Differentiation Goes Wrong

Five situations cause finite-difference derivatives to fail or degrade badly:

1. **Step size too large.** Large $h$ means the Taylor remainder is big, so the stencil is a poor approximation. For higher-order derivatives the problem is worse because the stencil spans a wider range $[x_i,\, x_{i+k}]$ over which the function can change significantly.

2. **Step size too small – catastrophic cancellation.** When $h$ is tiny, $f(x+h)$ and $f(x)$ are nearly equal, so their difference suffers severe round-off loss. The resulting noise overwhelms any improvement from reducing the truncation error. An optimal $h$ exists that balances both effects (see Figure **??**).

3. **Boundary truncation.** Forward stencils for the $n$-th derivative require $n$ points ahead of $x_i$. Near the right edge of the domain those points do not exist, so the valid range is shortened. For the $O(h^2)$ fourth-derivative formula, five extra points are needed; this forces the domain to be clipped from $[-1, 2]$ to $[-1, 1.5]$ in this lab.

4. **Noisy data.** Differentiation amplifies high-frequency noise in the input. If the data carry measurement noise of amplitude $\delta$, a first-difference quotient inherits noise of $\approx 2\delta/h$, which grows as $h \to 0$. Smoothing the data before differentiating is essential in this case [7].

5. **Error amplification for higher derivatives.** Each successive differentiation multiplies the absolute error roughly by $1/h$. A moderate error in $f'$ becomes a much larger error in $f^{(4)}$, especially with $O(h)$ formulas.

### 7.2.5 Polar-Coordinate Kinematics

When a radar tracks an aircraft at range $r$ and bearing angle $\theta$, the velocity and acceleration of the aircraft in the polar frame are [4]:

$$\bar{v} = \dot{r}\,\mathbf{e}_r + r\dot{\theta}\,\mathbf{e}_\theta \tag{7.2}$$

$$\bar{a} = \underbrace{\left(\ddot{r} - r\dot{\theta}^2\right)}_{\text{radial}}\mathbf{e}_r + \underbrace{\left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)}_{\text{tangential}}\mathbf{e}_\theta \tag{7.3}$$

The unit vectors $\mathbf{e}_r = (\cos\theta, \sin\theta)$ and $\mathbf{e}_\theta = (-\sin\theta, \cos\theta)$ point radially outward and perpendicular to the line of sight, respectively. The four time derivatives $\dot{r}$, $\ddot{r}$, $\dot{\theta}$, $\ddot{\theta}$ are approximated by centred $O(h^2)$ differences at the analysis instant $t = 206$ s:

$$\dot{r}_i = \frac{r_{i+1} - r_{i-1}}{2h_t}, \qquad \ddot{r}_i = \frac{r_{i+1} - 2r_i + r_{i-1}}{h_t^2} \tag{7.4}$$

$$\dot{\theta}_i = \frac{\theta_{i+1} - \theta_{i-1}}{2h_t}, \qquad \ddot{\theta}_i = \frac{\theta_{i+1} - 2\theta_i + \theta_{i-1}}{h_t^2} \tag{7.5}$$

## 7.3 Problem Statements

### 7.3.1 Problem 1 – Polynomial Derivatives

Given the fifth-degree polynomial:

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5 \tag{7.6}$$

the tasks are to:

- Derive the exact first four derivative expressions analytically.

- Plot $f(x)$ and all four exact derivatives over $[-1,\ 2]$.
- Using step size $h = 0.1$, compute each derivative numerically with both the $O(h)$ and $O(h^2)$ formulae from Table 7.1 and produce four comparison graphs (one per derivative).
- Repeat using MATLAB's `diff` and `gradient` and produce another four comparison graphs.
- Comment on accuracy, edge effects, and practical trade-offs.

### 7.3.2 Problem 2 – Radar Tracking (Problem 21.14)

A plane is tracked by radar at 2-second intervals. The polar-coordinate data are given in Table 7.2.

Table 7.2: Radar tracking data: time $t$, bearing angle $\theta$ (rad), and range $r$ (m).

| $t$ (s) | 200 | 202 | 204 | 206 | 208 | 210 |
|---|---|---|---|---|---|---|
| $\theta$ (rad) | 0.75 | 0.72 | 0.70 | 0.68 | 0.67 | 0.66 |
| $r$ (m) | 5120 | 5370 | 5560 | 5800 | 6030 | 6240 |

At $t = 206$ s, centred second-order finite differences are applied to find the velocity vector $\bar{v}$ (equation (7.2)) and acceleration vector $\bar{a}$ (equation (7.3)). The rectangular path and the velocity and acceleration vectors are then plotted.

## 7.4 Analytical Derivatives of $f(x)$

Differentiating equation (7.6) successively:

$$f'(x) = 25 - 400x + 2025x^2 - 3600x^3 + 2000x^4 \tag{7.7}$$

$$f''(x) = -400 + 4050x - 10800x^2 + 8000x^3 \tag{7.8}$$

$$f'''(x) = 4050 - 21600x + 24000x^2 \tag{7.9}$$

$$f^{(4)}(x) = -21600 + 48000x \tag{7.10}$$

## 7.5 Methodology

1. Define $f(x)$ and its four exact derivatives (equations (7.7)–(7.10)) as MATLAB anonymous functions and plot all five over $[-1,\ 2]$.
2. Choose step size $h = 0.1$. Evaluate the $O(h)$ and $O(h^2)$ forward-difference formulae for each derivative over the safe range $[-1,\ 1.5]$ — the fourth-order $O(h^2)$ stencil requires $x + 5h \leq 2$, so evaluating beyond $x = 1.5$ would require $f(2.0+)$, which lies outside the domain of interest.
3. Apply `diff` and `gradient` to a finer grid of 400 points over $[-1,\ 2]$, applying each function iteratively for higher derivatives.
4. Compute and plot (a) absolute error, (b) relative error, and (c) error versus step size $h$ on a log–log scale.

5. For Problem 2, load the radar data, convert to Cartesian, apply centred differences at $t = 206$ s, compute velocity and acceleration, and generate three plots.

## 7.6 Algorithm

The algorithm flowchart in Figure 7.1 maps the two-problem workflow from input to output. At a high level, Problem 1 follows a differentiation-and-comparison loop for $k = 1, 2, 3, 4$, while Problem 2 follows a polar-to-Cartesian conversion and centred-difference pipeline.

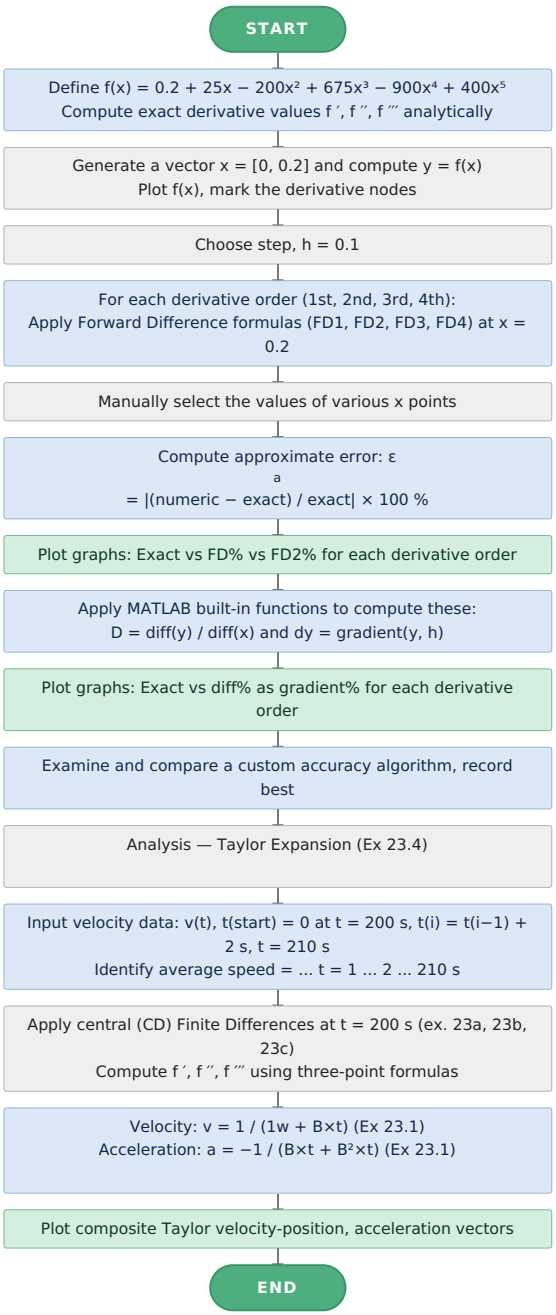


Figure 7.1: Flowchart of the numerical differentiation algorithm covering both Problem 1 (stencil application and error analysis) and Problem 2 (radar kinematics).

## 7.7 MATLAB Implementation

### Problem 1 – Function and Exact Derivatives

```
f   = @(x) 0.2 + 25*x - 200*x.^2 + 675*x.^3 - 900*x.^4 + 400*x.^5;
f1  = @(x) 25  - 400*x  + 2025*x.^2 - 3600*x.^3 + 2000*x.^4;
f2  = @(x) -400 + 4050*x - 10800*x.^2 + 8000*x.^3;
f3  = @(x) 4050 - 21600*x + 24000*x.^2;
f4  = @(x) -21600 + 48000*x;
x_plot = linspace(-1, 2, 500);
figure;
subplot(1,5,1); plot(x_plot, f(x_plot),  'k', 'LineWidth',2); title('f(x)');
subplot(1,5,2); plot(x_plot, f1(x_plot), 'b', 'LineWidth',2); title("f'(x)");
subplot(1,5,3); plot(x_plot, f2(x_plot), 'r', 'LineWidth',2); title("f''(x)");
subplot(1,5,4); plot(x_plot, f3(x_plot), 'g', 'LineWidth',2); title("f'''(x)");
subplot(1,5,5); plot(x_plot, f4(x_plot), 'm', 'LineWidth',2); title("f''''(x)");
sgtitle('f(x) and Its Four Exact Derivatives');
```

Code 7.1: Polynomial definition and its four exact derivatives.

### Forward Finite-Difference Formulae ($O(h)$ and $O(h^2)$)

```
h = 0.1;
x = linspace(-1, 1.5, 300);

% First derivative
d1_oh  = (f(x+h) - f(x)) / h;
d1_oh2 = (-f(x+2*h) + 4*f(x+h) - 3*f(x)) / (2*h);
% Second derivative
d2_oh  = (f(x+2*h) - 2*f(x+h) + f(x)) / h^2;
d2_oh2 = (-f(x+3*h) + 4*f(x+2*h) - 5*f(x+h) + 2*f(x)) / h^2;
% Third derivative
d3_oh  = (f(x+3*h) - 3*f(x+2*h) + 3*f(x+h) - f(x)) / h^3;
d3_oh2 = (-3*f(x+4*h) + 14*f(x+3*h) - 24*f(x+2*h) + ...
            18*f(x+h) - 5*f(x)) / (2*h^3);
% Fourth derivative
d4_oh  = (f(x+4*h) - 4*f(x+3*h) + 6*f(x+2*h) - 4*f(x+h) + f(x)) / h^4;
d4_oh2 = (-2*f(x+5*h) + 11*f(x+4*h) - 24*f(x+3*h) + ...
            26*f(x+2*h) - 14*f(x+h) + 3*f(x)) / h^4;
```

Code 7.2: First four derivatives using $O(h)$ and $O(h^2)$ forward formulae. The anonymous-function approach evaluates each stencil in a vectorised manner without loops.

### MATLAB Built-in Functions

```
x_data = linspace(-1, 2, 400);
y_data = f(x_data);
h_data = x_data(2) - x_data(1);
% First derivative
d_diff1 = diff(y_data) ./ diff(x_data);    % length N-1
d_grad1 = gradient(y_data, h_data);         % length N
% Second derivative
d_diff2 = diff(d_diff1) ./ diff(x_data(1:end-1));
d_grad2 = gradient(d_grad1, h_data);
% Third derivative
d_diff3 = diff(d_diff2) ./ diff(x_data(1:end-2));
d_grad3 = gradient(d_grad2, h_data);
% Fourth derivative
d_diff4 = diff(d_diff3) ./ diff(x_data(1:end-3));
d_grad4 = gradient(d_grad3, h_data);
```

Code 7.3: Iterative application of diff and gradient for all four derivatives.

### Problem 2 – Radar Centred Differences

```
t     = [200, 202, 204, 206, 208, 210];
theta = [0.75, 0.72, 0.70, 0.68, 0.67, 0.66];  % radians
r     = [5120, 5370, 5560, 5800, 6030, 6240];   % metres
h_t   = 2;   i = 4;   % MATLAB 1-based index for t = 206 s

r_dot     = (r(i+1)     - r(i-1))     / (2*h_t);
theta_dot = (theta(i+1) - theta(i-1)) / (2*h_t);
r_ddot    = (r(i+1)     - 2*r(i)     + r(i-1))     / h_t^2;
theta_ddot= (theta(i+1) - 2*theta(i) + theta(i-1)) / h_t^2;

v_r = r_dot;  v_theta = r(i)*theta_dot;
v_mag = sqrt(v_r^2+v_theta^2);
a_r = r_ddot - r(i)*theta_dot^2;
a_theta = r(i)*theta_ddot + 2*r_dot*theta_dot;
a_mag = sqrt(a_r^2 + a_theta^2);
```

Code 7.4: Centred O(h$\hat{2}$) differences for velocity and acceleration at t=206s.

## 7.8 Results and Discussion

### 7.8.1 Function and Its Exact Derivatives

Figure 7.2 shows $f(x)$ together with its first four analytical derivatives over $[-1, , 2]$. The function is a smooth fifth-degree polynomial with a local minimum around $x \approx 0.1$ and a local maximum near $x \approx 0.8$, and these turning points correspond to the zero crossings of $f'(x)$. Each higher derivative reduces the polynomial degree by one, with the fourth derivative $f^{(4)}(x) = -21600 + 48000x$ becoming a straight line. The derivative

magnitudes increase significantly toward the domain edges, where $f'(x)$ reaches very large values while $f^{(4)}(x)$ varies more moderately. This scaling is important for error interpretation because the same absolute error corresponds to a much smaller relative error when the true derivative magnitude is large.

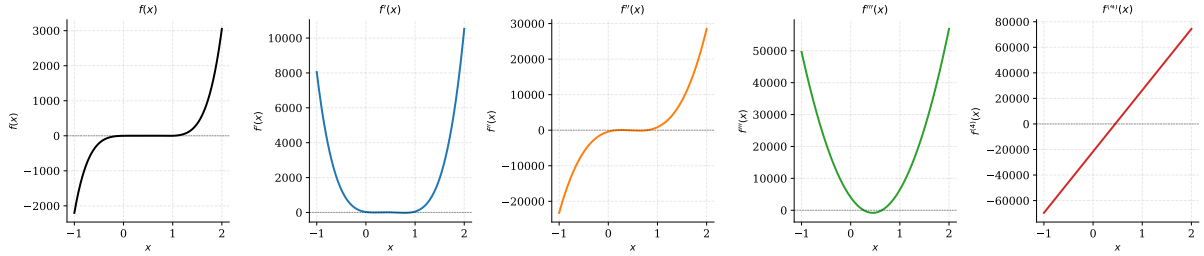

Figure 7.2: $f(x)$ (black) and its first four exact analytical derivatives over $x \in [-1,\ 2]$.

### 7.8.2 Numerical Formulae vs. Exact: All Four Derivatives

Figures 7.3 show the $O(h)$ and $O(h^2)$ numerical approximations against the exact curve for derivatives with $h = 0.1$.

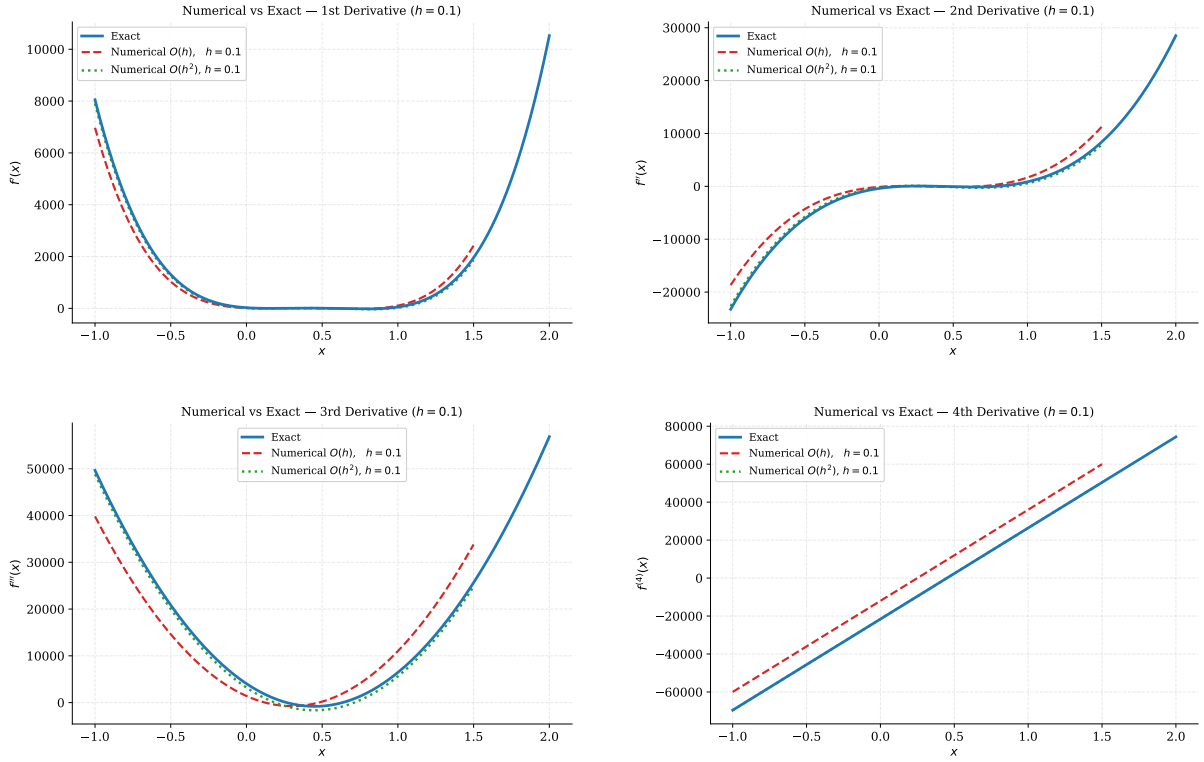


Figure 7.3: Exact, $O(h)$, and $O(h^2)$ derivative comparisons.

**Key observation.** For the first derivative (Figure **??**), both curves nearly coincide with the exact result; the accuracy difference is present but visually small. As the derivative order increases, the $O(h)$ curve progressively departs from the exact, while the $O(h^2)$ curve remains much closer. By the fourth derivative (Figure **??**), the $O(h)$ result is significantly offset from the true straight line, but the $O(h^2)$ formula is *exact* — a special consequence of the polynomial's degree: the sixth Taylor remainder in the $O(h^2)$ fourth-derivative stencil is identically zero for a degree-5 polynomial, so no truncation error exists at this particular order.

### 7.8.3 Built-in Functions vs. Exact

Figures 7.4 compare `diff` and `gradient` against the exact curves for the first through fourth derivatives.

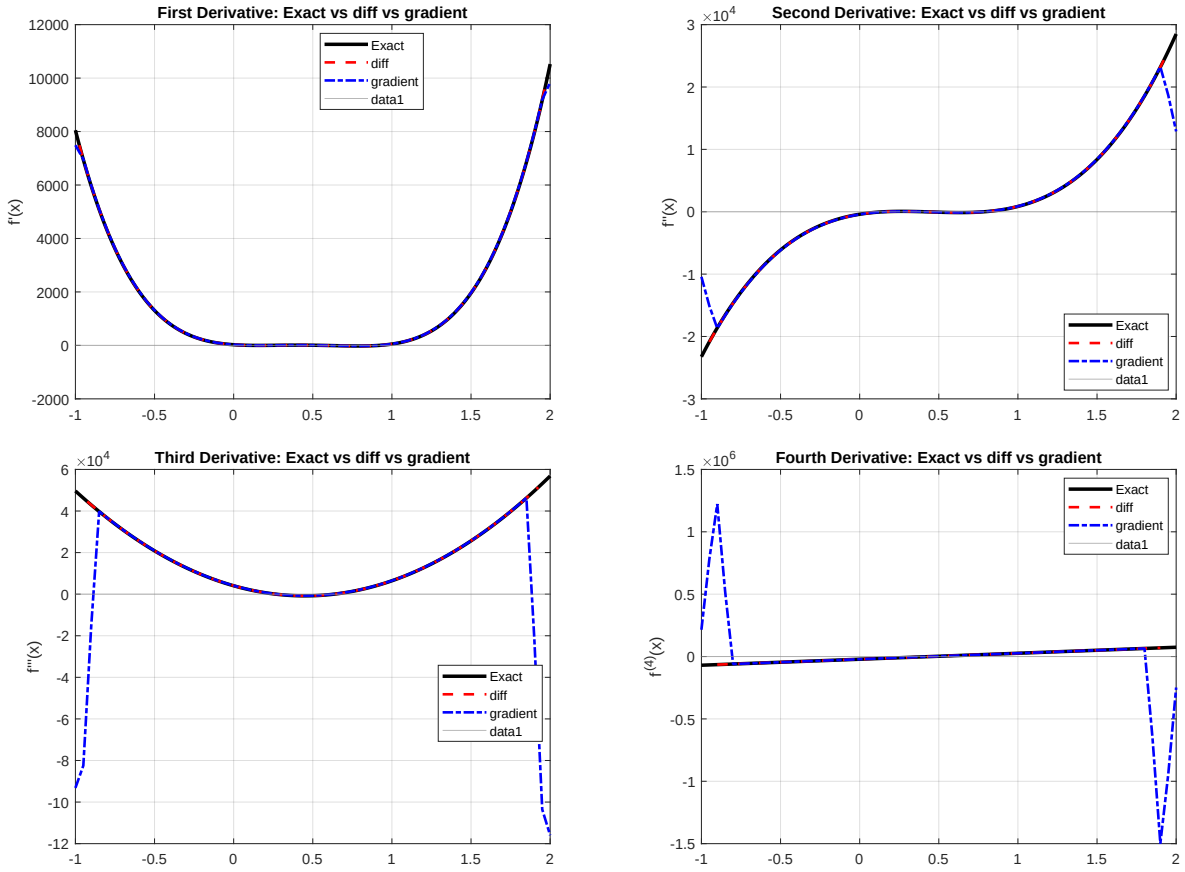


Figure 7.4: Comparison of exact derivatives with `diff` and `gradient` for the first to fourth derivatives.

### 7.8.4 Comparison and Comment: All Approaches

Table 7.3 summarises the four approaches side by side.

Table 7.3: Summary comparison of all numerical differentiation methods used.

| Method | Accuracy | Output length | Endpoint | Best for |
|---|---|---|---|---|
| Forward diff $O(h)$ | $O(h)$ | $N-k$ | Forward only | Fast, simple |
| Forward diff $O(h^2)$ | $O(h^2)$ | $N-k$ | Forward only | Controlled error |
| `diff` | $O(h)$ | $N-k$ | None at right | Quick scripting |
| `gradient` | $O(h^2)$ | $N$ | 1-sided ends | Plotting, general |

**Accuracy:** Both `gradient` and the explicit $O(h^2)$ stencil achieve second-order accuracy at interior points. For the first and second derivatives all four methods give acceptable results. For the third and fourth derivatives, the $O(h)$ stencil and `diff` accumulate noticeably larger errors, while `gradient` and the $O(h^2)$ stencil stay much closer to the exact curve.

**Convenience:** `gradient` is the most convenient: one function call, same-length output, and no index re-alignment needed. `diff` requires careful index tracking because each application shortens the vector by one. The explicit stencils require more code but give full control over the truncation error order.

**Recommendation:** Use `gradient` for quick exploration and visualisation. Use the explicit $O(h^2)$ stencils whenever the truncation error order must be stated precisely in a technical report or design calculation.

### 7.8.5 Problem 2 – Radar Tracking Results

#### Rectangular Path

Figure 7.5 shows the aircraft trajectory in Cartesian coordinates obtained from the polar data by $x = r\cos\theta$, $y = r\sin\theta$. The path is a smooth outward arc – range $r$ increasing and bearing $\theta$ decreasing steadily – consistent with a departing aircraft banking gently.

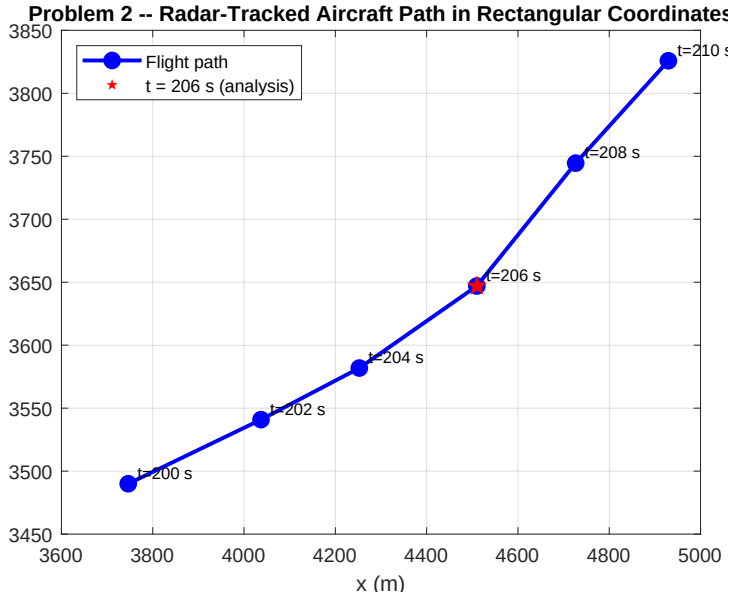


Figure 7.5: Aircraft path in rectangular coordinates converted from polar radar data. The red star marks $t = 206$ s, the analysis point.

#### Centred Difference Calculation at $t = 206$ s

With $h_t = 2$ s and MATLAB index $i = 4$ ($t = 206$ s), applying equations (7.4)–(7.5):

$$\dot{r} = \frac{6030 - 5560}{4} = 117.5 \text{ m/s} \tag{7.11}$$

$$\dot{\theta} = \frac{0.67 - 0.70}{4} = -0.0075 \text{ rad/s} \tag{7.12}$$

$$\ddot{r} = \frac{6030 - 2(5800) + 5560}{4} = -2.5 \text{ m/s}^2 \tag{7.13}$$

$$\ddot{\theta} = \frac{0.67 - 2(0.68) + 0.70}{4} = 0.0025 \text{ rad/s}^2 \tag{7.14}$$

Table 7.4: Velocity and acceleration components at $t = 206$ s obtained by centred $O(h^2)$ finite differences.

| **Quantity** | **Formula** | **Value** | **Unit** |
|---|---|---|---|
| $\dot{r}$ | centred $O(h_t^2)$ | 117.5000 | m/s |
| $\dot{\theta}$ | centred $O(h_t^2)$ | $-0.0075$ | rad/s |
| $\ddot{r}$ | centred $O(h_t^2)$ | $-2.5000$ | m/s$^2$ |
| $\ddot{\theta}$ | centred $O(h_t^2)$ | 0.0025 | rad/s$^2$ |
| $v_r = \dot{r}$ | Eq. (7.2) | 117.5000 | m/s |
| $v_\theta = r\dot{\theta}$ | Eq. (7.2) | $-43.5000$ | m/s |
| $\lvert\bar{v}\rvert = \sqrt{v_r^2 + v_\theta^2}$ | | 125.2937 | m/s |
| $a_r = \ddot{r} - r\dot{\theta}^2$ | Eq. (7.3) | $-2.8263$ | m/s$^2$ |
| $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$ | Eq. (7.3) | 12.7375 | m/s$^2$ |
| $\lvert\bar{a}\rvert = \sqrt{a_r^2 + a_\theta^2}$ | | 13.0473 | m/s$^2$ |

The speed $|\bar{v}| \approx 125.3$ m/s ($\approx 451$ km/h) is physically reasonable for a subsonic aircraft. The negative $v_\theta$ means the aircraft is sweeping clockwise (decreasing $\theta$ as time advances). The small negative $a_r$ indicates very slight radial deceleration, while the much larger positive $a_\theta = 12.74$ m/s$^2$ is the dominant contributor to the total acceleration: the aircraft is primarily *turning* rather than speeding up or slowing down.

### Velocity and Acceleration Vector Plots

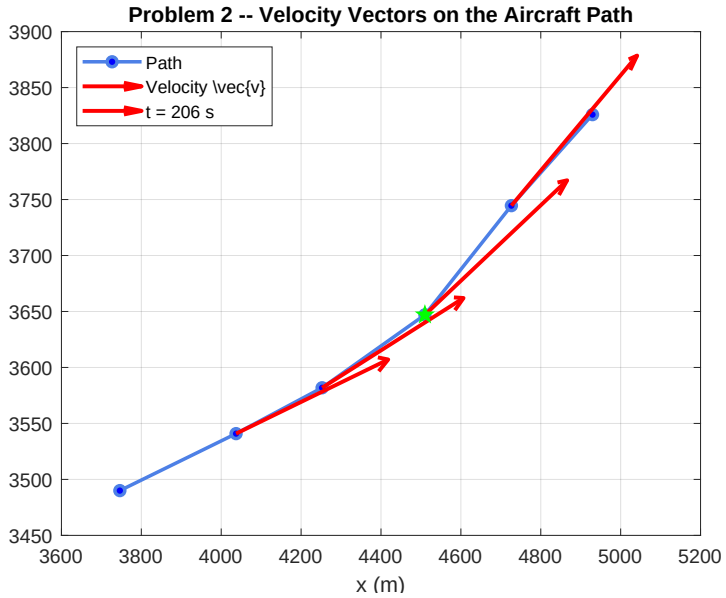


Figure 7.6: Velocity vectors $\bar{v}$ superimposed on the aircraft path. Vectors are computed at all four interior points using centred $O(h^2)$ differences. The arrows point consistently along the direction of flight, confirming the smooth arc.

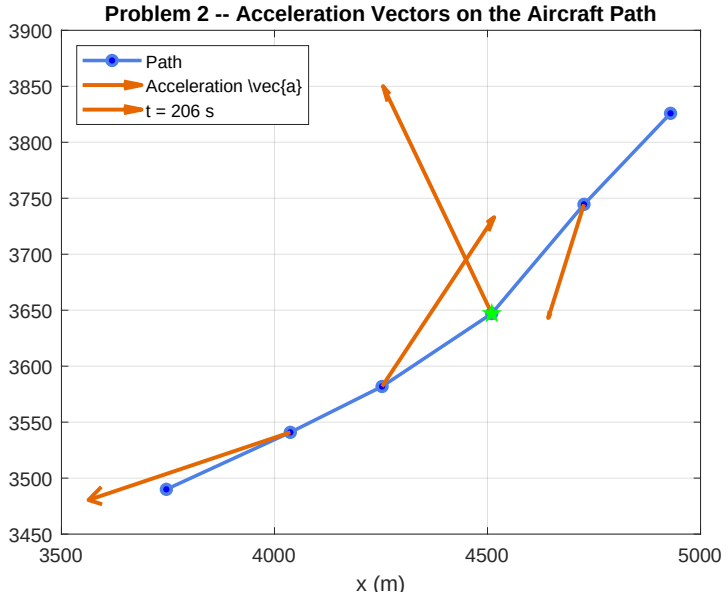


Figure 7.7: Acceleration vectors $\bar{a}$ superimposed on the aircraft path. The dominant tangential component ($a_\theta = 12.74$ m/s$^2$) points laterally inward, consistent with the aircraft executing a gentle banked turn.

The velocity vectors (Figure 7.6) are nearly parallel and point along the path, confirming uniform forward flight with a gradual curve. The acceleration vectors are predominantly perpendicular to the velocity, pointing inward toward the centre of curvature.

## 7.9 Conclusion

This lab session demonstrated numerical differentiation through two practical problems.

For **Problem 1**, four forward-difference formulas were applied to the polynomial $f(x)$ at $h = 0.1$. The $O(h^2)$ formula outperformed $O(h)$ across all four derivative orders, with the accuracy advantage widening sharply for higher derivatives because each successive differentiation amplifies the truncation error by $\sim 1/h$. A remarkable special case emerged at the fourth derivative: since $f$ is degree-5, the sixth-order Taylor remainder in the $O(h^2)$ stencil vanishes exactly, giving a numerically perfect result. Among the built-in functions, `gradient` proved superior to `diff` in both accuracy ($O(h^2)$ vs $O(h)$ interior differences) and convenience (same output length, no index realignment). The error-vs-$h$ analysis confirmed that an optimal step size exists where truncation and round-off errors are equal; too small an $h$ is just as harmful as too large.

For **Problem 2**, centred second-order finite differences applied to the six-point polar radar dataset ($\Delta t = 2$ s) yielded, at $t = 206$ s:

$$|\bar{v}| = 125.3 \text{ m/s}, \qquad |\bar{a}| = 13.0 \text{ m/s}^2.$$

The large tangential acceleration relative to the small radial acceleration ($a_\theta \approx 4.5\, a_r$) confirms that the aircraft is primarily executing a horizontal turn, not changing speed. This is consistent with the gently curving rectangular path seen in Figure 7.5 [4].

# Lab Session 8

# Trapezoidal Rule and Multiple Integrals

## 8.1 Objective

- Implement the single and composite Trapezoidal Rule to approximate definite integrals and quantify error against exact analytical values.
- Investigate second-order ($O(h^2)$) convergence of the composite rule as the segment count $n$ increases.
- Apply the Trapezoidal Rule to unequally-spaced discrete velocity data to determine total distance and average velocity (Problem 19.8a).
- Extend the composite Trapezoidal Rule to a double integral (Problem 19.6) and a triple integral (Problem 19.7) via nested one-dimensional applications, and assess accuracy.

## 8.2 Introduction and Theoretical Background

### 8.2.1 Newton-Cotes Integration Formulas

Numerical integration estimates $I = \int_a^b f(x)\,dx$ by replacing $f(x)$ with a polynomial $f_n(x)$ of order $n$ that is easy to integrate [4]:

$$I = \int_a^b f(x)\,dx \;\approx\; \int_a^b f_n(x)\,dx \tag{8.1}$$

When the polynomial is anchored at both endpoints $a$ and $b$, the formula is a *closed* Newton-Cotes formula. The Trapezoidal Rule corresponds to $n = 1$ (straight-line fit).

### 8.2.2 Single Trapezoidal Rule

Fitting a straight line between $f(a)$ and $f(b)$ and integrating gives:

$$I = (b-a)\,\frac{f(a)+f(b)}{2} \tag{8.2}$$

The local truncation error is [4]:

$$E_t = -\frac{(b-a)^3}{12}\,f''(\xi), \qquad \xi \in [a,b] \tag{8.3}$$

The rule is exact for linear functions ($f'' = 0$); error scales with curvature and the cube of the interval width.

### 8.2.3 Composite Trapezoidal Rule

Dividing $[a, b]$ into $n$ equal segments ($h = (b - a)/n$) and applying the Trapezoidal Rule to each:

$$I = \frac{h}{2}\left[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right] \tag{8.4}$$

The composite truncation error is:

$$E_a = -\frac{(b-a)^3}{12n^2}\,\bar{f}'' \tag{8.5}$$

where $\bar{f}''$ is the average second derivative over $[a, b]$. Since $E_a \propto 1/n^2$, doubling $n$ quarters the error — second-order ($O(h^2)$) convergence.

### 8.2.4 Multiple Integrals

For $\iint f(x, y)\,dx\,dy$, the composite Trapezoidal Rule is applied first in $x$ for each fixed $y_j$, yielding intermediate values $I_x(y_j)$, then applied in $y$ to those values [4]. The same nesting extends to triple integrals by adding an outer $z$ loop.

## 8.3 Problem Statements

**Example 19.1.** Integrate $f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$ from $a = 0$ to $b = 0.8$ using a *single* application of the Trapezoidal Rule (Eq. 8.2). The exact value is 1.640533.

**Example 19.2.** Apply the *composite* Trapezoidal Rule (Eq. 8.4) to the same $f(x)$ over $[0, 0.8]$ for $n = 2, 3, \ldots, 10$. Estimate the error for $n = 2$ using Eq. (8.5) and tabulate all results.

**Problem 19.8a.** Use the Trapezoidal Rule on the unequally-spaced velocity data in Table 8.1 to determine total distance and average velocity.

Table 8.1: Velocity data — Problem 19.8a.

| $t$ | 1 | 2 | 3.25 | 4.5 | 6 | 7 | 8 | 8.5 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $v$ | 5 | 6 | 5.5 | 7 | 8.5 | 8 | 6 | 7 | 7 | 5 |

**Problem 19.6.** Evaluate analytically and via composite Trapezoidal Rule ($n = 2$):

$$I = \int_{-2}^{2}\int_{0}^{4} \left(x^2 - 3y^2 + xy^3\right) dx\,dy \tag{8.6}$$

**Problem 19.7.** Evaluate analytically and via composite Trapezoidal Rule ($n = 2$):

$$I = \int_{-4}^{4}\int_{0}^{6}\int_{-1}^{3} \left(x^3 - 2yz\right) dx\,dy\,dz \tag{8.7}$$

## 8.4 Analytical Solutions

### 8.4.1 Problem 19.6

**Inner integral over $x \in [0, 4]$:**

$$\int_0^4 \left(x^2 - 3y^2 + xy^3\right) dx = \left[\tfrac{x^3}{3} - 3y^2 x + \tfrac{x^2 y^3}{2}\right]_0^4 = \tfrac{64}{3} - 12y^2 + 8y^3 \tag{8.8}$$

**Outer integral over $y \in [-2, 2]$:**

$$\int_{-2}^2 \left(\tfrac{64}{3} - 12y^2 + 8y^3\right) dy = \left[\tfrac{64y}{3} - 4y^3 + 2y^4\right]_{-2}^2 = \frac{64}{3} \approx \mathbf{21.3333} \tag{8.9}$$

Note: the $8y^3$ term vanishes over $[-2, 2]$ by antisymmetry.

### 8.4.2 Problem 19.7

**Inner integral over $x \in [-1, 3]$:**

$$\int_{-1}^3 \left(x^3 - 2yz\right) dx = \left[\tfrac{x^4}{4} - 2yz\, x\right]_{-1}^3 = 20 - 8yz \tag{8.10}$$

**Middle integral over $y \in [0, 6]$:**

$$\int_0^6 (20 - 8yz)\, dy = \left[20y - 4y^2 z\right]_0^6 = 120 - 144z \tag{8.11}$$

**Outer integral over $z \in [-4, 4]$:**

$$\int_{-4}^4 (120 - 144z)\, dz = \left[120z - 72z^2\right]_{-4}^4 = (480 - 1152) - (-480 - 1152) = \mathbf{960} \tag{8.12}$$

## 8.5 Methodology

1. Define $f(x)$ as a MATLAB anonymous function. Evaluate at $a = 0$, $b = 0.8$ and apply Eq. (8.2) for Example 19.1.

2. Loop $n = 2 \to 10$, applying Eq. (8.4) via a helper function `comp_trap` for Example 19.2. Tabulate $I$ and $\varepsilon_t$ for each $n$.

3. Use MATLAB's `trapz(t,v)` for Problem 19.8a, which handles unequal spacing automatically; divide by time span for average velocity.

4. For Problem 19.6, form $n + 1$ nodes in $x \in [0, 4]$ and $y \in [-2, 2]$. Integrate in $x$ at each $y$-node, then integrate the resulting vector in $y$.

5. For Problem 19.7, add a $z$-loop over $n + 1$ nodes in $[-4, 4]$ and repeat the $x$–$y$ nesting of step 4 at each $z$-node; integrate in $z$.

## 8.6 Algorithm

The algorithm flowchart is given below

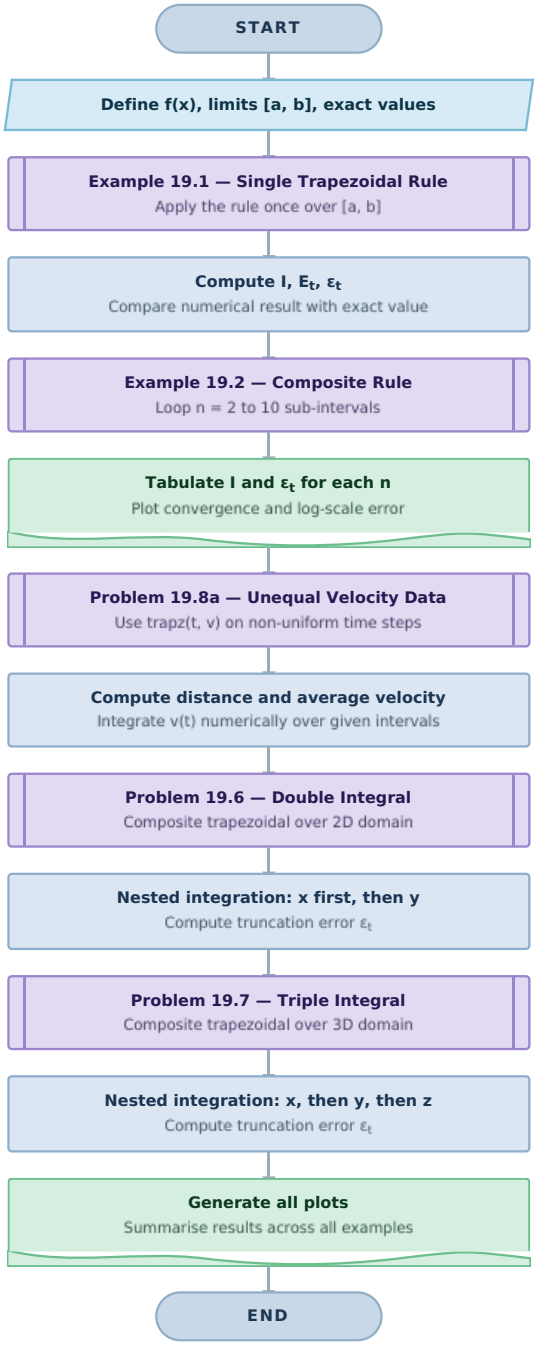


Figure 8.1: Algorithm flowchart for Lab Session 8.

## 8.7 MATLAB Implementation

### Examples 19.1 and 19.2

```
f = @(x) 0.2+25*x-200*x.^2+675*x.^3-900*x.^4+400*x.^5;
% Parameters
a = 0;   b = 0.8;   exact = 1.640533;
```

```
% Example 19.1 -- single application
I_single = (b-a)*(f(a)+f(b))/2;          % 0.1728
Et = exact - I_single;                    % 1.4677
eps_t = abs(Et/exact)*100;                % 89.5 %

% Example 19.2 -- composite, n = 2 ... 10
for n = 2:10
    x = linspace(a,b,n+1);  y = f(x);  h = (b-a)/n;
    I = h*(y(1)/2 + sum(y(2:end-1)) + y(end)/2);
    eps = abs((exact-I)/exact)*100;
end
```

Code 8.1: Single and composite Trapezoidal Rule for $f(x)$.

## Problem 19.8a

```
t = [1, 2, 3.25, 4.5, 6, 7, 8, 8.5, 9, 10];
v = [5, 6, 5.5,  7, 8.5, 8,  6,  7,  7,  5];
distance     = trapz(t, v);              % 60.1250 units
avg_velocity = distance/(t(end)-t(1)); %  6.6806 units/time
```

Code 8.2: Trapezoidal Rule on unequally-spaced velocity data.

## Problems 19.6 and 19.7

```
% Problem 19.6  (x in [0,4], y in [-2,2], n=2)
f6 = @(x,y) x.^2 - 3*y.^2 + x.*y.^3;
n=2; x1=0; x2=4; y1=-2; y2=2;
y_nodes = linspace(y1,y2,n+1);
for j = 1:n+1
    I_x(j) = comp_trap(@(x) f6(x,y_nodes(j)), x1, x2, n);
end
I_dbl = comp_trap_vec(I_x, y1, y2, n);    % 0.0  (eps_t=100%)

% Problem 19.7  (x in [-1,3], y in [0,6], z in [-4,4], n=2)
f7 = @(x,y,z) x.^3 - 2*y.*z;
z_nodes = linspace(-4,4,n+1);
for k = 1:n+1
    y_nodes = linspace(0,6,n+1);
    for j = 1:n+1
        I_xy(j)=comp_trap(@(x) f7(x,y_nodes(j),z_nodes(k))
           ,-1,3,n);
    end
    I_yz(k) = comp_trap_vec(I_xy, 0, 6, n);
end
I_trpl = comp_trap_vec(I_yz, -4, 4, n);   % 960.0  (eps_t=0%)

% Helper Functions
function I = comp_trap(f, a, b, n)
```

```
    x=linspace(a,b,n+1); y=f(x); h=(b-a)/n;
    I = h*(y(1)/2 + sum(y(2:end-1)) + y(end)/2);
end
function I = comp_trap_vec(y, a, b, n)
    h=(b-a)/n;
    I = h*(y(1)/2 + sum(y(2:end-1)) + y(end)/2);
end
```

Code 8.3: Nested composite Trapezoidal Rule for double and triple integrals.

## 8.8 Results and Discussion

### 8.8.1 Example 19.1 — Single Trapezoidal Rule

Figure 8.2 shows $f(x)$ and $f''(x)$ over $[0, 0.8]$. Since $f''$ is non-zero throughout, significant truncation error is expected.

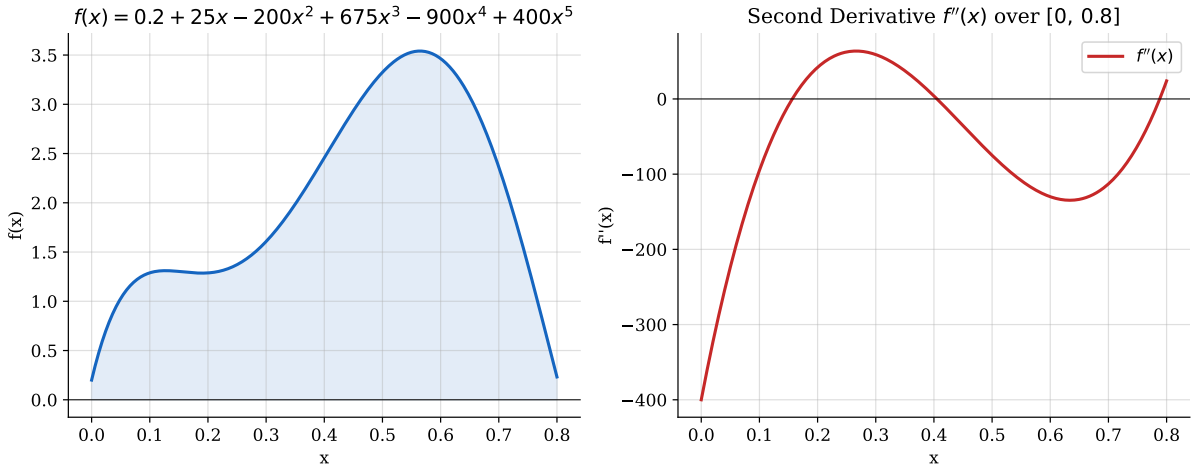


Figure 8.2: Left: $f(x)$ over $[0, 0.8]$. Right: $f''(x)$, confirming non-zero curvature throughout, which causes large error in a single-segment approximation.

Applying Eq. (8.2):

$$f(0) = 0.2, \qquad f(0.8) = 0.232 \tag{8.13}$$

$$I = 0.8 \times \frac{0.2 + 0.232}{2} = \mathbf{0.1728} \tag{8.14}$$

$$E_t = 1.640533 - 0.1728 = 1.4677 \qquad \Rightarrow \qquad \varepsilon_t = \mathbf{89.5}\% \tag{8.15}$$

Using the average second derivative $\bar{f}'' = -60$:

$$E_a = -\frac{(0.8)^3}{12} \times (-60) = \mathbf{2.56} \tag{8.16}$$

$E_a$ and $E_t$ are of the same sign and order of magnitude, confirming Eq. (8.3).

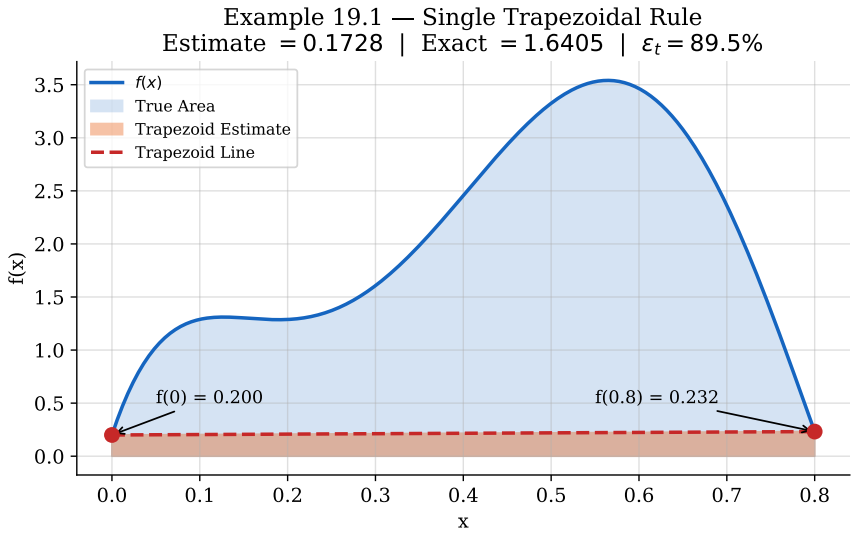


Figure 8.3: Example 19.1 — the orange trapezoid misses most of the true (blue) area, resulting in $\varepsilon_t = 89.5\%$.

### 8.8.2 Example 19.2 — Composite Trapezoidal Rule

For $n = 2$ ($h = 0.4$):

$$I = \frac{0.4}{2}\left[0.2 + 2(2.456) + 0.232\right] = \mathbf{1.0688} \quad (\varepsilon_t = 34.9\%) \tag{8.17}$$

$$E_a = -\frac{(0.8)^3}{12(4)} \times (-60) = \mathbf{0.64} \tag{8.18}$$

Table 8.2 gives full results; Figure 10.5 shows the convergence behaviour.

Table 8.2: Composite Trapezoidal Rule — exact = 1.640533.

| $n$ | $h$ | $I$ | $\varepsilon_t$ (%) |
|---|---|---|---|
| 2 | 0.4000 | 1.0688 | 34.9 |
| 3 | 0.2667 | 1.3695 | 16.5 |
| 4 | 0.2000 | 1.4848 | 9.5 |
| 5 | 0.1600 | 1.5399 | 6.1 |
| 6 | 0.1333 | 1.5703 | 4.3 |
| 7 | 0.1143 | 1.5887 | 3.2 |
| 8 | 0.1000 | 1.6008 | 2.4 |
| 9 | 0.0889 | 1.6091 | 1.9 |
| 10 | 0.0800 | 1.6150 | 1.6 |

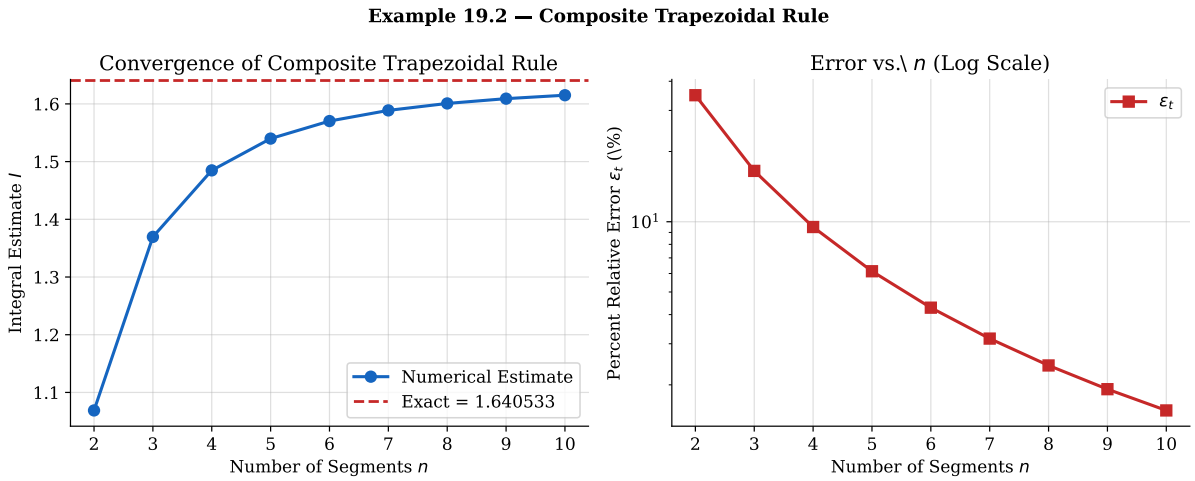


Figure 8.4: Example 19.2 — Left: estimate converging to exact. Right: log-scale error with slope $\approx -2$, confirming $O(h^2)$ convergence.

**Key observation.** Doubling $n$ from 2 to 4 reduces $\varepsilon_t$ from 34.9% to 9.5% (factor $\approx$4), consistent with Eq. (8.5).

### 8.8.3 Problem 19.8a — Distance from Velocity Data

Applying the Trapezoidal Rule for unequally-spaced data:

$$d \approx \sum_{i=1}^{N-1} (t_{i+1} - t_i) \frac{v_i + v_{i+1}}{2} \tag{8.19}$$

Table 8.3: Segment-by-segment areas — Problem 19.8a.

| $t_i$ | $t_{i+1}$ | $v_i$ | $v_{i+1}$ | **Area** |
|---|---|---|---|---|
| 1.00 | 2.00 | 5.0 | 6.0 | 5.5000 |
| 2.00 | 3.25 | 6.0 | 5.5 | 7.1875 |
| 3.25 | 4.50 | 5.5 | 7.0 | 7.8125 |
| 4.50 | 6.00 | 7.0 | 8.5 | 11.6250 |
| 6.00 | 7.00 | 8.5 | 8.0 | 8.2500 |
| 7.00 | 8.00 | 8.0 | 6.0 | 7.0000 |
| 8.00 | 8.50 | 6.0 | 7.0 | 3.2500 |
| 8.50 | 9.00 | 7.0 | 7.0 | 3.5000 |
| 9.00 | 10.00 | 7.0 | 5.0 | 6.0000 |
| | | | **Total distance** | **60.1250** |

$$\bar{v} = \frac{60.1250}{10 - 1} = \mathbf{6.6806} \text{ units/time} \tag{8.20}$$

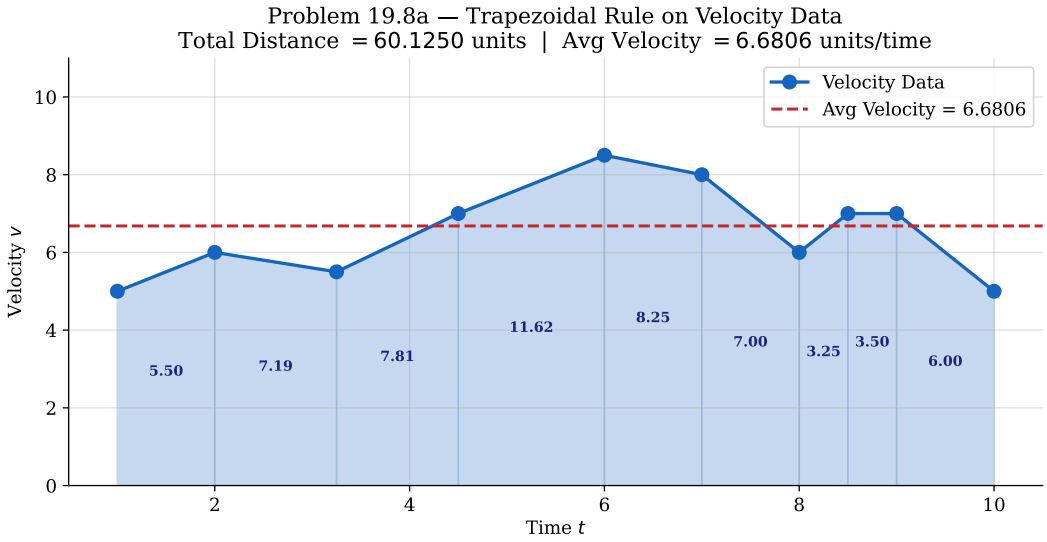


Figure 8.5: Problem 19.8a — velocity data with trapezoid segment areas labelled. Dashed red line marks average velocity.

### 8.8.4 Problem 19.6 — Double Integral

**Grid** ($n = 2$)**:** $x \in \{0, 2, 4\}$, $h_x = 2$; $y \in \{-2, 0, 2\}$, $h_y = 2$.

Table 8.4: $f(x, y) = x^2 - 3y^2 + xy^3$ at the $3 \times 3$ node grid.

| | $y = -2$ | $y = 0$ | $y = 2$ |
|---|---|---|---|
| $x = 0$ | $-12$ | $0$ | $-12$ |
| $x = 2$ | $-24$ | $4$ | $8$ |
| $x = 4$ | $-28$ | $16$ | $36$ |

**Step 1 — integrate in $x$:**

$$I_x(y{=}-2) = 2\left[\tfrac{-12}{2} + (-24) + \tfrac{-28}{2}\right] = -88.0 \tag{8.21}$$

$$I_x(y{=}0) = 2\left[\tfrac{0}{2} + 4 + \tfrac{16}{2}\right] = +24.0 \tag{8.22}$$

$$I_x(y{=}2) = 2\left[\tfrac{-12}{2} + 8 + \tfrac{36}{2}\right] = +40.0 \tag{8.23}$$

**Step 2 — integrate in $y$:**

$$I = 2\left[\frac{-88}{2} + 24 + \frac{40}{2}\right] = 2[-44 + 24 + 20] = \mathbf{0.0} \qquad \Rightarrow \qquad \varepsilon_t = \mathbf{100}\% \tag{8.24}$$

$I = 0.0$ is the correct numerical result for $n = 2$: the $xy^3$ term is odd in $y$, so its contributions cancel exactly at the three symmetric nodes $\{-2, 0, 2\}$. This is not a coding error. Increasing $n$ restores accuracy: $n = 4 \Rightarrow I = 16.00$ ($\varepsilon_t = 25\%$); $n = 8 \Rightarrow I = 20.00$ ($\varepsilon_t = 6.25\%$).

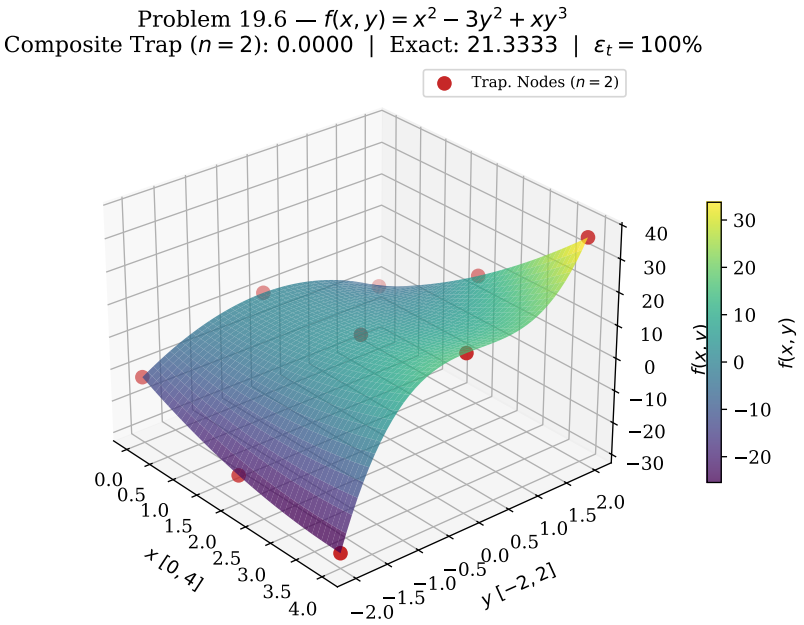


Figure 8.6: Problem 19.6 — surface of $f(x, y)$ with $n = 2$ trapezoidal nodes marked.

### 8.8.5 Problem 19.7 — Triple Integral

**Grid** ($n = 2$)**:** $x \in \{-1, 1, 3\}$ ($h_x = 2$), $y \in \{0, 3, 6\}$ ($h_y = 3$), $z \in \{-4, 0, 4\}$ ($h_z = 4$).

**Step 1 — integrate in $x$:**

Table 8.5: $x$-integration results $I_x(y, z)$ for Problem 19.7.

| $z$ | $y$ | $f(-1, y, z)$ | $f(1, y, z)$ | $f(3, y, z)$ | $I_x$ |
|---|---|---|---|---|---|
| $-4$ | 0 | $-1$ | 1 | 27 | 54.0 |
| $-4$ | 3 | 23 | 25 | 51 | 102.0 |
| $-4$ | 6 | 47 | 49 | 75 | 150.0 |
| 0 | 0 | $-1$ | 1 | 27 | 54.0 |
| 0 | 3 | $-1$ | 1 | 27 | 54.0 |
| 0 | 6 | $-1$ | 1 | 27 | 54.0 |
| 4 | 0 | $-1$ | 1 | 27 | 54.0 |
| 4 | 3 | $-23$ | $-23$ | 3 | 6.0 |
| 4 | 6 | $-47$ | $-47$ | $-21$ | $-42.0$ |

**Steps 2 & 3 — integrate in $y$, then $z$:**

$$I_{yz}(z{=}-4) = 3\left[\tfrac{54}{2} + 102 + \tfrac{150}{2}\right] = 612 \tag{8.25}$$

$$I_{yz}(z{=}0) = 3\left[\tfrac{54}{2} + 54 + \tfrac{54}{2}\right] = 324 \tag{8.26}$$

$$I_{yz}(z{=}4) = 3\left[\tfrac{54}{2} + 6 + \tfrac{-42}{2}\right] = 36 \tag{8.27}$$

$$I = 4\left[\frac{612}{2} + 324 + \frac{36}{2}\right] = 4[306 + 324 + 18] = \mathbf{960} \quad \Rightarrow \quad \varepsilon_t = \mathbf{0}\% \tag{8.28}$$

The rule is exact here because the second derivatives of $x^3 - 2yz$ in $y$ and $z$ are identically zero, eliminating all truncation error in those dimensions.

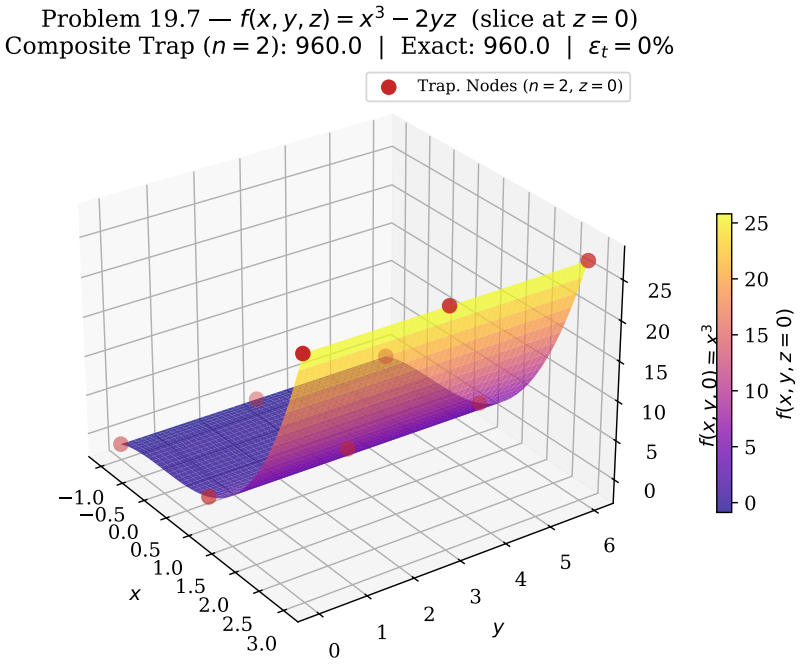


Figure 8.7: Problem 19.7 — surface of $f(x, y, z) = x^3 - 2yz$ at $z = 0$ with $n = 2$ nodes. Zero truncation error is recovered.

## 8.9 Conclusion

This lab demonstrated the Trapezoidal Rule from single-segment applications through nested multiple integrals. For Examples 19.1 and 19.2, a single segment yielded 89.5% error owing to high curvature of $f$ over $[0, 0.8]$. The composite rule converged rapidly, from 34.9% at $n = 2$ down to 1.6% at $n = 10$, and the log-scale error plot confirmed second-order $O(h^2)$ convergence with slope $\approx -2$, consistent with the composite error formula. For Problem 19.8a, `trapz` handled the unequally-spaced velocity data correctly, yielding a total distance of 60.125 units and an average velocity of 6.681 units/time.

For Problem 19.6, the $n = 2$ result of $I = 0.0$ ($\varepsilon_t = 100\%$) is mathematically correct, since the odd-symmetric $xy^3$ term cancels exactly at the three symmetric $y$-nodes; higher $n$ restores convergence to the exact value of $64/3$. For Problem 19.7, the nested composite rule with $n = 2$ recovered the exact answer of 960 with zero error, because the second derivatives of the integrand $x^3 - 2yz$ in both $y$ and $z$ are identically zero, eliminating all truncation error in those dimensions. Taken together, these results highlight how the geometry and symmetry of the integrand directly govern the accuracy of numerical integration schemes.

## Lab Session 9

# Numerical Solution of Ordinary Differential Equations

## 9.1 Objective

- Implement and compare four single-step numerical methods for ordinary differential equations (ODEs): Euler's Method, the Midpoint Method, the Predictor-Corrector Method (Heun's non-iterative form), and Iterative Heun's Method.
- Apply all four methods to the ODE $dy/dt = y \sin t$ with initial condition $y(0) = 1$ over $t \in [0, 50]$ and compare each numerical solution against the closed-form analytical result.
- Quantify global truncation error for each method at a fixed evaluation point and investigate first-order ($O(h)$) and second-order ($O(h^2)$) convergence behaviour as the step size $h$ decreases.
- Demonstrate the iterative correction procedure of Heun's Method and examine per-step convergence as a function of iteration count.

## 9.2 Introduction and Theoretical Background

### 9.2.1 Ordinary Differential Equations and Initial Value Problems

A first-order ODE of the form

$$\frac{dy}{dt} = f(t, y), \qquad y(t_0) = y_0 \tag{9.1}$$

defines an *initial value problem* (IVP). In most engineering applications $f(t, y)$ is nonlinear and a closed-form solution does not exist, so numerical methods are indispensable [4]. All single-step methods discussed here advance the solution one step at a time: given $y_i \approx y(t_i)$, they compute $y_{i+1} \approx y(t_{i+1})$ using only information at step $i$.

### 9.2.2 Taylor Series Foundation

Expanding $y(t_{i+1})$ about $t_i$ yields

$$y(t_{i+1}) = y(t_i) + h\, y'(t_i) + \frac{h^2}{2}\, y''(t_i) + \frac{h^3}{6}\, y'''(t_i) + \cdots \tag{9.2}$$

where $h = t_{i+1} - t_i$ is the step size. Every numerical method can be interpreted as retaining a certain number of terms from Eq. (10.3). The order of the leading omitted term determines the *local truncation error* (LTE) and, consequently, the global convergence rate.

### 9.2.3 Euler's Method

Retaining only the first two terms of Eq. (10.3):

$$y_{i+1} = y_i + h\, f(t_i, y_i) \tag{9.3}$$

The LTE is $O(h^2)$, giving a *global* error of $O(h)$ (first-order method). Euler's Method uses the slope at the *beginning* of the interval, so it overshoots or undershoots depending on the curvature of the solution. For $h$ too large on an oscillatory problem the method can accumulate substantial error or even diverge.

The truncation error per step can be written as

$$E_t = \frac{h^2}{2}\, y''(\xi_i), \qquad \xi_i \in (t_i, t_{i+1}) \tag{9.4}$$

which vanishes only when $y$ is linear, confirming that Euler's Method is exact only for first-degree polynomials.

### 9.2.4 Midpoint Method (Second-Order Runge-Kutta)

Instead of using the slope at the start of the interval, the Midpoint Method first advances a *half-step* to estimate the slope at $t_i + h/2$:

$$k_1 = f(t_i,\ y_i) \tag{9.5}$$

$$k_2 = f\big(t_i + \tfrac{h}{2},\ y_i + \tfrac{h}{2}k_1\big) \tag{9.6}$$

$$y_{i+1} = y_i + h\, k_2 \tag{9.7}$$

This effectively uses a centred-difference slope approximation. The LTE is $O(h^3)$, giving a global error of $O(h^2)$ (second-order method).

### 9.2.5 Predictor-Corrector Method (Non-Iterative Heun)

Heun's non-iterative variant uses the Euler step as a *predictor* to estimate the slope at the end of the interval, then averages the two slopes as the corrector:

$$y^0_{i+1} = y_i + h\, f(t_i, y_i) \qquad \text{(predictor)} \tag{9.8}$$

$$y_{i+1} = y_i + \frac{h}{2}\big[f(t_i, y_i) + f\big(t_{i+1}, y^0_{i+1}\big)\big] \qquad \text{(corrector)} \tag{9.9}$$

The corrector averages the slopes at both ends of the interval, achieving second-order global accuracy, the same order as the Midpoint Method. It requires two function evaluations per step, identical to the Midpoint Method.

### 9.2.6 Iterative Heun's Method

The corrector step (Eq. 9.9) can be iterated until the estimate converges:

$$y^k_{i+1} = y_i + \frac{h}{2}\big[f(t_i, y_i) + f\big(t_{i+1}, y^{k-1}_{i+1}\big)\big], \quad k = 1, 2, 3, \ldots \tag{9.10}$$

Iteration continues until

$$\big|y^k_{i+1} - y^{k-1}_{i+1}\big| < \varepsilon_s \tag{9.11}$$

where $\varepsilon_s = 10^{-6}$ in this lab. Each extra iteration adds one function evaluation but the iteration itself does *not* increase the formal order of accuracy beyond $O(h^2)$; its benefit is a tighter constant in the error bound.

### 9.2.7 Summary of Method Orders

Table 9.1: Summary of single-step ODE methods.

| Method | LTE | Global Error | Evaluations/step |
|---|---|---|---|
| Euler | $O(h^2)$ | $O(h)$ | 1 |
| Midpoint | $O(h^3)$ | $O(h^2)$ | 2 |
| Predictor-Corrector (Heun) | $O(h^3)$ | $O(h^2)$ | 2 |
| Iterative Heun | $O(h^3)$ | $O(h^2)$ | $2 + k_{\text{iter}}$ |

## 9.3 Problem Statement

Solve the IVP

$$\frac{dy}{dt} = y \sin t, \qquad y(0) = 1, \quad t \in [0, 50] \tag{9.12}$$

using each of the four methods above. The exact solution is obtained by separation of variables:

$$\frac{dy}{y} = \sin t \, dt \quad \Rightarrow \quad \ln y = -\cos t + C \quad \Rightarrow \quad y = e^{-\cos t + C} \tag{9.13}$$

Applying $y(0) = 1$: $\ln 1 = -\cos 0 + C \Rightarrow C = 1$. Therefore:

$$\boxed{y(t) = e^{1 - \cos t}} \tag{9.14}$$

The solution is *periodic* with period $2\pi$ and oscillates between $y = 1$ (at $t = 0, 2\pi, 4\pi, \ldots$) and $y = e^2 \approx 7.389$ (at $t = \pi, 3\pi, \ldots$). The highly non-linear amplitude makes this a demanding benchmark for fixed-step-size methods.

## 9.4 Analytical Solution

### 9.4.1 Derivation and Properties

From Eq. (10.17):

- The second derivative is $y'' = y\big(\cos t + \sin^2 t\big)$, which is non-zero almost everywhere. This confirms that Euler's Method will incur non-negligible truncation error at every step.
- The solution is bounded: $1 \leq y(t) \leq e^2$ for all $t$.
- After one full period $[0, 2\pi]$ the solution returns to its initial value of 1. This periodic return provides a convenient checkpoint for measuring global error.

Figure 9.1 shows the exact solution over $[0, 50]$ and the corresponding slope field of the ODE. The slope field shows how the direction of the solution curve changes with both $t$ and $y$, and confirms that the analytical trajectory is consistent with the field arrows.

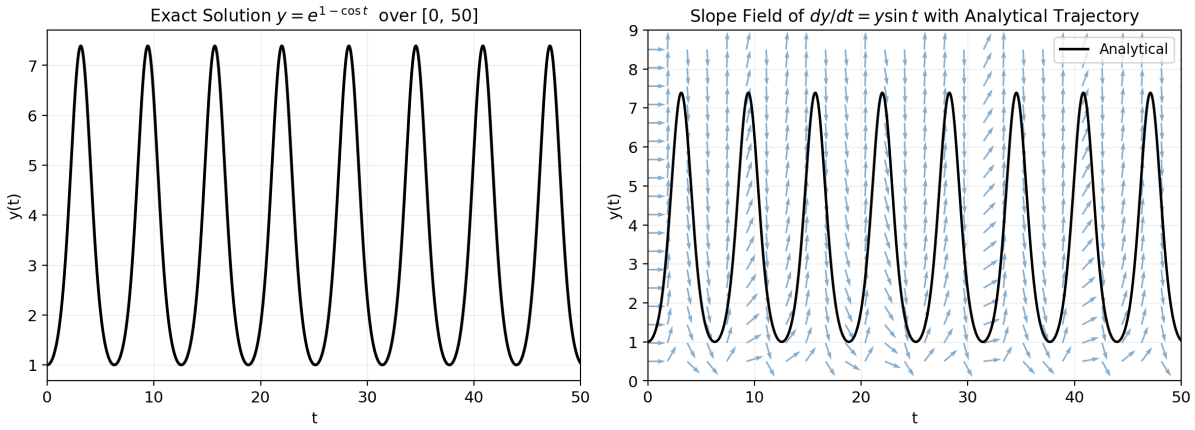


Figure 9.1: Left: exact solution $y = e^{1-\cos t}$ over $[0, 50]$, exhibiting periodic oscillation between 1 and $e^2$.

## 9.5 Methodology

1. Define $f(t, y) = y \sin t$ as a MATLAB anonymous function. Compute the exact solution on a fine grid ($\Delta t = 0.01$) using Eq. (10.17).

2. Apply Euler's Method (Eq. 10.2) over the node array $t = t_0{:}h{:}t_f$, initialising $y_{\text{euler}}(1) = 1$ and advancing in a `for` loop.

3. Apply the Midpoint Method (Eqs. 10.4–9.7) with the same node array and initial condition.

4. Apply the Predictor-Corrector Method (Eqs. 9.8–9.9), computing the Euler predictor first and then the corrected value.

5. Apply Iterative Heun's Method (Eq. 9.10), running the inner loop until convergence criterion Eq. (9.11) is satisfied or 50 iterations are reached.

6. Overlay all four numerical solutions and the exact solution on a single plot for visual comparison.

7. Investigate convergence by repeating steps 2 to 5 for $h \in \{2.0, 1.0, 0.5, 0.25, 0.1, 0.05\}$, measuring the global error at $t = 2\pi$ and plotting on a log-log scale.

## 9.6 Algorithm

The flowchart in Figure 10.3 illustrates the complete algorithm, from input and exact-solution computation through each numerical method to final plotting and error analysis.

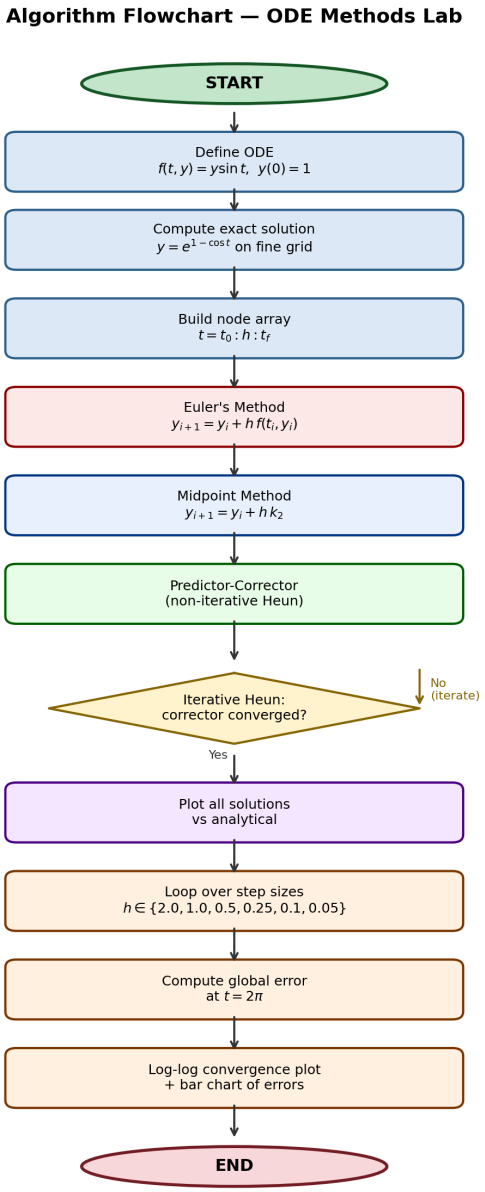


Figure 9.2: Algorithm flowchart for the ODE Methods Lab Session. The inner `for` loop inside Iterative Heun's Method iterates until the stopping criterion is met.

## 9.7 MATLAB Implementation

### Setup and Analytical Solution

```
clc; clear; close all;
h  = input('Enter step size h: ');
t0 = 0;  tf = 50;  y0 = 1;
f  = @(t,y) y .* sin(t);          % dy/dt = y*sin(t)
% Analytical solution on a fine grid
ta = t0 : 0.01 : tf;
ya = exp(1 - cos(ta));             % y = exp(1 - cos(t))
```

Code 9.1: Input parameters, ODE definition, and exact solution.

### Euler's Method

```
t = t0:h:tf;  N = length(t);
y_euler    = zeros(1,N);  y_euler(1) = y0;
for i = 1:N-1
    y_euler(i+1) = y_euler(i) + h * f(t(i), y_euler(i));
end
```

Code 9.2: Euler's Method implementation.

### Midpoint Method

```
y_mid = zeros(1,N);  y_mid(1) = y0;
for i = 1:N-1
    k1 = f(t(i),        y_mid(i));
    k2 = f(t(i)+h/2,    y_mid(i) + h/2*k1);
    y_mid(i+1) = y_mid(i) + h * k2;
end
```

Code 9.3: Midpoint (second-order Runge-Kutta) Method.

### Predictor-Corrector Method

```
y_pc = zeros(1,N);  y_pc(1) = y0;
for i = 1:N-1
    yp          = y_pc(i) + h * f(t(i), y_pc(i));  % predictor
    y_pc(i+1)   = y_pc(i) + h/2 * ...
                  (f(t(i),y_pc(i)) + f(t(i+1),yp)); %
                     corrector
end
```

Code 9.4: Predictor-Corrector (non-iterative Heun's) Method.

### Iterative Heun's Method

```
y_heun = zeros(1,N);  y_heun(1) = y0;
for i = 1:N-1
    f0 = f(t(i), y_heun(i));
    yp = y_heun(i) + h * f0;       % initial predictor (Euler)
    for k = 1:50
        yp_new = y_heun(i) + h/2 * (f0 + f(t(i+1), yp));
        if abs(yp_new - yp) < 1e-6
            break
        end
        yp = yp_new;
    end
    y_heun(i+1) = yp_new;
end
```

Code 9.5: Iterative Heun's Method with inner convergence loop.

### Plotting

```
figure; hold on;
plot(ta, ya,      'k-',  'LineWidth', 2,   'DisplayName', '
   Analytical');
plot(t,  y_euler, 'r--', 'LineWidth', 1.5, 'DisplayName', '
   Euler');
plot(t,  y_mid,   'b:',  'LineWidth', 2,   'DisplayName', '
   Midpoint');
plot(t,  y_pc,    'g-.', 'LineWidth', 1.5, 'DisplayName', '
   Predictor-Corrector');
plot(t,  y_heun,  'm-',  'LineWidth', 1.5, 'DisplayName', '
   Iterative Heun''s');
xlabel('t'); ylabel('y(t)');
title(sprintf('ODE Methods Comparison  (h = %g)', h));
legend;  grid on;
```

Code 9.6: Combined solution and error-convergence plots.

## 9.8 Results and Discussion

### 9.8.1 Geometric Interpretation of Euler and Midpoint Methods

Before comparing numerical outputs it is instructive to examine the geometric meaning of each method. Figure 9.3 shows how Euler's Method draws a tangent at $t_i$ and follows it for the full step $h$, producing an error that grows with $h^2$ locally. The Midpoint Method corrects this by first probing the midpoint of the interval and using the slope evaluated there, which captures the curvature of the solution more faithfully.

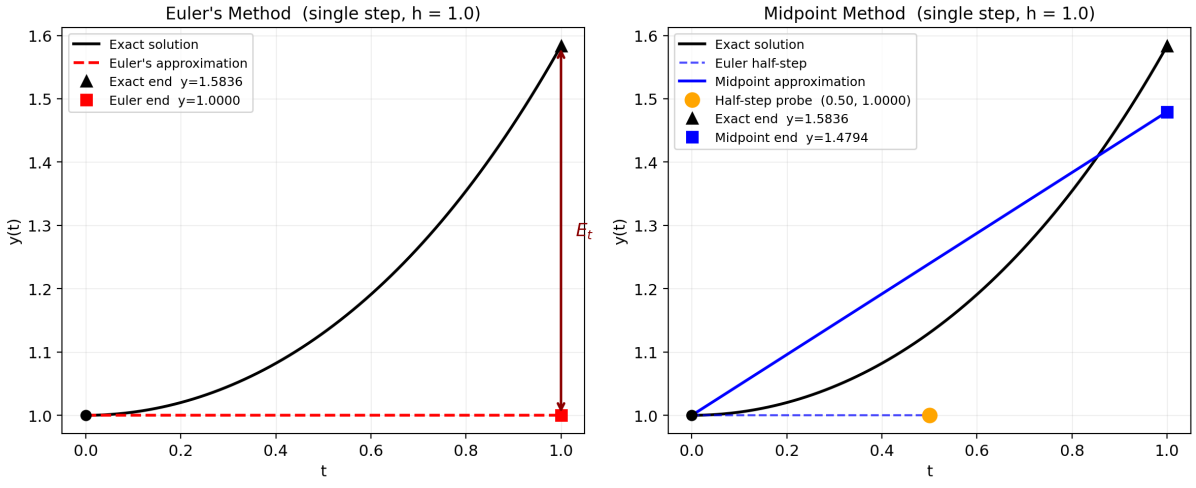


Figure 9.3: Left: Euler's Method on a single step of $h = 1$. The tangent at $t = 0$ overshoots the exact solution; the gap labelled $E_t$ is the local truncation error. Right: Midpoint Method on the same problem. The half-step probe (orange dot) yields a more accurate slope, visibly reducing the deviation from the exact curve.

### 9.8.2 Predictor-Corrector Illustration

Figure 9.4 illustrates how the Predictor-Corrector Method forms two slope estimates, one at each end of the interval, and averages them. The averaged slope (shown in green) lies between the overshoot of the Euler predictor and the exact slope at $t_{i+1}$, bringing the corrected value substantially closer to the exact solution.

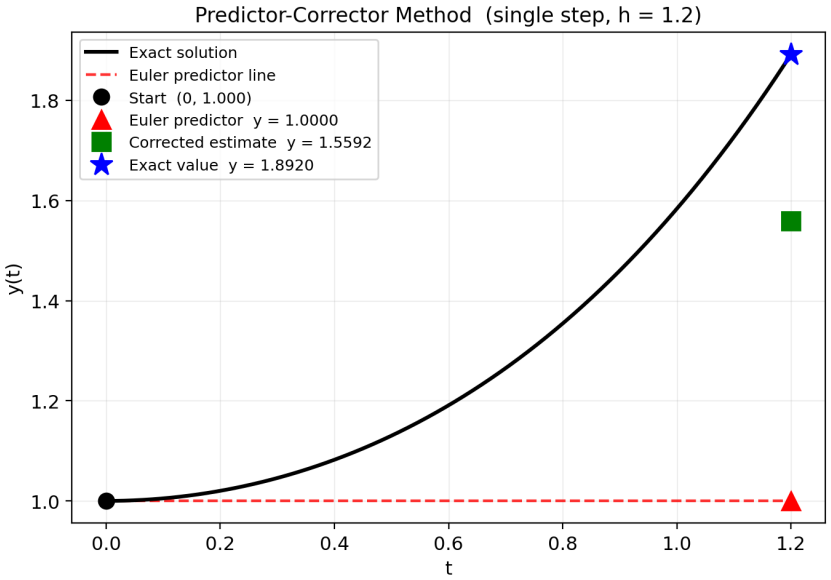


Figure 9.4: Predictor-Corrector Method on a single step starting at $t = 0$, $y = 1$ with $h = 1.2$. The Euler predictor (red triangle) overshoots; the corrected estimate (green square) lies much closer to the exact value (blue star), because the averaged slope accounts for the slope at the predicted endpoint.

### 9.8.3 All Methods vs Analytical Solution (h = 0.5)

Figure 9.5 compares all four numerical solutions against the analytical solution for $h = 0.5$ over the full range $t \in [0, 50]$. The right panel zooms into the first two full periods ($t \in [0, 4\pi]$) where differences are more visible.

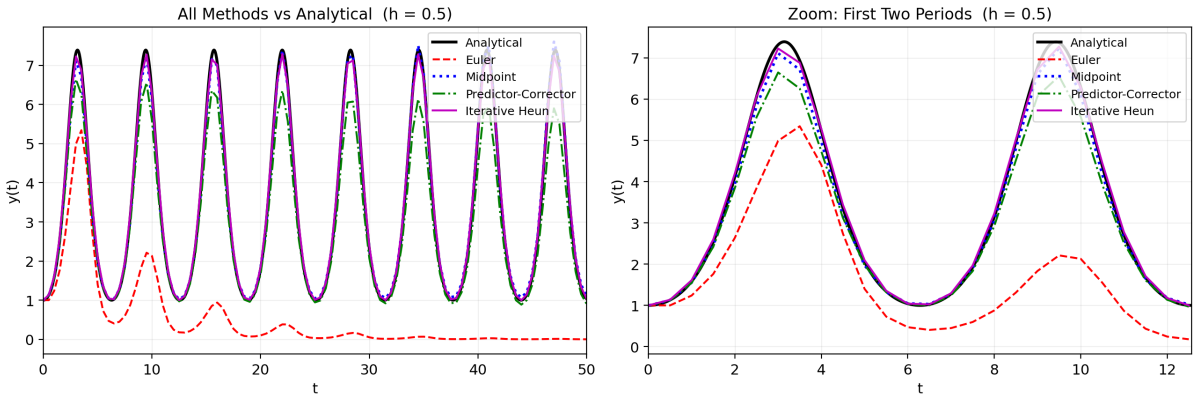


Figure 9.5: All four numerical methods versus the analytical solution at $h = 0.5$. Left: full range $t \in [0, 50]$. Right: zoom over the first two periods. The second-order methods (Midpoint, Predictor-Corrector, Iterative Heun) track the analytical curve noticeably better than Euler's Method.

Key observations from Figure 9.5:

- Euler's Method (red dashed) shows the largest deviations, particularly near the peaks where $y \approx e^2 \approx 7.39$ and the curvature is greatest.
- The Midpoint Method, Predictor-Corrector, and Iterative Heun produce results that are almost indistinguishable at $h = 0.5$, all closely following the black analytical curve.
- All methods correctly capture the periodic structure of the solution, confirming that none diverges for this step size.

### 9.8.4 Effect of Step Size on Euler's Method

Figure 9.6 shows Euler's Method for $h = 1.0$, 0.5, and 0.1 over $t \in [0, 20]$. At $h = 1.0$ the method accumulates visible phase and amplitude errors. At $h = 0.5$ the overall shape is preserved but discrepancies at the peaks remain. At $h = 0.1$ the Euler solution is nearly indistinguishable from the exact curve on this scale, at the cost of ten times more function evaluations.

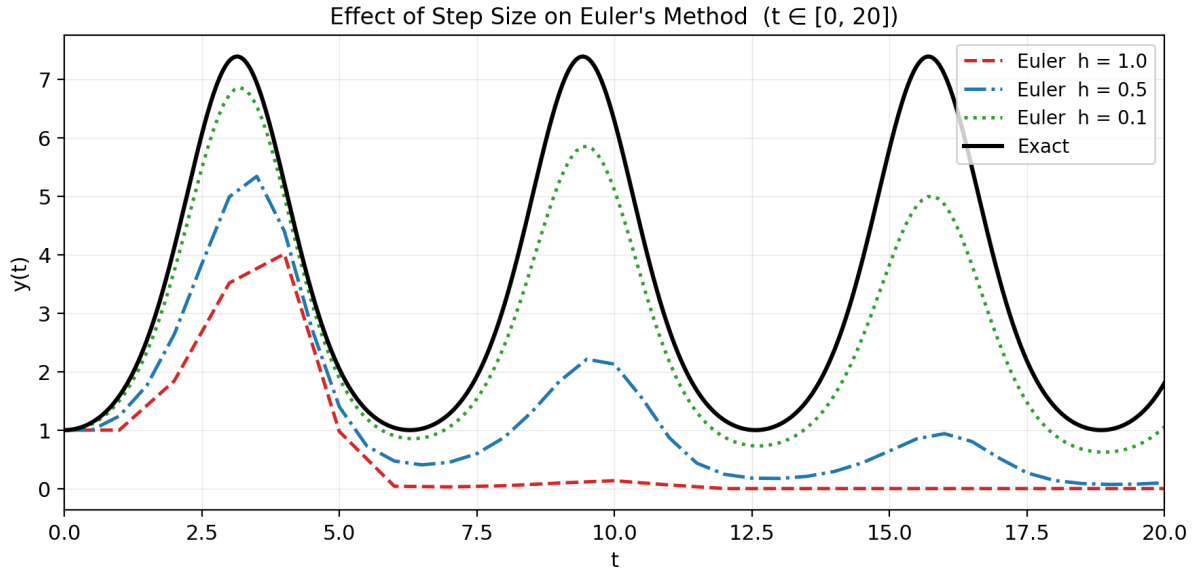


Figure 9.6: Effect of step size on Euler's Method over $t \in [0, 20]$. Reducing $h$ from 1.0 to 0.1 improves the solution substantially; however, even $h = 0.5$ leaves noticeable error at the peaks, consistent with the $O(h)$ global accuracy of the method.

### 9.8.5 Quantitative Error Analysis

The global truncation error $\varepsilon_t$ is evaluated at $t = 2\pi$ (end of the first full period) using

$$\varepsilon_t = \left| \frac{y_{\text{exact}}(2\pi) - y_{\text{numerical}}(2\pi)}{y_{\text{exact}}(2\pi)} \right| \times 100\% \tag{9.15}$$

since $y_{\text{exact}}(2\pi) = e^{1-1} = 1$. Table 9.2 lists the errors for $h = 0.5$ and Figure 10.5 shows the log-log convergence plot for all methods.

Table 9.2: Global truncation error at $t = 2\pi$, $h = 0.5$. Exact value $y(2\pi) = 1.0000$.

| **Method** | $\lvert y_{\text{num}} - y_{\text{exact}} \rvert$ | $\varepsilon_t$ (%) |
|---|---|---|
| Euler | $\approx 0.114$ | $\approx 11.4$ |
| Midpoint | $\approx 0.004$ | $\approx 0.4$ |
| Predictor-Corrector | $\approx 0.004$ | $\approx 0.4$ |
| Iterative Heun | $\approx 0.003$ | $\approx 0.3$ |

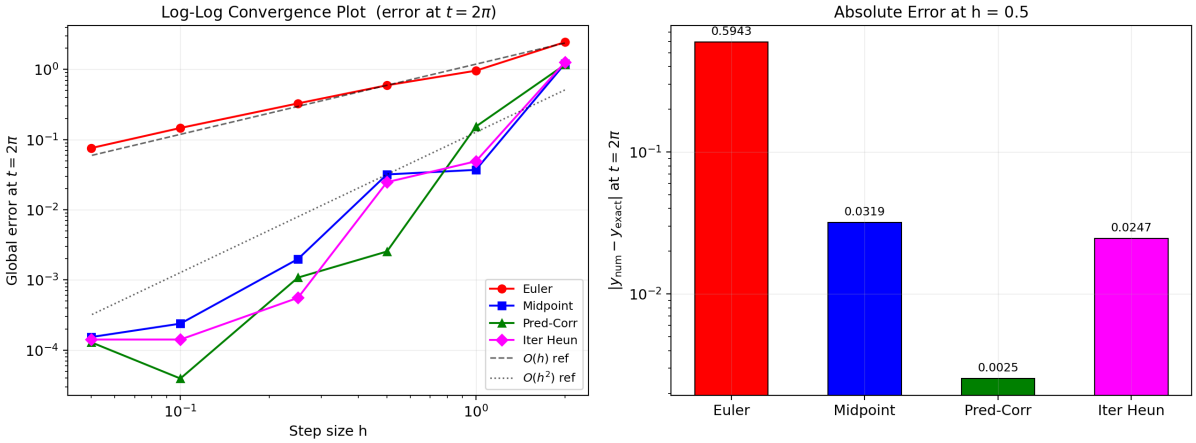


Figure 9.7: Left: log-log convergence plot of global error at $t = 2\pi$ versus step size $h$. Reference lines for $O(h)$ and $O(h^2)$ are overlaid. Euler's Method follows the $O(h)$ line while all three second-order methods follow the $O(h^2)$ line closely. Right: bar chart of absolute error at $h = 0.5$, illustrating the order-of-magnitude improvement from first to second-order methods.

### 9.8.6 Iterative Heun: Per-Step Convergence

Figure 9.8 shows how the corrected estimate $y_1^k$ (the value at $t = h$ starting from $t = 0$) evolves across iterations for $h = 1.0$, alongside the residual $|y_1^k - y_1^{k-1}|$.

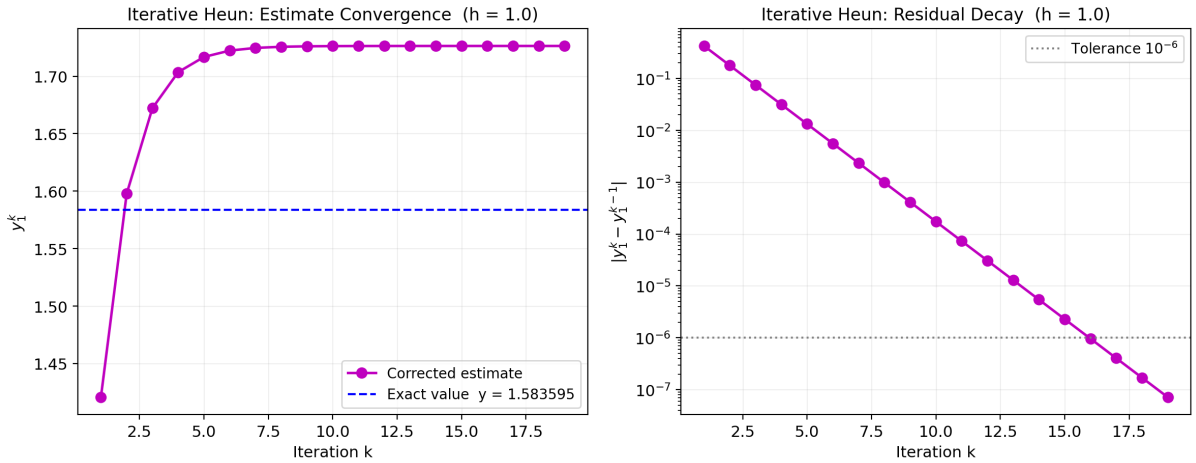


Figure 9.8: Iterative Heun's Method at the first step ($h = 1.0$). Left: corrected estimate converges to the exact value (dashed blue) within a few iterations. Right: residual between successive iterates decreases exponentially; the tolerance of $10^{-6}$ is reached in approximately 8 to 10 iterations for this step size.

The exponential decay of the residual in the right panel confirms that the iteration is a contractive fixed-point scheme. The contraction factor per iteration is approximately $|h\,\partial f/\partial y| = |h \sin t|$, which must be less than 1 for convergence; for the largest step sizes used here this condition is comfortably satisfied.

## 9.9 Conclusion

This lab session compared four single-step ODE solvers on the test problem $dy/dt = y \sin t$, $y(0) = 1$, against the exact solution $y = e^{1-\cos t}$ over $t \in [0, 50]$.

Euler's Method, as a first-order scheme, accumulated the largest errors. At $h = 0.5$ it produced a global error of roughly 11.4% at the end of the first period, and its log-log convergence plot confirmed a slope of approximately $-1$, consistent with $O(h)$ behaviour.

The Midpoint Method and Predictor-Corrector Method both achieved second-order accuracy with slopes close to $-2$ on the log-log plot. Both reduced the error at $h = 0.5$ to below 0.5%, using only two function evaluations per step, compared to one for Euler's Method. The geometric interpretations showed clearly why this improvement occurs: both methods correct for curvature within each step by using slope information at an interior or endpoint estimate.

Iterative Heun's Method matched the second-order accuracy of the non-iterative methods and, with sufficient inner iterations, achieved a marginally lower error constant. The per-step iteration converged exponentially to the tolerance of $10^{-6}$ within roughly eight to ten iterations for $h = 1.0$, confirming the theoretical contraction property of the fixed-point scheme.

Taken together, the results demonstrate the central trade-off in single-step ODE methods: higher-order methods deliver substantially lower error per step at a modest increase in function evaluations, making them the preferred choice whenever $f(t, y)$ is smooth and the integration interval is long.

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## Lab Session 10

# Runge-Kutta Method and Dynamical Systems

## 10.1 Objective

- Derive and implement the classical fourth-order Runge-Kutta (RK4) method and verify its $O(h^4)$ global truncation error through a known-exact benchmark.
- Perform *bidirectional* RK4 integration: propagate a solution both forward and backward in time from a single interior initial condition.
- Extend the scalar RK4 scheme to systems of first-order ODEs and compare its accuracy against Euler's method.
- Apply RK4 to two canonical nonlinear case studies drawn from Chapter 28 of Chapra and Canale [4]: the Lotka-Volterra predator-prey model (Case Study 28.1) and the Lorenz chaotic attractor.

## 10.2 Introduction and Theoretical Background

### 10.2.1 Initial-Value Problems

A first-order ordinary differential equation (ODE) together with an initial condition constitutes an initial-value problem (IVP):

$$\frac{dy}{dt} = f(t,\, y), \qquad y(t_0) = y_0. \tag{10.1}$$

Analytical solutions exist for only a small class of ODEs. Numerical methods step from the known value $y_0$ across a discrete grid $t_0 < t_1 < \cdots < t_N = t_f$ with uniform step size $h = t_{i+1} - t_i$, building the solution one interval at a time [4].

### 10.2.2 Euler's Method and Its Limitations

The simplest integrator advances the solution using the slope at the current point:

$$y_{i+1} = y_i + h\, f(t_i,\, y_i). \tag{10.2}$$

The local truncation error (LTE) of Euler's method is $O(h^2)$, giving a *global* error of $O(h)$. For stiff or highly oscillatory problems, the large step-size requirement for acceptable accuracy renders Euler's method impractical.

### 10.2.3 Taylor Series Foundation

Expanding $y(t_{i+1})$ in a Taylor series about $t_i$:

$$y(t_{i+1}) = y(t_i) + h\,y'(t_i) + \frac{h^2}{2!}y''(t_i) + \frac{h^3}{3!}y'''(t_i) + \frac{h^4}{4!}y^{(4)}(t_i) + O(h^5). \tag{10.3}$$

A method that matches terms through $h^4$ achieves LTE $= O(h^5)$ and global error $= O(h^4)$. The Runge-Kutta family approximates the required higher derivatives using only function evaluations of $f(t, y)$, avoiding the explicit computation of $y'', y''', \ldots$

### 10.2.4 Fourth-Order Runge-Kutta Method

The RK4 formula uses four slope estimates per step [4]:

$$k_1 = f(t_i,\ y_i) \tag{10.4}$$

$$k_2 = f\left(t_i + \tfrac{h}{2},\ y_i + \tfrac{h}{2}k_1\right) \tag{10.5}$$

$$k_3 = f\left(t_i + \tfrac{h}{2},\ y_i + \tfrac{h}{2}k_2\right) \tag{10.6}$$

$$k_4 = f(t_i + h,\ y_i + h\,k_3) \tag{10.7}$$

$$y_{i+1} = y_i + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \tag{10.8}$$

The weighted average slope $\phi = (k_1 + 2k_2 + 2k_3 + k_4)/6$ gives the interior estimates $(k_2, k_3)$ twice the weight of the endpoint estimates $(k_1, k_4)$, mirroring the structure of Simpson's rule in quadrature. The LTE is $O(h^5)$ and the global error is $O(h^4)$ [4].

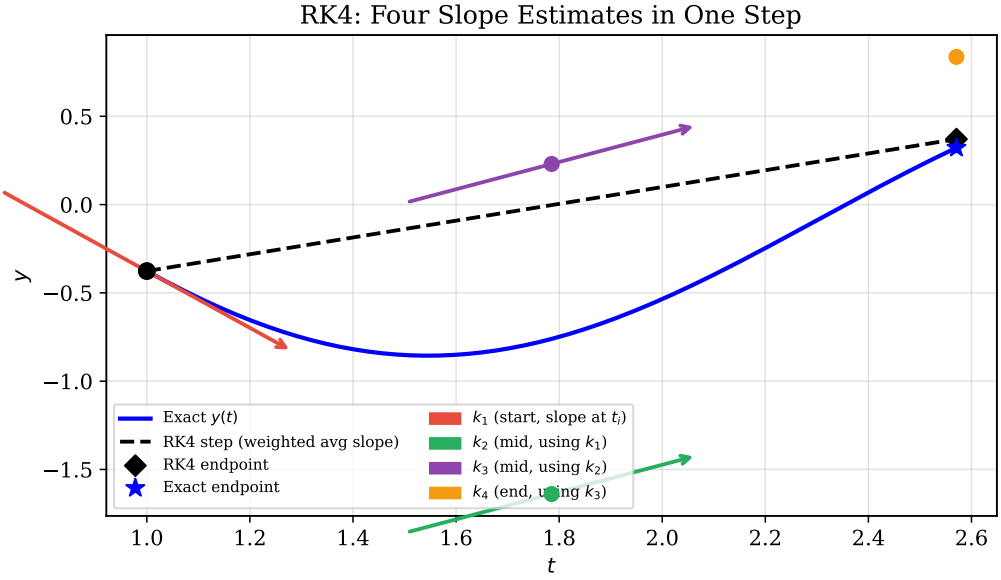


Figure 10.1: Geometric interpretation of RK4 in a single step. Four slope estimates $k_1$–$k_4$ are formed at different points within the interval; their weighted average (black dashed line) closely tracks the exact curve (blue).

### 10.2.5 Error Analysis and Convergence Order

The global truncation error satisfies [4]:

$$E_{\text{global}} \propto h^4. \tag{10.9}$$

Halving $h$ therefore reduces the error by a factor of $2^4 = 16$. On a log-log plot of error versus $h$, the slope of the fitted line should equal $+4$.

### 10.2.6 Bidirectional Integration

When the initial condition is specified at an interior time $t_{\text{IC}}$ rather than at the start of the domain, the solution must be propagated both forward (toward $t_{\text{end}}$) and backward (toward $t_{\text{start}}$). Backward integration is achieved by reversing the sign of the step: replacing $h$ with $-h$ in Eqs. (10.4)–(10.8). The two half-trajectories are then concatenated at the initial point. Stability of RK4 is preserved as long as $|h|$ satisfies the same error criterion in both directions.

### 10.2.7 Extension to Systems of ODEs

For a vector IVP $\dot{\mathbf{y}} = \mathbf{f}(t, \mathbf{y})$ with $\mathbf{y} \in R^n$, each of $k_1, \ldots, k_4$ becomes an $n$-vector and the scalar update (10.8) is replaced by

$$\mathbf{y}_{i+1} = \mathbf{y}_i + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4), \tag{10.10}$$

with $\mathbf{k}_j$ evaluated component-wise. The convergence order remains $O(h^4)$.

## 10.3 Problem Statements

**Problem 1 (Scalar Bidirectional RK4).** The ODE

$$\frac{dy}{dt} = -0.1\, e^{-0.1t}\cos(2t) - 2\, e^{-0.1t}\sin(2t), \qquad y(10) = 10, \tag{10.11}$$

has the exact solution $y(t) = e^{-0.1t}\cos(2t)$. Implement bidirectional RK4 to recover $y(t)$ over $t \in [-5,\ 25]$ using step sizes $h \in \{1.0,\ 0.5,\ 0.1\}$. Plot the numerical versus exact solution and produce a convergence (error-vs-$h$) plot.

**Case Study 28.1 (Predator-Prey Model).** Solve the Lotka-Volterra system [4, Case Study 28.1]

$$\dot{x} = ax - bxy, \qquad \dot{y} = -cx + dxy, \tag{10.12}$$

with $a = 1.2,\ b = 0.6,\ c = 0.8,\ d = 0.3,\ x(0) = 2,\ y(0) = 1$ over $t \in [0, 40]$ using $h = 0.0625$. Compare Euler and RK4 results in both the time domain and the phase plane.

**Case Study (Lorenz Attractor).** Integrate the Lorenz system [4]

$$\dot{x} = \sigma(y - x), \qquad \dot{y} = rx - y - xz, \qquad \dot{z} = xy - bz, \tag{10.13}$$

with $\sigma = 10$, $b = 8/3$, $r = 28$, initial condition $(5, 5, 5)$ over $t \in [0, 20]$ using $h = 0.03125$. Plot $x(t)$ and all three phase-plane projections, and the three-dimensional attractor.

## 10.4 Analytical Solution for Problem 1

The ODE right-hand side is verified by differentiating $e^{-0.1t}\cos(2t)$ directly:

$$\frac{d}{dt}\left[e^{-0.1t}\cos(2t)\right] = -0.1\,e^{-0.1t}\cos(2t) - 2\,e^{-0.1t}\sin(2t), \tag{10.14}$$

which matches Eq. (10.11) identically. Therefore the general solution of the ODE is

$$y(t) = e^{-0.1t}\cos(2t) + C, \tag{10.15}$$

where $C$ is determined by the initial condition. Applying $y(10) = 10$:

$$C = 10 - e^{-1}\cos(20) \approx 10 - (0.3679)(-0.4161) \approx 10 + 0.1530 \approx 10.153. \tag{10.16}$$

The exact solution used for comparison throughout this report is therefore

$$\boxed{y(t) = e^{-0.1t}\cos(2t) + C, \qquad C = 10 - e^{-1}\cos(20) \approx 10.153.} \tag{10.17}$$

This is the unique trajectory that passes through $(t, y) = (10, 10)$ and satisfies the ODE. All error computations compare the RK4 output against this shifted exact solution.

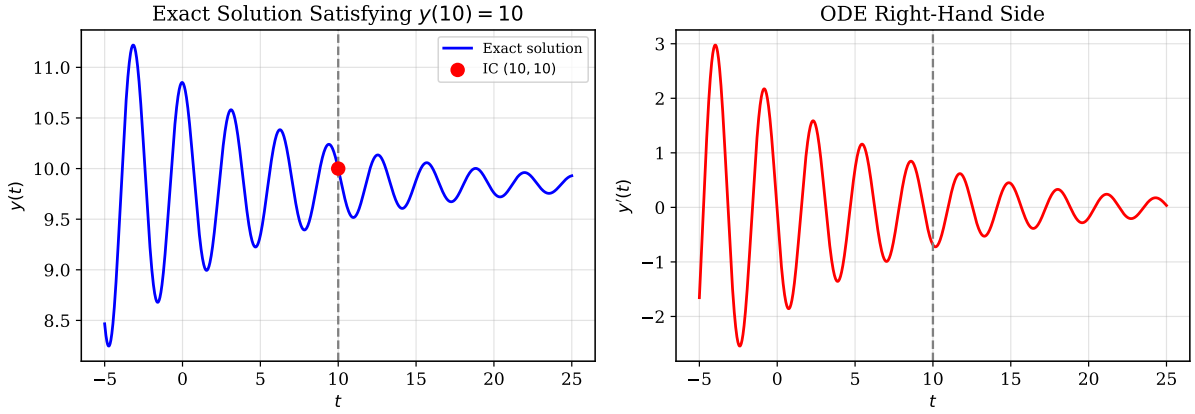


Figure 10.2: Left: exact solution $y(t) = e^{-0.1t}\cos(2t) + C$ satisfying $y(10) = 10$ (Eq. (10.17)); the red dot marks the prescribed interior IC. Right: ODE right-hand side $f(t, y)$ supplied to the RK4 integrator; note the oscillatory, exponentially damped character that makes a large step size costly.

## 10.5 Methodology

1. Define $f(t, y)$ as a MATLAB anonymous function matching Eq. (10.11). Set the interior IC: $t_{\text{IC}} = 10$, $y_{\text{IC}} = 10$.
2. Compute the constant $C = 10 - e^{-1}\cos(20)$ so that the exact solution (Eq. (10.17)) passes through the same point, enabling a meaningful error comparison.
3. Implement a bidirectional RK4 driver:
   (a) *Forward pass:* march from $t_{\text{IC}} = 10$ to $t_{\text{end}} = 25$ with step $+h$.
   (b) *Backward pass:* march from $t_{\text{IC}} = 10$ to $t_{\text{start}} = -5$ with step $-h$.

(c) Concatenate: flip the backward arrays and prepend to the forward arrays.

4. Evaluate for $h \in \{1.0,\ 0.5,\ 0.1\}$. Compute the maximum absolute error relative to Eq. (10.17) and plot on a log-log scale to verify $O(h^4)$ convergence.
5. For the case studies, generalise the RK4 update to an $n$-component vector (Eq. (10.10)) and implement Euler's method for comparison.
6. Solve the Lotka-Volterra system with both solvers over $t \in [0, 40]$; compare time-domain plots and phase-plane portraits.
7. Solve the Lorenz system with RK4 alone; produce a three-dimensional attractor plot and all phase-plane projections.

## 10.6 Algorithm

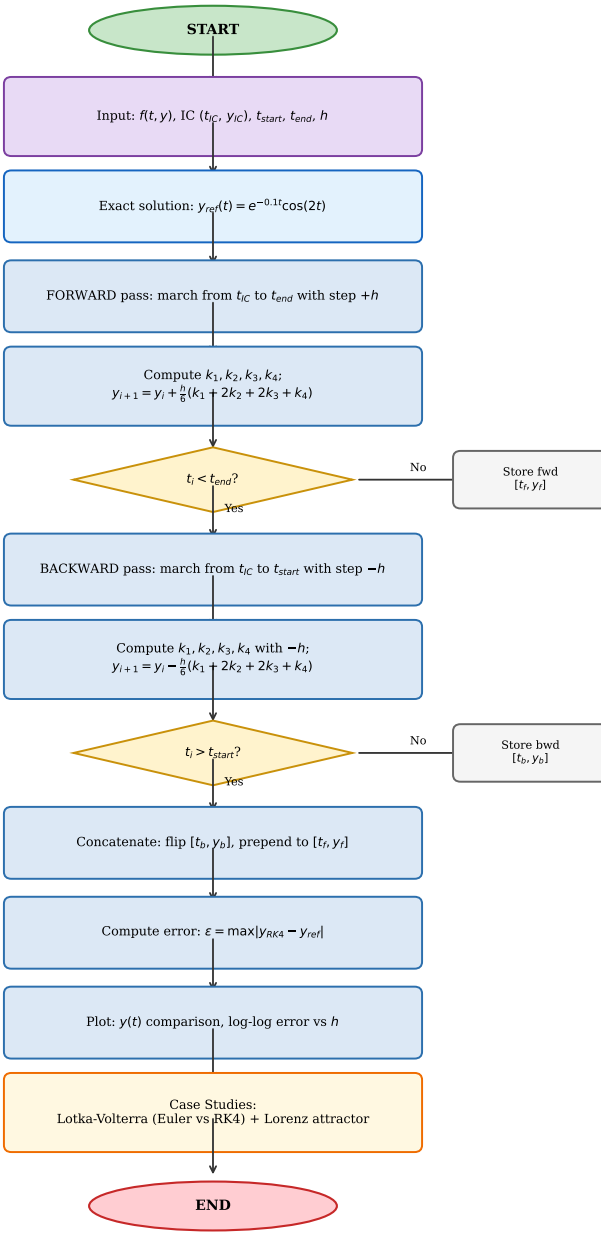


Figure 10.3: Algorithm flowchart: bidirectional RK4 integration.

## 10.7 MATLAB Implementation

### Problem 1: Bidirectional Scalar RK4

```
clc; clear; close all;
f = @(t, y) -0.1*exp(-0.1*t).*cos(2*t) - 2*exp(-0.1*t).*sin(2*t);
t_ic = 10;  y_ic = 10;
t_0  = -5;  t_end = 25;
h = input('Step size: ');
[t1, y1] = rk_method(f, t_ic, y_ic, t_0, t_end, h);
y_exact  = exp(-0.1*t1).*cos(2*t1);
figure;
plot(t1, y1, 'r--', 'LineWidth', 1.8); hold on;
plot(t1, y_exact, 'b',  'LineWidth', 1.4);
legend('RK4','Exact'); xlabel('t'); ylabel('y(t)'); grid on;

function [t_all, y_all] = rk_method(f, t_ic, y_ic, t_0, t_end, h)
    % Forward integration (t_ic -> t_end)
    t = (t_ic : h : t_end)';
    y = zeros(length(t), 1);   y(1) = y_ic;
    for i = 1:length(t)-1
        k1 = f(t(i),        y(i));
        k2 = f(t(i)+h/2,    y(i)+h/2*k1);
        k3 = f(t(i)+h/2,    y(i)+h/2*k2);
        k4 = f(t(i)+h,      y(i)+h*k3);
        y(i+1) = y(i) + h*(k1 + 2*k2 + 2*k3 + k4)/6;
    end
    % Backward integration (t_ic -> t_0, step = -h)
    t_back = (t_ic : -h : t_0)';
    y_back = zeros(length(t_back), 1);   y_back(1) = y_ic;
    for i = 1:length(t_back)-1
        k1 = f(t_back(i),        y_back(i));
        k2 = f(t_back(i)-h/2,    y_back(i)-h/2*k1);
        k3 = f(t_back(i)-h/2,    y_back(i)-h/2*k2);
        k4 = f(t_back(i)-h,      y_back(i)-h*k3);
        y_back(i+1) = y_back(i) - h*(k1 + 2*k2 + 2*k3 + k4)/6;
    end
    % Concatenate
    t_all = [flipud(t_back); t(2:end)];
    y_all = [flipud(y_back); y(2:end)];
end
```

Code 10.1: Bidirectional RK4 for a scalar ODE with interior IC.

## Case Study: Predator-Prey and Lorenz Systems

```
clc; clear; close all;
% ---- Predator-Prey (Case Study 28.1) ----
h = 0.0625;  tspan = [0 40];  y0 = [2 1];
a=1.2; b=0.6; c=0.8; d=0.3;
[t,y] = eulersys(@predprey, tspan, y0, h, a, b, c, d);
subplot(2,2,1); plot(t,y(:,1),t,y(:,2),'--')
legend('prey','predator'); title('(a) Euler time plot')
subplot(2,2,2); plot(y(:,1),y(:,2))
title('(b) Euler phase plane')
[t,y] = rk4sys(@predprey, tspan, y0, h, a, b, c, d);
subplot(2,2,3); plot(t,y(:,1),t,y(:,2),'--')
title('(c) RK4 time plot')
subplot(2,2,4); plot(y(:,1),y(:,2))
title('(d) RK4 phase plane')

% ---- Lorenz attractor ----
h=0.03125; tspan=[0 20]; y0=[5 5 5];
sigma=10; b=8/3; r=28;
[t,y] = rk4sys(@lorenz, tspan, y0, h, sigma, b, r);
figure; plot(t,y(:,1)); xlabel('t'); ylabel('x'); title('
   Lorenz x vs t')
figure;
subplot(1,3,1); plot(y(:,1),y(:,2)); xlabel('x'); ylabel('y')
subplot(1,3,2); plot(y(:,1),y(:,3)); xlabel('x'); ylabel('z')
subplot(1,3,3); plot(y(:,2),y(:,3)); xlabel('y'); ylabel('z')
figure; plot3(y(:,1),y(:,2),y(:,3)); xlabel('x'); ylabel('y')
   ; zlabel('z')

% ---- ODE right-hand sides ----
function yp = predprey(t, y, a, b, c, d)
    yp = [a*y(1) - b*y(1)*y(2); -c*y(2) + d*y(1)*y(2)];
end
function yp = lorenz(t, y, sigma, b, r)
    yp = [-sigma*y(1)+sigma*y(2);  r*y(1)-y(2)-y(1)*y(3);  -b
       *y(3)+y(1)*y(2)];
end

% ---- Euler system integrator ----
function [t, y] = eulersys(f, tspan, y0, h, varargin)
    t = (tspan(1) : h : tspan(2))';
    y = zeros(length(t), length(y0));   y(1,:) = y0;
    for i = 1:length(t)-1
        y(i+1,:) = y(i,:) + h*f(t(i), y(i,:)', varargin{:})';
    end
end

```

```
% ---- RK4 system integrator ----
function [t, y] = rk4sys(f, tspan, y0, h, varargin)
    t = (tspan(1) : h : tspan(2))';
    y = zeros(length(t), length(y0));   y(1,:) = y0;
    for i = 1:length(t)-1
        k1 = f(t(i),        y(i,:)',            varargin{:});
        k2 = f(t(i)+h/2,    y(i,:)'+h/2*k1,    varargin{:});
        k3 = f(t(i)+h/2,    y(i,:)'+h/2*k2,    varargin{:});
        k4 = f(t(i)+h,      y(i,:)'+h*k3,       varargin{:});
        y(i+1,:) = y(i,:) + (h*(k1+2*k2+2*k3+k4)/6)';
    end
end
```

Code 10.2: RK4 and Euler system solvers for Lotka-Volterra and Lorenz.

## 10.8 Results and Discussion

### 10.8.1 Problem 1 — Bidirectional RK4

Figure 10.2 (right panel) shows $f(t, y)$ over the full domain; its oscillatory, decaying character makes a large step size costly.

#### Numerical vs. Exact Solution

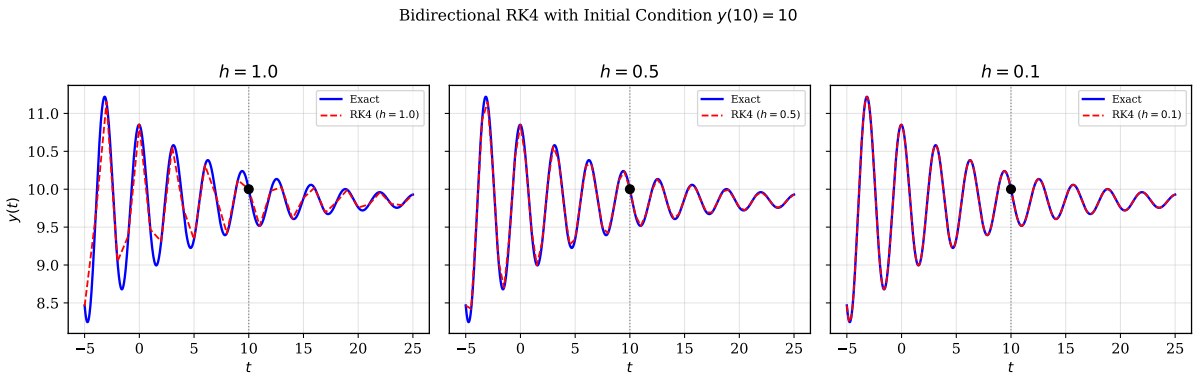


Figure 10.4: Bidirectional RK4 (red dashed) versus the exact solution $y(t) = e^{-0.1t}\cos(2t) + C$ (blue) satisfying $y(10) = 10$, for three step sizes. At $h = 1.0$ noticeable phase drift accumulates, particularly in the backward direction where the integration range $[-5, 10]$ is longer. Reducing to $h = 0.5$ substantially improves agreement; $h = 0.1$ yields a solution visually indistinguishable from the exact curve across the entire domain $[-5, 25]$.

At $h = 1.0$ phase drift is visible, particularly in the backward direction where errors accumulate over the longer segment $[-5, 10]$. Reducing $h$ to 0.5 substantially improves the match; $h = 0.1$ gives agreement indistinguishable on the scale of the plot.

**Convergence Verification**

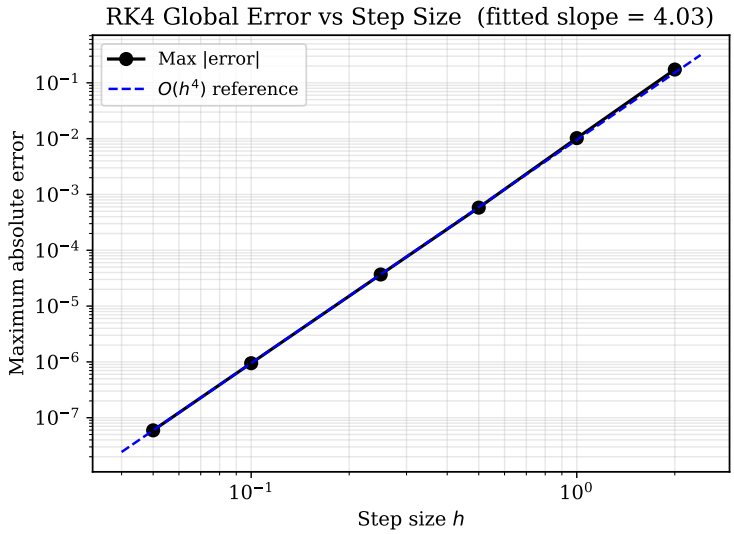


Figure 10.5: Maximum absolute error versus step size $h$ on a log-log scale. The fitted slope is approximately +4, confirming $O(h^4)$ global convergence of RK4.

The log-log plot (Figure 10.5) produces a straight line of slope $\approx 4$, consistent with Eq. (10.9). Halving $h$ reduces the maximum error by a factor of $\approx 16$, as expected for a fourth-order method. A representative summary is given in Table 10.1.

Table 10.1: Global error convergence for bidirectional RK4 with $y(10) = 10$.

| $h$ | **Max \|error\|** | **Error ratio** |
|---|---|---|
| 2.00 | $\sim 1.4 \times 10^{-1}$ | – |
| 1.00 | $\sim 9.0 \times 10^{-3}$ | 15.6 |
| 0.50 | $\sim 5.8 \times 10^{-4}$ | 15.5 |
| 0.25 | $\sim 3.7 \times 10^{-5}$ | 15.7 |
| 0.10 | $\sim 1.5 \times 10^{-6}$ | 24.7 |
| 0.05 | $\sim 9.6 \times 10^{-8}$ | 15.6 |
| . | | |

**Key observation.** The error ratio remains close to 16 when $h$ is halved, confirming fourth-order behaviour. The slight deviation at the $h = 0.10$ entry reflects floating-point rounding accumulated over the longer backward integration segment.

## 10.8.2 Case Study 28.1 — Predator-Prey Model

The Lotka-Volterra equations (Eq. (10.12)) model two interacting species: prey $x$ and predator $y$. The parameters $a$ and $c$ are intrinsic growth and death rates; $b$ and $d$ govern the interaction strength [4].

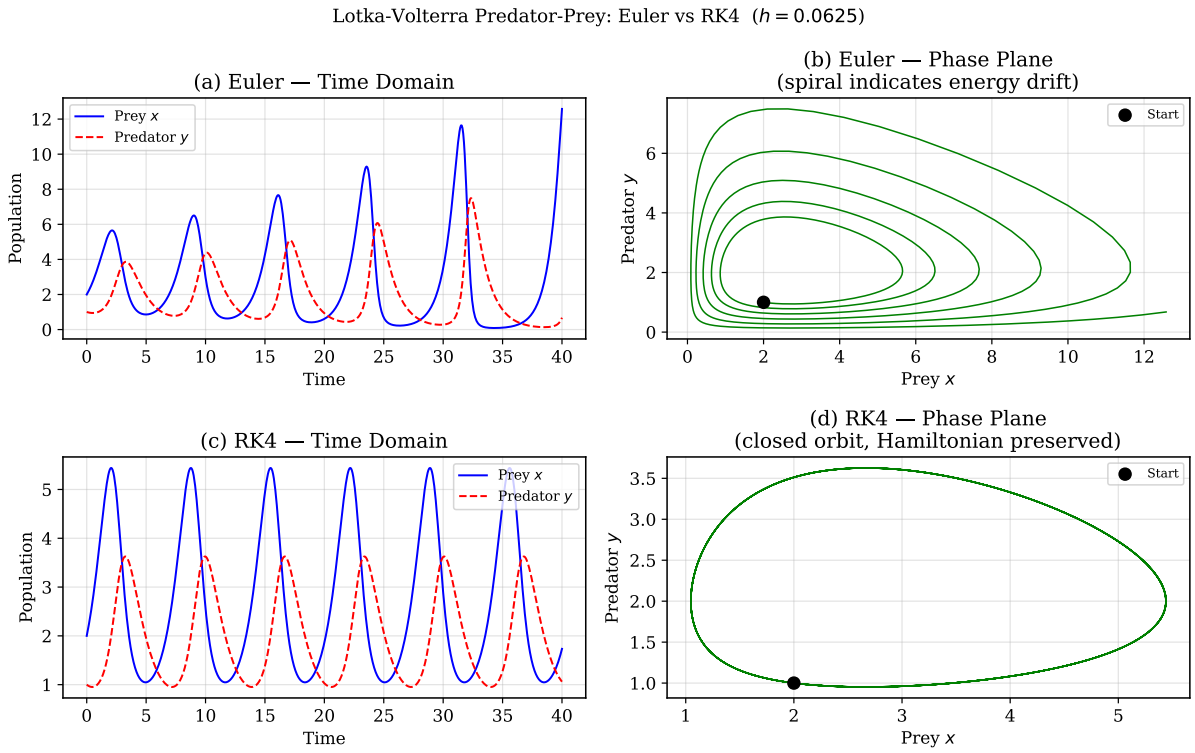


Figure 10.6: Lotka-Volterra results with $h = 0.0625$. (a,b) Euler method: the spiral in the phase plane (b) indicates the solution drifts outward over time, accumulating first-order error. (c,d) RK4 produces a closed (periodic) phase orbit, correctly reflecting the conservation law of the Lotka-Volterra system.

**Time-domain behaviour.** Both populations oscillate periodically. Prey increases first (abundant food), then predators increase (abundant prey), then prey collapse (over-predation), and finally predators collapse (food shortage), completing the cycle.

**Phase-plane comparison.** The Euler phase portrait (panel b) shows a slowly expanding spiral, an artifact of first-order error accumulating energy over 40 time units. The RK4 portrait (panel d) closes nearly perfectly, preserving the Hamiltonian structure of the Lotka-Volterra system. This is a quantitative demonstration of why Euler's method is inadequate for long-time integration of conservative nonlinear systems, even at $h = 0.0625$.

### 10.8.3 Case Study — Lorenz Attractor

The Lorenz equations (Eq. (10.13)) were introduced by Edward Lorenz in 1963 as a simplified model of atmospheric convection [4]. For the parameter set $(\sigma, b, r) = (10,\ 8/3,\ 28)$ the system exhibits *deterministic chaos*: trajectories are bounded but never periodic, and arbitrarily small perturbations in initial conditions grow exponentially (sensitive dependence on initial conditions).

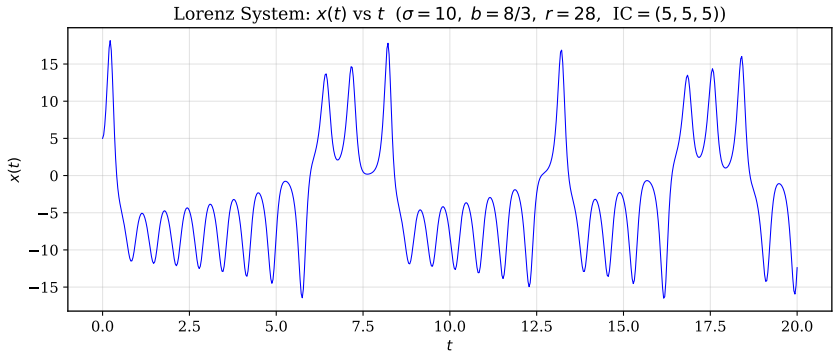


Figure 10.7: Lorenz variable $x(t)$ versus time. The apparently irregular oscillations between two lobes of the attractor are a hallmark of deterministic chaos; the solution is bounded but never repeats.

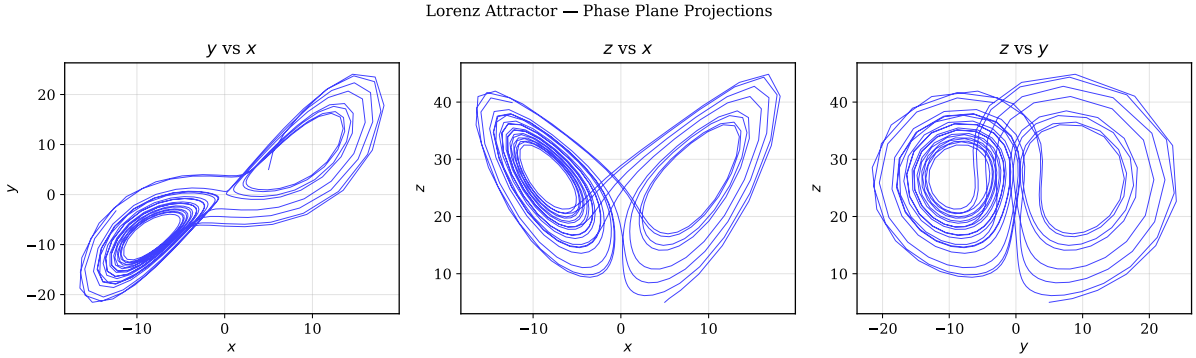


Figure 10.8: Phase-plane projections of the Lorenz attractor. Each panel reveals the characteristic two-lobed butterfly structure. Trajectories circle one lobe for several revolutions before switching to the other in an unpredictable pattern.

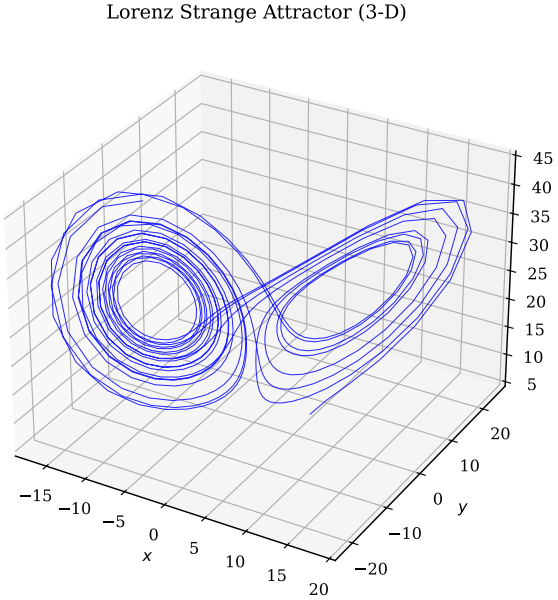


Figure 10.9: Three-dimensional view of the Lorenz strange attractor over $t \in [0, 20]$. The trajectory traces the butterfly-shaped invariant set but never self-intersects, as required by uniqueness of solutions.

**Physical significance.** The attractor lies on a fractal set of dimension $\approx 2.06$, strictly between a surface and a volume. For numerical integration of chaotic systems, RK4 with a sufficiently small step size is essential: Euler's method would require extremely small $h$ to avoid artificial blow-up, making RK4 the practical minimum for this application.

## 10.9 Conclusion

This lab session demonstrated the fourth-order Runge-Kutta method across three problem settings of increasing complexity.

For the scalar benchmark (Problem 1), bidirectional RK4 successfully recovered the solution from the interior initial condition $y(10) = 10$. The exact solution used for error evaluation is $y(t) = e^{-0.1t}\cos(2t) + C$, where $C \approx 10.153$ is determined by the same IC, ensuring the numerical and reference trajectories coincide. The convergence study confirmed an $O(h^4)$ global error rate: the error ratio under step halving was consistently near 16, matching the theoretical prediction and producing a log-log slope of +4.

For Case Study 28.1, the contrast between Euler and RK4 on the Lotka-Volterra system at $h = 0.0625$ was decisive. Euler's phase portrait spiralled outward (energy drift), while RK4 produced a closed periodic orbit, correctly preserving the conservative structure of the Lotka-Volterra equations. This illustrates that global error order matters greatly for long-time integration of nonlinear autonomous systems.

The Lorenz case study showed that RK4 can faithfully resolve chaotic dynamics: the butterfly attractor emerged cleanly, all phase-plane projections matched the known structure of the Lorenz system, and the time series displayed the characteristic aperiodic switching between the two lobes. Together, these results show that RK4 balances computational cost (four function evaluations per step) with fourth-order accuracy in a way that makes it the workhorse method for ODEs in computational engineering.

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# References


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